OPTIMIZING MANGROVE PRESERVATION USING CELLULAR EVOLUTIONARY ALGORITHM AND ENHANCED SPATIAL ANALYSIS FROM REMOTE SENSING DATA

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DEDICATION

This thesis is lovingly dedicated to the people who laid the foundation for our success and fueled our perseverance. To our dearest parents

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X. M. M. B.

A. P. A

ABSTRACT

Mangrove forests are essential coastal ecosystems that protect shorelines from erosion and extreme weather while supporting diverse plant and animal species. However, these ecosystems are increasingly threatened by human activities and environmental change, making effective monitoring and targeted conservation strategies necessary. This study presents a computational approach to optimizing mangrove preservation in Pagbilao, Quezon by integrating satellite-based remote sensing data with a Cellular Evolutionary Algorithm (CEA). Thirty-six cloud-free Sentinel-2 images with a spatial resolution of 10 meters, acquired from 2019 to 2024, were processed to extract vegetation indicators, with the Normalized Difference Vegetation Index (NDVI) used as the primary measure of mangrove health and density. The CEA was implemented in the NetLogo platform to simulate spatial dynamics and identify priority conservation zones through neighborhood-based optimization. Model performance was assessed using accuracy, F1 score, Kappa coefficient, Intersection over Union (IoU), Root Mean Square Error (RMSE), and computation time. The results demonstrate strong model performance, achieving an accuracy of 0.94, an F1 score of 0.93, a Kappa value of 0.87, an IoU of 0.91, and a low RMSE of 0.022. Independent-samples t-test results confirmed statistically significant improvements over the baseline approach ($p < .001$). Overall, the integration of remote sensing and evolutionary computation enhanced spatial coherence and provided a reliable, data-driven basis for sustainable mangrove management.

Keywords - *Mangrove Preservation, Cellular Evolutionary Algorithm, Remote Sensing, Normalized Difference Vegetation Index, Spatial Optimization, NetLogo, Sentinel-2*

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DEFINITION OF TERMS

Cellular Evolutionary Algorithm	Uses a spatially structured population to evolve solutions, often applied in environmental and ecological modeling.
Data Pre-processing	Utilized images to augment the dataset and increase the precision of disease detection.
Intersection over Union	IoU metric for evaluating the accuracy of spatial classifications by comparing the overlap between predicted and reference areas.
Mangrove Ecosystem	Coastal forests made up of salt-tolerant trees and shrubs that provide ecological, social, and economic benefits.
NetLogo	Programming language and environment designed for agent-based modeling
Normalized Difference Vegetation Index	NDVI derived from satellite imagery that measures vegetation health based on reflectance in the red and near-infrared bands.
Sentinel-2	European Space Agency satellite mission providing high-resolution optical imagery for environmental monitoring.
Simulation	Representation of a real-world process, system, or phenomenon over time by means of a model.
Fitness Function	A mathematical formulation used in the Cellular Evolutionary Algorithm to evaluate and rank candidate solutions based on their effectiveness in classifying or optimizing mangrove spatial patterns.

Mangrove Degradation	The decline in mangrove ecosystem structure, health, or extent caused by natural stressors or human activities such as deforestation, coastal development, or pollution
Spatial–Temporal Analysis	An analytical approach that examines changes in mangrove distribution and condition across both space and time using satellite imagery and computational models.
Vegetation Index	A numerical indicator derived from spectral reflectance values used to assess vegetation characteristics such as density, vigor, and canopy health.
Conservation Priority Zones	Specific geographic areas identified through spatial modeling and analysis as requiring immediate protection, monitoring, or restoration to prevent further mangrove loss.
Spatial Resolution	The size of the smallest distinguishable unit in satellite imagery, determining the level of detail at which mangrove features and spatial patterns can be detected and analyzed.
Remote Sensing	The Science and art of obtaining information about an object or phenomenon without making physical contact with it. Satellite imagery is a primary form of remote sensing.

CHAPTER I

INTRODUCTION

Background of the Study

Mangrove forests are among the most important coastal ecosystems, providing essential ecological and socio-economic benefits such as shoreline stabilization, carbon sequestration, and habitat support for numerous aquatic and terrestrial species (Friess et al., 2016). These salt-tolerant trees thrive in intertidal zones, where they trap sediments, reduce coastal erosion, and contribute to nutrient cycling within nearshore environments (Friess et al., 2019). In addition to their role in coastal protection, mangroves store large amounts of carbon in both biomass and soil, making them significant contributors to climate regulation and ecosystem resilience.

Despite their importance, mangrove ecosystems are experiencing rapid degradation worldwide. Human-driven activities such as deforestation, aquaculture expansion, and coastal development have resulted in extensive habitat loss, while climate change has intensified these pressures through sea-level rise and increased storm intensity. These combined stressors reduce biodiversity, weaken natural coastal defenses, and threaten the livelihoods of communities that depend on mangrove-related ecosystem services.

In the Philippines, mangrove degradation remains a persistent environmental challenge. Large areas of mangroves have been converted into fishponds, settlements, and other land uses, leading to significant ecological decline (Buitre et al., 2019; Melana et al., 2011). Although rehabilitation programs have been implemented, many initiatives suffer from limited

monitoring and inadequate site selection, reducing their long-term effectiveness. Traditional mangrove monitoring methods, which often rely on manual interpretation of satellite imagery, are time-consuming and susceptible to human bias, particularly in complex coastal landscapes.

Advancements in remote sensing technology provide opportunities to improve mangrove monitoring and assessment. High-resolution satellite imagery, such as Sentinel-2 data, allows for the extraction of vegetation indices like the Normalized Difference Vegetation Index (NDVI), which is widely used to evaluate vegetation health and density (Chitradevi et al., 2007). When combined with automated and computational approaches, remote sensing enables more accurate detection of spatial and temporal changes in mangrove ecosystems.

Cellular Evolutionary Algorithms (CEA) offer a promising computational framework for addressing spatial optimization problems in environmental management. By simulating evolutionary processes within localized neighborhoods, CEAs can model spatial dynamics and identify priority areas for conservation (Alba & Troya, 2000; Tomassini, 2010). Integrating CEA with remote sensing data allows for adaptive and data-driven analysis of mangrove ecosystems, supporting improved classification accuracy and informed decision-making. Addressing the limited application of this approach in the Philippine context, this study applies a CEA-based spatial analysis model using Sentinel-2 imagery to enhance mangrove monitoring and preservation efforts in Pagbilao, Quezon. This integrated approach aims to provide a more efficient, objective, and scalable method for identifying mangrove conditions and priority conservation zones. The results of this study are expected to

support local government units and environmental agencies in developing evidence-based strategies for sustainable mangrove management and long-term coastal resilience.

Objectives of the Study

To develop and implement a Cellular Evolutionary Algorithm (CEA) that enhances the spatial analysis of mangrove ecosystems using remote sensing data, with the aim of optimizing preservation strategies by identifying critical areas for conservation, monitory degradation patterns, in particular, this research seeks to:

1. Collect and preprocess remote sensing data that focuses on the distribution, health, and changes of mangrove forests over time, utilizing satellite imagery and geospatial analysis.
2. Develop and implement CEA model using NetLogo, enabling the spatial visualization and analysis of the dynamics within the mangrove ecosystem.
3. Evaluate the performance of the proposed CEA-based model using accuracy, F1 score, Kappa coefficient, Intersection over Union (IoU), Root Mean Square Error (RMSE), and Computation Time.
4. Validate the effectiveness of a CEA model in optimizing mangrove management strategies by comparing its outcomes against a baseline.

Significance of the Study

This research develops an optimized computational model using Cellular Evolutionary Algorithms (CEA) and remotely sensed information to enhance the spatial analysis of mangrove forests. The primary aim is to move beyond traditional analysis to identify highly efficient, spatially targeted

conservation strategies that ensure the long-term sustainability and health of these vital ecosystems.

Additionally, the study directly supports several Sustainable Development Goals (SDGs) by providing a more effective, data-driven approach to mangrove management:

SDG 13: Climate Action: Mangroves are critical "blue carbon" ecosystems. The findings will maximize their potential for carbon sequestration and storage, thereby mitigating climate change impacts and enhancing coastal resilience.

SDG 14: Life Below Water: By preserving these important habitats, which serve as essential nursery grounds and food sources for marine life, the research contributes directly to the health of coastal fisheries and overall marine biodiversity.

SDG 15: Life on Land: As essential components of both terrestrial and coastal systems, the enhanced spatial analysis facilitates the sustained conservation of mangroves' unique biodiversity and the crucial ecosystem services they provide.

By providing a more targeted approach to conservation, this study promotes healthier coastal ecosystems and yields significant societal and environmental benefits, including improved water quality, enhanced coastal defense, and better community welfare.

Environmental Agencies and Local Government Agencies: These organizations will gain efficient tools for determining priority areas for conservation, tracking ecosystem health, and assessing restoration success. This information will empower them to implement targeted conservation

policies and improve land-use planning decisions that directly benefit human populations and biodiversity.

Coastal Communities: Communities relying on mangroves will benefit from improved ecosystem health, which provides natural filtration for cleaner water and acts as a strong buffer against coastal erosion and storm surges, safeguarding livelihoods, infrastructure, and the health of residents.

Future Researchers and Scientists: This research advances the body of scientific knowledge by pioneering a new application of CEA in spatial ecological analysis. The methods created for integrating CEA with remote sensing will serve as a valuable framework for future studies in ecological modeling, spatial optimization for conservation, and the use of advanced computational methods in environmental science.

Scope and Limitation

This research project focuses on developing and applying a Cellular Evolutionary Algorithm (CEA) to optimize strategies or parameters relevant to mangrove preservation, utilizing spatial data derived from remote sensing of mangrove forests. The study will involve defining a representation for potential solutions and formulating a fitness function that evaluates the effectiveness of these solutions based on spatial analysis of mangrove characteristics extracted from satellite imagery. The simulation and testing of these evolved strategies will be implemented using NetLogo, a modeling platform that allows researchers to visualize and analyze how mangrove patterns respond to different preservation strategies over time.

The Cellular Evolutionary Algorithm (CEA) will be employed to search through the solution space, iteratively improving solutions based on their

fitness. The research will explore different CEA operators and parameter settings to identify optimal or near-optimal preservation strategies or parameters for the chosen mangrove ecosystems using NetLogo. The performance of the evolved solutions will be assessed by simulating their impact on mangrove spatial patterns using established spatial analysis techniques and by evaluating their potential to enhance mangrove resilience and long-term survival.

The study area, Pagbilao, Quezon, was selected due to its high biodiversity importance, exposure to anthropogenic pressures, and relevance to Department of Environment and Natural Resources (DENR) monitoring initiatives, making it an ideal site for modeling mangrove preservation and optimization strategies through computational methods.

Algorithmic Constraints:

- Fitness Function
- Parameter Tuning
- Computational Cost
- Convergence Behavior
- Exploration and Exploitation Balance
- Algorithm Scalability
- Stochastic Variability
- Simplified Ecological Modeling

The performance of the Cellular Evolutionary Algorithm (CEA) in optimizing mangrove preservation strategies is subject to several algorithmic constraints that may affect its effectiveness, reliability, and practical applicability. One of the main constraints is the fitness function, which serves

as the guiding metric for evaluating candidate solutions. Designing a robust and ecologically meaningful fitness function is challenging, as it must integrate multiple ecological indicators such as vegetation density, spatial distribution, and ecosystem resilience to accurately reflect the health and sustainability of mangroves.

Another key constraint is parameter tuning, including variables such as mutation rate, population size, and iteration limits, which directly influence the search behavior and convergence of the algorithm. Inappropriate settings may result in slow progress, sub-optimal solutions, or inconsistent results. Computational cost is also significant, as evaluating large numbers of potential solutions over high-resolution or multi-temporal datasets requires substantial processing power and time, which may limit the number of scenarios that can be tested. The algorithm's convergence behavior can present limitations if it prematurely settles on local optima, reducing the exploration of potentially better solutions, while maintaining an appropriate exploration-exploitation balance is crucial to efficiently searching the solution space without missing high-quality outcomes.

Additionally, stochastic variability inherent in evolutionary algorithms means that results may differ between runs, necessitating multiple simulations for reliability. Finally, the simplified ecological modeling used in NetLogo, though useful for visualization and testing, may not fully capture complex real-world ecological processes, environmental interactions, or species-specific responses, which can limit the accuracy, ecological relevance, and interpretability of the algorithm's outputs. These constraints highlight the need

for careful design, calibration, and interpretation when applying the CEA to real-world ecosystem management.

Furthermore, the scope of this study is limited to the use of remotely sensed spatial data and computational modeling, without direct field validation or ground-truthing of mangrove conditions. While satellite imagery provides valuable large-scale and temporal insights, it may not fully capture fine-scale ecological factors such as soil conditions, species composition, hydrological dynamics, and human disturbances at the local level. The preservation strategies generated by the Cellular Evolutionary Algorithm (CEA) are therefore intended as decision-support tools rather than definitive management prescriptions. As a result, the findings should be interpreted as exploratory and complementary to empirical ecological studies and expert-driven conservation planning, rather than as substitutes for on-site assessments and policy implementation.

Moreover, this study is constrained by the availability and temporal resolution of Sentinel-2 satellite imagery, which may be affected by cloud cover, atmospheric conditions, and data gaps that influence the accuracy of vegetation indices and spatial classifications. The analysis focuses on a specific time period and geographic location, limiting the generalization of the results to other mangrove ecosystems with different environmental conditions or management contexts. Consequently, the findings are best interpreted within the defined spatial and temporal scope of Pagbilao, Quezon, and future studies may extend this framework by incorporating multi-seasonal datasets, higher-resolution imagery, field validation, and comparative analyses across multiple sites.

CHAPTER II

THEORETICAL FRAMEWORK

This chapter presents the relevant literature and studies acquired from both local and international sources. Literature is from books, and professional journal articles. This foundation will inform the methodology, analysis, and interpretation of findings in the following sections, providing a solid theoretical groundwork for addressing research problems.

All the research literature and studies are cited through a thorough review by the researchers in connection with various ideas, concepts, generalization, conclusion and developments related to the study. Furthermore, the information in this chapter serves as a guide and helps the researchers understand details that correspond to and are like current research.

Cellular Evolutionary Algorithms in Environmental Modeling

(Hassan et al.,2020) define Cellular Evolutionary Algorithms (CEAs) as bio-inspired computational techniques that mimic mechanisms of natural selection within a cellular framework to facilitate evolutionary processes and local interactions. These algorithms have been applied to various environmental modeling contexts, including land use mapping, habitat suitability modeling, and resource allocation. Their capability to handle complex, high-dimensional datasets makes them particularly useful for simulating dynamic mangrove ecosystem behavior.

In computational optimization, evolutionary algorithms have extensively solved multi-objective problems. In mangrove conservation, CEAs are especially appealing due to their local search mechanism that partitions the

problem space into neighborhoods, thus effectively utilizing local information and preserving population diversity (Ding & Zheng, 2015). (Biswas et al. 2015) demonstrated that incorporating niching behaviors ensures the optimization process identifies multiple sustainable solutions simultaneously-a key consideration when addressing the multi-modality of environmental systems. (Jiang and Yang 2017) further showed that evolutionary algorithms' adaptability enables integration with spatial data to dynamically update conservation models as new remote sensing data are acquired.

As a variant of traditional evolutionary algorithms, CEAs have proven particularly effective for spatial problem-solving. Individuals (solutions) in CEAs are positioned on a lattice structure, with relationships between points making them especially suitable for modeling spatial phenomena found in ecological systems. The spatial structure maintains population diversity while searching for global solutions, avoiding common difficulties in genetic algorithms such as stagnation or premature convergence. CEAs have addressed various environmental challenges, including habitat rehabilitation and land use planning. For instance, CEAs have simultaneously optimized multiple objectives, such as maximizing biodiversity while minimizing economic development costs (Rodríguez & Broussard, 2016). These results can be integrated with Geographic Information Systems (GIS) to create real-time conservation status models. When combined with remote sensing data, CEAs can continuously update conservation models based on changing environmental conditions, making them valuable for long-term management challenges like mangrove preservation where ecological circumstances vary due to tides, storms, or human activities.

Unlike classical linear optimization methods, CEAs are more appropriate for biological and ecological applications where interactions are localized and dynamic. This is particularly useful in mangrove conservation, where landscape connectivity and ecological zoning must be considered (Ding & Zheng, 2015). Integrating CEAs with real-time positional information enhances their performance. (Jiang and Yang 2017) implemented CEAs with remote sensing inputs and found that conservation models can dynamically respond to regional environmental variations. This flexibility is essential in rapidly changing coastal zones, where dynamics such as erosion, flooding, and vegetation loss can change rapidly.

Environmental planning and land use optimization represent prominent CEA applications. (Rodríguez and Broussard 2016) implemented a CEA to optimize land use allocation in deforestation-prone areas. Their model balanced ecological sustainability and economic needs, demonstrating that CEAs can effectively conduct spatial assessments and generate robust land use scenarios. In water resource and hydrological modeling, cellular algorithms have calibrated models with high-dimensional, spatially sensitive input parameters. (Chen et al. 2018) demonstrated CEA efficacy in optimizing parameters of a distributed hydrological model (SWAT) for sediment efficiency prediction, improving model accuracy and calibration efficiency compared to traditional global search methods.

While numerous studies have demonstrated CEAs' potential in environmental modeling, most focus on general land-use or hydrological applications. Limited research has explored CEA integration with remote sensing data for mangrove ecosystem analysis, particularly within the

Philippine context. This presents a clear research gap for developing localized, adaptive, and data-driven conservation models.

Mangrove Forest and Their Preservation

Mangrove forests are valuable coastal ecosystems providing numerous ecological services including shoreline protection, carbon sequestration, and support for marine and terrestrial biodiversity (Heumann, 2011; Abdel-Hamid et al., 2018). These salt-tolerant woody plants inhabit tropical and subtropical intertidal zones but face substantial deforestation due to demands for wood and forest products, aquaculture conversion, and urban coastal development, necessitating advanced monitoring and preservation strategies.

Mangrove forests represent crucial coastal ecosystems worldwide. They provide significant benefits including shoreline stabilization, fisheries habitat provision, and carbon storage. Unfortunately, many mangrove areas face severe threats from land use changes, pollution, and climate change. With increasing anthropogenic pressure, researchers are investigating advanced technologies to improve mangrove conservation and restoration efforts.

Globally, mangroves play significant roles in carbon sequestration, biodiversity maintenance, and coastal resilience. Despite their importance, they continue to decline due to human activities and climate-induced changes. In the Philippines, the total mangrove area has decreased from approximately 450,000 hectares in 1920 to 311,400 hectares in recent years (DENR, 2020). Addressing this decline, the government has implemented several programs including the Mangrove and Beach Forest Development Project (MBFDP), the

National Mangrove Conservation and Rehabilitation Program, and the Integrated Coastal Management (ICM) initiative led by the Department of Environment and Natural Resources (DENR).

Another government initiative, the Philippine Plan of Action for Sustainable Fisheries (PPASF), represents a comprehensive program for preserving and restoring mangrove ecosystems-essential habitats for numerous fish species. It seeks to guarantee fishery resource sustainability, which is crucial for economic stability and food security. Additionally, The National Greening Program (NGP) aims to plant 1.5 billion trees, including mangroves, from 2011 to 2028. This initiative employs a participatory approach, engaging government and civil society to halt Philippine forest and mangrove loss while mitigating climate change impacts (Lu, 2024). Given the extreme weather conditions currently affecting the Philippines and globally, this conservation efforts are paramount.

Mangrove conservation also involves protected area networks and legislation (Climate Change Commission, 2024). Legal basis for mangrove conservation and management is provided by several laws and administrative directives, including Presidential Proclamation No. 2151, which declared small islands indented by mangroves as Wilderness Areas, and Presidential Proclamation No. 2152, declaring Palawan and other mangrove parcels totaling 74,267 hectares as Mangrove Swamp Forest Reserves.

Complementing government initiatives, recent local research has begun leveraging technology for mangrove management. Studies have utilized Google Earth Engine for mapping mangrove extent in specific Philippine bays (Pamungkas, 2023) and explored machine learning for

classification (Ambarwari & Husni, 2023). However, these studies primarily focus on mapping and monitoring. The present study builds upon this foundation by moving beyond descriptive mapping to predictive and optimization modeling using CEAs, addressing a more complex layer of conservation planning currently underrepresented in local literature.

Remote Sensing for Mangrove Monitoring

Remote sensing has become a cornerstone in environmental monitoring due to its ability to collect frequent, large-scale, and high-resolution spatial data (Xia et al., 2020). It enables analysis of mangrove extent, biomass, and health through multispectral, hyperspectral, and radar-based imagery. Studies have shown that integrating multiple sensing technologies improves classification accuracy significantly from 76% with single sensors to over 91% when combined (Ghorbanian et al., 2022; Abdel-Hamid, 2018).

Remote sensing technologies have revolutionized mangrove forest monitoring and investigation by providing detailed spatial and temporal data from challenging environments. Previous research has examined numerous remote sensing applications for mangrove ecosystems, underscoring the effectiveness of optical sensors, radar systems, and LiDAR in capturing diverse aspects of mangrove structure and health. Results demonstrate that integrating multiple sensing technologies enhances classification accuracy from 76% with individual sensors to over 91% with combined techniques.

Remote sensing has emerged as a significant monitoring tool for mangrove forests, allowing for multi-temporal, large-scale, and high-resolution spatial data collection (Heumann, 2011; Xia et al., 2020). Methods based on multispectral and hyperspectral imagery and vegetation indices such as NDVI

have successfully mapped mangrove cover, health, and spatial distribution changes.

Combining advanced computational approaches with remote sensing has created promising avenues for mangrove habitat preservation and management. Mangroves are crucial for coastal sustainability; however, they face rapid degradation due to anthropogenic stresses and climate change (Heumann et al, 2011). (Xia et al. 2020) noted that remote sensing methods have been extensively used to monitor these ecosystems because of their ability to provide frequent, high-resolution spatial data over large areas. Several studies have demonstrated the feasibility of various image classification approaches powered by machine learning algorithms to extract vital information about mangrove extent and health (Ghorbanian et al., 2022; Abdel-Hamid et al, 2018).

Several studies have utilized remote sensing for mangrove monitoring. (Blanchard et al. 2024) used Sentinel-2 data to monitor the production and dissemination of products and services useful for local-scale mangrove monitoring. Their online platform, MangMap, is an end-user driven portal designed to support scientific studies and help institutional actors and stakeholders involved in mangrove conservation and management. MangMap enhances understanding of mangrove dynamics through up-to-date mapping and monitoring tools. (De Souza et al. 2024) found that the Mamba model, recognized in remote sensing for efficient inference with high-resolution images in downstream tasks (Yuan et al., 2021; Chen et al., 2024; Wang et al., 2025), surpasses convolutional and transformer models. These findings support using cutting-edge deep learning methods in environmental remote

sensing, providing a strong foundation for enhancing comprehension of global mangrove dynamics.

(Giri et al., 2015) identified remote sensing as one of the most effective approaches for understanding and monitoring mangroves. Satellites and drones have accelerated mangrove mapping processes, enabling rapid change detection and health assessment over the past decade. (Pham et al., 2019) confirmed these tools help conservation practitioners more effectively monitor mangrove ecosystem status.

(Pham et al., 2019) noted these advances are particularly valuable for nations like the Philippines, where vast, remote coastlines traditionally make direct monitoring difficult. Using vegetation indices like NDVI and machine learning classification of satellite data, remote sensing helps fill critical data gaps. Tools such as MangMap, introduced by (Blanchard et al. 2024), provide stakeholders access to frequently updated mangrove data online. Similarly, deep learning models like Mamba have proven more accurate than earlier computer vision and transformer models in automatically analyzing high-definition aerial images for environmental assessments (de Souza et al., 2024).

While remote sensing techniques have advanced mangrove mapping, few studies have coupled these datasets with optimization algorithms like CEAs to simulate future mangrove conditions or predict degradation patterns. This integration represents a novel contribution of the present study.

Optimization Algorithms in Ecosystem Preservation

According to (Li et al., 2018), Particle Swarm Optimization (PSO) has shown significant promise for land use optimization and resource planning.

PSO optimized ecological corridors in China's urbanizing areas to effectively preserve habitat connectivity with minimal land purchase costs, demonstrating the algorithm's simplicity and scalability.

Nature-inspired computational optimization techniques have gained popularity in environmental management. Evolutionary algorithms, including Genetic Algorithms (GA) and Particle Swarm Optimization (PSO), have effectively resolved complex resource allocation and conservation planning problems.

(Zhang et al. 2017) utilized GA to identify optimal locations for protected mangrove areas, balancing ecological importance with socio-economic considerations. Similarly, (Li and Wang 2020) used PSO to prioritize restoration sites based on land suitability and threat levels.

Hybrid optimization models combining techniques like GA-PSO or ACO-GA have emerged as powerful tools. Zhao (2022) proposed a hybrid GA-PSO framework for optimizing marine protected areas, integrating satellite imagery and bathymetric data to support adaptive marine conservation. These hybrids offer enhanced convergence rates and solution accuracy.

Multi-Objective Evolutionary Algorithms (MOEAs) such as NSGA-II and MOEA/D have recently addressed ecosystem services trade-offs. Chan et al. (2020) developed an MOEA framework for watershed land-use planning while concurrently optimizing flood risk reduction, carbon sequestration, and biodiversity conservation. This approach enables decision-makers to select solutions balancing different ecological priorities.

Current literature rarely integrates CEAs, remote sensing, and socio-ecological variables in a unified conservation framework. The present study

addresses this gap by proposing a hybrid, multi-dimensional model for adaptive mangrove management in the Philippines.

Importance of Mangroves in Coastal Ecosystems

Mangroves represent biodiversity havens in tropical and subtropical regions while providing vital ecosystems services essential for fisheries resources. They perform crucial functions including wave energy dissipation, shoreline protection from storm surges, and sediment stabilization.

According to (Gilma et al., 2008), mangroves act as significant carbon sinks and support diverse biological communities. However, their decline threatens ecological stability and fishery resources. Destruction also removes timber and fuelwood resources upon which coastal communities depend. Mangrove forest degradation represents a global problem driven by urban development, aquaculture expansion, and unsustainable resource extraction. In Southeast Asia and the Philippines specifically, mangroves have experienced rampant deforestation and misguided land-use conversion for decades (Primavera et al., 2019). This continuing trend demands data-driven, pragmatic models for sustainable ecosystem management.

Challenges and Future Directions

(Dahdouh-Guebas et al., 2021) identified challenges in transforming raw spectral data into ecologically meaningful indicators. The computational complexity of handling high-resolution datasets presents significant barriers. Innovations in edge computing and automated feature extraction may help address these limitations.

(Ward et al., 2020) proposed integrating local ecological knowledge with computational models to develop conservation plans that are both

scientifically sound and culturally appropriate. Similarly, models incorporating human decision-making processes and policy contexts provide better guidance for practical implementation (Richards & Friess et al., 2019). Ultimately, researchers need to enhance these tools for rapid decision-making support. As mangrove conservation urgency increases, integrating diverse disciplines including ecological science, satellite remote sensing, and computational intelligence has become imperative rather than optional.

Despite encouraging progress, significant challenges remain in combining CEAs with remote sensing for mangrove preservation. (Dahdouh-Guebas et al., 2021) identified limitations including difficulties transforming spectral data into biologically meaningful parameters and computational complexity when processing high-resolution spatial data in evolutionary models. They suggested that automated feature extraction and edge computing developments could potentially address these constraints.

(Ward et al., 2020) proposed a methodology integrating CEAs with remote sensing and local ecological knowledge to generate more robust preservation designs. Their research demonstrated how utilizing multiple knowledge sources could enhance model accuracy while incorporating necessary socio-ecological concerns for successful conservation. Similarly, (Richards & Friess et al., 2019) emphasized that models must incorporate policy scenarios and human intervention alongside biological processes to provide actionable recommendations for decision-makers.

Merely conducting mapping exercises proves insufficient for effective conservation. To prioritize areas for protection and restoration, sophisticated data analysis and informed decision-making are essential. Optimization

algorithms provide critical support in this context. In recent years, cellular evolutionary algorithms (CEAs) have gained recognition for solving complex, spatially distributed problems characteristic of mangrove management. Unlike traditional optimization methods, CEAs operate by creating solution neighborhoods on a grid structure, enabling exploration of multiple possibilities while preserving spatial landscape configuration.

Existing literature rarely considers multi-dimensional integration-combining CEAs, remote sensing data, and socio-ecological factors-for adaptive mangrove management. This interdisciplinary approach represents the novel contribution of the present study.

Simulation using NetLogo

NetLogo continues to hold a foundational role in agent-based modeling (ABM) as an exceptional tool for rapid prototyping and educational instruction. Its core architecture, built upon a simplified programming syntax and an integrated graphical environment, allows for the immediate visualization of model outcomes, creating a direct feedback loop between hypothesis and result (Cocho-Bermejo, A. et al., 2025). This capability is crucial for iterative model development, where understanding emerges from the process of building and experimenting rather than from pre-defined outcomes. The platform's extensive Models Library, consistently updated with new contributions, serves as a vital repository of cross-disciplinary "thought experiments," enabling students and researchers to deconstruct and build upon established models from fields ranging from epidemiology to social dynamics (Wilensky & Rand, 2015). As of 2024, the active maintenance of NetLogo, with its most recent version 6.4.0 available through the Center for

Connected Learning and Computer-Based Modeling, ensures its continued relevance in academic curricula focused on computational thinking and complex systems science

NetLogo provides an agent-based simulation framework suitable for rapid prototyping. Its user-friendly design incorporates a built-in graphical interface and robust visualization tools, enabling immediate feedback on model behavior. This environment facilitates development and exploration of complex system dynamics. Additionally, NetLogo's extensive Models Library contains pre-made, well-documented examples from numerous disciplines that serve as excellent starting points (Umlauft, M., & Schranz, M. 2024) , accelerating prototyping through ready-to-use components and concepts. This combination of simplicity, interactive features, and extensive resources makes NetLogo an accessible platform for creating and refining agent-based simulations. Agent-based modeling is a unique supplement to a more conventional system wide framework (Collins, B., & Liang, C.-L. 2025).

Literature Matrix

Table 1. Matrix of Review of Related Literature

AUTHOR (S)	YEAR	TITLE	METHOD/ ALGORITHM	FINDINGS
Blanchard, Catry, Delbar, Marland, Sonko, Mouquet, Delaître, & Faure	2024	MangMap: A New Platform for Mangrove Mapping and Monitoring Using Earth Observation Data	Integration of Sentinel-2 Data in monitoring the production and dissemination of specific products and services useful for mangrove monitoring at local scales.	Provides services that calculate differences from date to date or between temporal composites of indexes; statistics calculation: changes overtime in index values (mean, quantiles, and standard deviation); vector difference calculation: estimation of the spatial evolution of

				the mangrove between 2 dates.
de Souza, L. J., Zreik, I. V., Sermanet, A. S., Seghouani, N., Pourchier, L.	2024	A Deep Learning-Based Approach for Mangrove Monitoring	Introduced open-source dataset, MagSet-2, incorporating mangrove annotations from the Global Mangrove Watch and satellite images from Sentinel-2, from mangrove positions all over the world. Then benchmark three architectural groups, namely convolutional, transformer, and mamba models, using the created dataset.	The experimental outcomes further validate the deep learning community's interest in the Mamba model, which surpasses other architectures in all metrics.
Fatima Belhabib, Mohamed Benslimane, Karim El Moutaouaki	2024	Advanced Big Data Analytics: Integrating Fuzzy C-Means, Encoder-Decoder CNNs, and Genetic Algorithms for Efficient Clustering and Classification.	Integration of fuzzy C-means clustering, encoder–decoder CNNs, and genetic algorithms for data clustering and classification.	Demonstrated that the marriage of evolutionary algorithms and deep CNNs significantly improves classification and clustering accuracy in big data analysis, and provides a methodological basis for combining evolutionary methods (e. g., Cellular Evolutionary Algorithms) with CNNs for spatial optimization
Berlanga-Robles	2024	Trends in mangrove canopy and cover in the Teacapan–Agua Brava Lagoon System (Marismas Nacionales), Mexico: An approach using open-access geospatial data	Multi-temporal analysis using open-access satellite data	Detailed multi-temporal trends revealed cyclical and long-term changes in canopy structure and cover, emphasizing the importance of using diverse satellite sources for comprehensive monitoring.
Tian, Hu, Zhou	2023	Mangrove Biodiversity Assessment Using UAV Lidar and Hyperspectral Data in China’s	UAV-based LiDAR and hyperspectral imaging for assessing spatial structure and spectral diversity	Offered a detailed framework for biodiversity assessment that can quantify species variations and

		Pinglu Canal Estuary		structural parameters in mangrove forests.
Ambarwari & Husni	2023	Studying How Machine Learning Maps Mangroves in Moderate-Resolution Satellite Images	Multiple machine learning algorithms (e. g., various ML classifiers applied in a structured mapping workflow).	Compared and contrasted ML approaches on medium-resolution images, with a focus on the significance of algorithm choice towards better mangrove classification performance.
M. Rondón, Ewane Basil Ewane, Meshal M. Abdullah	2023	Remote sensing-based assessment of mangrove ecosystems in the Gulf Cooperation Council countries: A systematic review	Systematic review of multi-scale remote sensing approaches across diverse RS sensors	Summarized trends, challenges, and potentials in using remote sensing for monitoring mangrove health and identifying stressors, emphasizing gaps that could benefit from optimization algorithms.
Pamungkas	2023	Analysis Of Vegetation Index For NDVI, EVI-2, and SAVI For Mangrove Forest Density Using Google Earth Engine In Lembar Bay, Lombok Island	Vegetation index analysis (NDVI, EVI-2, SAVI) computed using Google Earth Engine for estimating mangrove density	Provided reliable estimation of mangrove forest density and health status, reinforcing the utility of spectral indices in monitoring mangrove ecosystems.
Xue, Fatoyinbo, Thomas	2021	Large-Scale High-Resolution Coastal Mangrove Forests Mapping Across West Africa with Machine Learning Ensemble and Satellite Big Data	Machine learning ensemble methods (e. g., Random Forest, other ensemble classifiers)	Showed that ensemble ML methods can produce enhanced accuracy for high-resolution satellite big data coastal mangrove mapping, providing a framework for large-scale spatial.
Neda Bihamta Toosi, A. Soffianian, Sima Fakheran	2020	Land Cover Classification in Mangrove Ecosystems Based on VHR Satellite Data and Machine Learning—An Upscaling	Integration of very high resolution (VHR) satellite data with machine learning for land cover classification	Achieved enhanced mapping accuracy and scalability in delineating mangrove extents, providing insights into spatial heterogeneity.

		Approach		
Samy Ismail Elmahdy, Tarig Ali	2020	Incorporating the Plant Phenological Trajectory into Mangrove Species Mapping with Dense Time Series Sentinel-2 Imagery and the Google Earth Engine Platform	Use of dense time series Sentinel-2 imagery combined with NDVI time series and Random Forest classification, implemented on the Google Earth Engine platform	Showed the feasibility of phenology-based classification to resolve species-level distinctions in mangroves and improve temporal monitoring.
Tien Dat Pham, Naoto Yokoya, Dieu Tien Bui	2019	Remote Sensing Approaches for Monitoring Mangrove Species, Structure, and Biomass: Opportunities and Challenges	Review of various remote sensing techniques and state-of-the-art machine learning algorithm	Given a detailed account of remote sensing applications in mangrove monitoring, identifying the main limitations and proposing future integration with advanced algorithms.
Alejandro Baldominos, Yago Sáez, Pedro Isasi	2019	Hybridizing Evolutionary Computation and Deep Neural Networks: An Approach to Handwriting Recognition Using Committees and Transfer Learning	Hybrid methodology integrating evolutionary computation (genetic algorithms) with deep CNN architectures	While the current study is concerned with handwriting recognition, it illustrates that hybrid evolutionary algorithms can optimize convolutional neural networks for better performance; the underlying method can be used to solve other spatial optimization problems like mangrove mapping, although direct application requires additional research.
Tien Dat Pham, Naoto Yokoya, Dieu Tien Bui	2019	Remote Sensing Approaches for Monitoring Mangrove Species, Structure, and Biomass: Opportunities and Challenges	Comprehensive review discussing the use of various remote sensing techniques and sensors	Identified limitations and challenges in monitoring biomass and carbon stocks, recommending integration of advanced computational methods to improve accuracy.
Nathan Thomas, Peter Bunting, Richard Lucas	2018	Mapping Mangrove Extent and Change: A	Standardized remote sensing analysis using	Provided a global-applied approach for mapping the extent

et al.		Globally Applicable Approach	Landsat scenes and quantifiable accuracy assessments	of mangroves with an emphasis on the utility of remote sensing observations for monitoring temporal change in area
Benjamin Heumann	W. 2011	An Object-Based Classification of Mangroves Using a Hybrid Decision Tree—Support Vector Machine Approach	Hybrid decision tree–support vector machine (SVM) approach for object-based classification	Attained greater accuracy in mangrove mapping by removing spectral ambiguity and spatial heterogeneity, warranting the use of hybrid classifiers in remote sensing.

Synthesis and Relevance of the Study

Mangrove ecosystems play a critical role in coastal protection, carbon storage, and biodiversity conservation, yet they remain highly vulnerable to both anthropogenic pressures and environmental change. Previous studies consistently emphasize the ecological importance of mangroves, particularly their function in stabilizing shorelines, supporting fisheries, and mitigating climate impacts (Heumann, 2011; Primavera et al., 2019). Despite this recognition, global and local assessments indicate a continued decline in mangrove coverage, highlighting the need for more effective monitoring and management approaches.

Remote sensing has emerged as a widely accepted tool for mangrove monitoring due to its ability to provide large-scale, multi-temporal, and high-resolution spatial data. Numerous studies have demonstrated the effectiveness of vegetation indices, such as the Normalized Difference Vegetation Index (NDVI), in assessing mangrove health, density, and spatial distribution (Xia et al., 2020; Ghorbanian et al., 2022). The integration of multispectral satellite data, particularly from Sentinel-2, has further improved

classification accuracy and change detection capabilities. However, much of the existing literature remains focused on descriptive mapping and retrospective analysis rather than predictive or optimization-based approaches.

In recent years, evolutionary and optimization algorithms have been increasingly applied in environmental and land-use planning contexts. Cellular Evolutionary Algorithms (CEAs) have demonstrated strong potential for solving spatially complex and multi-objective problems due to their localized neighborhood interactions and ability to preserve population diversity (Alba & Troya, 2000; Tomassini, 2010). Studies applying CEAs to environmental modeling have shown improved spatial coherence and adaptability compared to traditional global optimization techniques (Ding & Zheng, 2015; Rodríguez & Broussard, 2016). However, applications of CEAs specifically for mangrove ecosystem analysis remain limited, especially within the Philippine context.

The reviewed literature reveals a clear research gap at the intersection of remote sensing, evolutionary computation, and mangrove conservation. While advanced remote sensing techniques provide detailed spatial information, and evolutionary algorithms offer robust optimization capabilities, few studies have effectively combined these approaches to support adaptive mangrove management. Addressing this gap, the present study integrates Sentinel-2–derived vegetation indices with a Cellular Evolutionary Algorithm to move beyond static mapping toward spatial optimization and predictive analysis. This approach contributes a novel, data-driven framework for identifying conservation-priority zones and supporting sustainable mangrove management in Pagbilao, Quezon.

Conceptual Framework

The conceptual framework developed in this study tries to present the design and implementation of a Cellular Evolutionary Algorithm (CEA) to enhance spatial evaluation of mangrove ecosystems based on remote sensing data. The framework will examine the use of cellular automata concepts in enhancing conservation by identifying priority areas for conservation and monitoring degradation trends with advanced spatial analysis techniques.

The model development for this study consists of three phases: data acquisition and preprocessing, CEA development and selection, and model evaluation. Data acquisition and preprocessing are crucial steps in the study, during which the analysis must commence. This study concentrates on acquiring imagery of mangrove ecosystems from trusted remote sensing platforms, including Sentinel-2. The diversity of input data from the different platforms aids in ensuring the efficiency and effectiveness of the study.

Data augmentation techniques to improve the analysis and avoid over generalization are used. These techniques enrich the datasets with anthropogenic and natural factors responsible for the stress on mangrove ecosystems. Then, processed data goes through feature extraction and selection to retrieve relevant spatial patterns for subsequent analysis tasks.

Transitioning to the CEA development phase, design and implementation of CEA were carried out. The CEA methodology was considered for modeling the evolution of spatial dynamics and evolutionary patterns of mangrove ecosystems with much emphasis put on degradation mechanisms and resilience factors. After the CEA design, spatial analytical

tools were used for optimizing the decision-making processes for conservation purposes.

These tools were concerned with prioritizing conservation efforts while monitoring degradation patterns through higher-order spatial analytic approaches. The last activity in this phase is to formulate management strategies concerning design strategies for the management of Mangrove ecosystems in relation to the recognized spatial patterns and evolutionary trends.

After CEA development, the performance evaluation stage weighs the performance metrics such as IoU (Intersection over Union), F1 score, Kappa coefficient, and RMSE (Root Mean Square Error) against the model. These metrics inform how well CEA is working in the application and the usefulness of each configuration for the specific research objective. In the validation presentation, model outputs will classify mangrove ecosystem health status into healthy, degraded, or critical. Improvements in the spatial decision making for mangrove conservation will also be validated by comparing CEA optimally positioned conservation strategy with the conventional ones by Independent T-test analysis.

In addition to quantitative validation, the framework focuses on utilizing spatial visualization tools for better interpretation of model results. Thematic maps, heatmaps, and spatial overlays facilitate easy navigation between the patterns of mangrove condition, and CEA optimized classifications can be visually compared against baseline remote sensing interpretations. These plots do not only increase the interpretation of the results of the algorithm, but also provide tools for an evidence-based policy (e. g., identifying ecological

hotspots, transition zones and anthropogenically pressured areas). Such geospatial insights may help decision-makers to better predict ecological threats and to plan site-specific conservation interventions.

The conceptual framework highlights the importance of translating computational findings into practical, policy-aligned conservation strategies. The results produced by the CEA model, coupled with statistical validation, are intended to bridge scientific analysis with real-world environmental management. By identifying priority zones for protection, restoration, or monitoring, the framework positions the CEA as a strategic tool for local government units, environmental agencies such as the DENR, and community-based coastal managers. This reinforces the idea that advanced spatial models are not only technical innovations but also essential components in adaptive and sustainable mangrove ecosystem management.

Furthermore, the conceptual framework emphasizes the iterative and adaptive nature of the proposed CEA-based system, wherein feedback from model evaluation and validation continuously informs subsequent refinements in data processing, rule calibration, and spatial transition parameters. This feedback loop allows the model to dynamically adjust to changing environmental conditions and emerging anthropogenic pressures, ensuring long-term applicability and robustness. By incorporating iterative learning and adaptive calibration, the framework supports sustainable decision-making and enhances the capacity of stakeholders to respond proactively to evolving mangrove ecosystem dynamics. This adaptive framework ensures that conservation strategies remain responsive to both short-term disturbances and long-term ecological trends. As a result, the CEA model serves not only

as a predictive tool but also as a continuous decision-support system for sustainable mangrove ecosystem management.

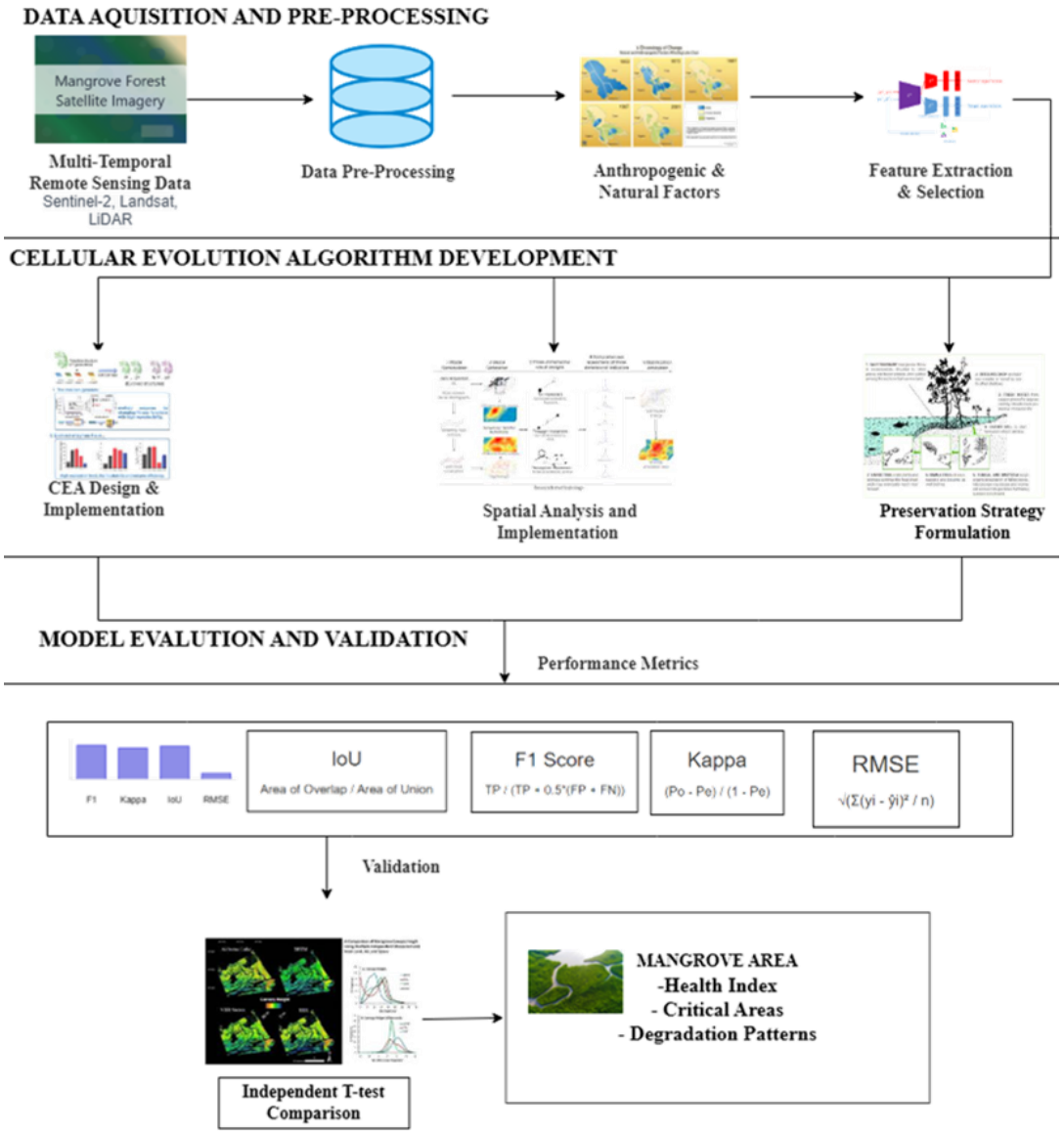


Figure 1. Conceptual Framework Integrating Remote Sensing and Cellular Evolutionary Algorithm (CEA) for Mangrove Preservation

CHAPTER III

METHODOLOGY

This chapter outlines the methodological framework employed to develop and apply a Cellular Evolutionary Algorithm for optimizing mangrove management strategies, utilizing spatial data derived from remote sensing. It details the specific study area, the materials used in the research, and the software employed for data processing, analysis, and algorithm implementation.

STUDY AREA



Figure 2. Map Illustrating Mangroves Zones in Pagbilao, Quezon

Pagbilao, Quezon, presents a significant and relevant study area for investigating mangrove ecosystems within the Philippine archipelago. Pagbilao hosts notable mangrove stands that contribute substantially to the

ecological health and resilience of the Quezon province coastline. These mangrove forests provide critical habitat for diverse marine and terrestrial life, supporting local fisheries and contributing to the overall biodiversity of the region. Understanding the dynamics and health of Pagbilao's mangroves is crucial for local conservation efforts and for recognizing their role within the broader context of Philippine coastal ecosystems.

The coastal communities of Pagbilao directly benefit from the ecosystem services provided by these mangroves. They act as natural protection against storm surges and coastal erosion, which are increasing concerns in this area. The mangroves also play a role in nutrient cycling and sediment trapping, contributing to water quality and the health of adjacent marine environments. Recognizing these vital functions underscores the importance of effective mangrove management in Pagbilao.

Studying mangrove management in Pagbilao offers a valuable opportunity to understand local challenges and conservation strategies. Investigating mangrove cover changes and health using innovative approaches like Cellular Evolutionary Algorithm and spatial data derived from remote sensing can provide crucial insights for local environmental agencies and communities involved in conservation efforts. Understanding the specific threats facing Pagbilao's mangroves is essential for developing tailored and effective management strategies. The application of spatial analysis techniques, leveraging available remote sensing data, can significantly enhance the ability to monitor, assess, and ultimately ensure the long-term sustainability of Pagbilao's valuable mangrove ecosystems.

A total of 36 cloud-free Sentinel-2 Level-2A images were collected from

January 2019 to December 2024 through the Google Earth Engine (GEE) platform. Each image provides 10-meter spatial resolution across visible and near-infrared bands used for NDVI computation. The study area, Pagbilao, Quezon, Philippines, is located at approximately 13°58′45″ N latitude and 121°43′27″ E longitude, covering an estimated 1,500 hectares of mangrove and adjacent coastal zones. Cloud masking was applied using the Sentinel-2 QA60 band to ensure accurate reflectance values.

MATERIALS

The study utilized a specific set of materials, tools, and resources essential in completing the model, encompassing satellite imagery for spatial data extraction, field data for ground truthing and validation, and computational resources for algorithm implementation and analysis.

Python

Python as the primary programming language for this research was driven by its robust ecosystem of scientific and geospatial libraries, its relatively gentle learning curve, and its widespread adoption within the remote sensing, ecological modeling, and evolutionary computation communities. Python's open-source nature fosters a large and active community, resulting in extensive documentation, readily available online resources, and a wealth of pre-built packages that significantly streamline complex tasks.

Specifically, libraries such as GDAL and Rasterio provide powerful tools for handling and processing large raster datasets associated with satellite imagery, while GeoPandas facilitates efficient manipulation and analysis of vector-based geospatial data like coastlines and existing maps. Furthermore, Python’s capabilities in numerical computing, offered by NumPy and SciPy,

are essential for the mathematical operations underpinning both the spatial analysis of remote sensing data and the implementation of the Evolutionary Algorithm. The DEAP (Distributed Evolutionary Algorithms in Python) framework offers a flexible and well-documented platform for designing and executing evolutionary computations, which form a core component of this study's methodology. Finally, the rich data visualization capabilities of Matplotlib and Seaborn enable effective communication of findings through informative charts, graphs, and maps.

Visual Studio Code

VS Code is a lightweight yet powerful and highly extensible code editor that offers excellent support for Python development, extensive marketplace of extensions offers a wide range of tools that enhance productivity for scientific computing and data science workflows. Also, a widely adopted and actively supported IDE within the data science and software development communities, ensuring ample online resources, documentation, and community support were readily available should any issues or questions arise during the research. Its cross-platform compatibility also ensured a consistent development experience regardless of the underlying operating system.

Netlogo

Netlogo is a flexible, user-friendly system for developing agent-based models, especially for modelling complex systems, and in ecosystem and environmental models. With encrypted innate abilities and a friendly interface, this simulation was implemented in the cellular evolutionary algorithm used in the current study to simulate and optimize the mangrove preservation.

Academics and researchers around the world use NetLogo which has strong documentation, an active community supporting it, and thousands of example models, helping across all stages in the research workflow.

Sentinel-2

Sentinel-2 is essential for identifying mangrove areas that provide not only visibility in the blue, green, and red spectral bands, which allows for the initial detection of vegetation, but more importantly, it offers a range of additional spectral bands crucial for accurately distinguishing mangroves from other land cover types and assessing their characteristics.

NumPy

NumPy is the cornerstone of numerical computing in Python, efficient in handling multi-dimensional arrays and its extensive library of high-level mathematical functions. Satellite imagery, the primary data source, is inherently structured as raster data, which NumPy's array operations could process and manipulate with speed and efficacy. This capability was crucial for tasks ranging from accessing and modifying pixel values to performing complex band calculations necessary for deriving vegetation indices and other spatial metrics.

SciPy

SciPy played a crucial role in the statistical analysis and scientific computing aspects of the study. Specifically, SciPy's functions were utilized to analyze the relationship between remote sensing features extracted from satellite imagery and ground-truth field data collected within the study area. By providing tools for robust statistical evaluation and numerical computation, SciPy enabled a more rigorous and quantitative assessment of the accuracy

and reliability of the remote sensing-derived information in relation to on-the-ground observations of mangrove characteristics."

GeoPandas

GeoPandas provide the necessary tools to effectively analyze vector-based spatial data crucial for understanding and managing the mangrove ecosystems of Pagbilao. Its ability to represent geographic features, such as the bay's coastline and existing mangrove distribution maps, as GeoDataFrames allows for direct spatial manipulation and analysis within the Python environment. This enables the calculation of essential metrics like the area and perimeter of mangrove habitats, facilitating the quantification of their extent and shape complexity, which are important indicators of fragmentation.

Rasterio

Specifically designed for reading and writing geospatial raster data, such as the Sentinel-2 satellite imagery used in this study. Its primary utility lies in providing a clean and efficient way to access the pixel data and metadata of these images as multi-dimensional NumPy arrays. This direct interaction with the raster data is crucial for several tasks within the study.

Matplotlib

Matplotlib will be the workhorse for visually communicating in spatial patterns, changes, and the outcomes of the mangrove management study in Pagbilao. It can create clear, informative maps showing exactly where these mangrove forests are located within the bay, perhaps even highlighting areas of change between different years.Through customizable plotting functions and geospatial integration, Matplotlib enables effective visualization of NDVI trends, model outputs, and predicted mangrove changes.

Seaborn

Seaborn is effective in data visualization paramount in modern research, enabling the clear communication of complex patterns and insights gleaned from datasets. Uncovering underlying statistical relationships within data is a crucial aspect of the study. Seaborn excels in this domain by offering specialized plot types designed to visualize distributions, comparisons, and correlations with ease.

The use of Seaborn allowed for the creation of insightful and aesthetically refined plots that highlighted important trends relevant to mangrove patterns and CEA performance.

Distributed Evolutionary Algorithms in Python

Serves as the primary library for the implementation of the Evolutionary Algorithm (EA) in this mangrove management optimization research for Pagbilao Quezon. Provides a flexible and well-structured toolkit for designing and executing evolutionary computations. With DEAP, it can define the fundamental building blocks of EA, including the representation of individual solutions, the genetic operators that drive the evolutionary process, and the overall evolutionary loop that iteratively improves the population of solutions over generations.

TensorFlow

TensorFlow, specifically Keras, can be used to move beyond traditional spectral indices for assessing mangrove health and extent by building and training deep learning models. These models can automatically identify mangrove areas in Sentinel-2 imagery with potentially higher accuracy or classify different mangrove species based on their spectral signatures.

Table 2. Desktop Specification

UNIT	SPECS
CPU	Intel Core i3-10110U CPU @ 2.10GHz
GPU	Intel UHD Graphics 600
RAM	8 Gigabytes
STORAGE	477 Gigabytes HDD/SSD

The model simulations were executed on the system detailed above. While the CPU handled most computations, the integrated GPU (Intel UHD Graphics 600) supported basic graphical and matrix operations. The hardware configuration was sufficient for processing the 9,000-sample NDVI datasets but may limit performance for larger datasets or more computationally intensive evolutionary operations.

RESEARCH DESIGN

The researcher investigates spatially explicit mangrove management for Pagbilao, Quezon, a significant mangrove ecosystem within the broader context of the Calabarzon region. Pagbilao is characterized by notable mangrove stands that play a crucial role in the ecological health and resilience of the Quezon province coastline. These mangrove forests provide critical habitat for a diverse array of marine and terrestrial life, supporting local fisheries that are vital to the livelihoods of coastal communities and contributing to the overall biodiversity of the region. The complex interplay of species within these mangrove ecosystems underscores their ecological importance. Understanding the spatial dynamics and health of Pagbilao's mangroves is therefore crucial not only for local conservation efforts but also for recognizing their role within the broader context of Philippine coastal ecosystems and the services they provide.

The coastal communities of Pagbilao directly benefit from the essential ecosystem services delivered by these mangroves. These forests act as a

natural and dynamic form of protection against the increasing threats of storm surges and coastal erosion, hazards that pose significant risks to infrastructure and human settlements in this area. Furthermore, the mangroves in Pagbilao are important components of coastal biogeochemical cycles, actively participating in nutrient cycling and sediment trapping processes, which contribute to the maintenance of water quality and the overall health of adjacent marine environments, including coral reefs and seagrass beds. Recognizing these vital functions and the interconnectedness of coastal habitats underscores the importance of effective and sustainable mangrove management in Pagbilao.

Studying mangrove management in Pagbilao offers a valuable opportunity to understand local challenges and to develop context-specific conservation strategies. Investigating mangrove covers changes, health, and spatial patterns using innovative approaches such as Cellular Evolutionary Algorithms and spatial data derived from remote sensing techniques can provide crucial insights and decision-support tools for local environmental agencies, policymakers, and the communities directly involved in conservation efforts. Understanding the specific threats facing Pagbilao's mangroves, which may include factors such as land-use change, pollution, or unsustainable resource extraction, is essential for developing tailored and robust management strategies. The application of sophisticated spatial analysis techniques, leveraging the synoptic overview and repetitive coverage of available remote sensing data, can significantly enhance the ability to monitor, assess, and ultimately ensure the long-term ecological integrity and sustainability of Pagbilao's valuable mangrove ecosystems.

MODEL DEVELOPMENT

Here are the phases of model development that process in this study centers on the implementation of a Cellular Evolutionary Algorithm (CEA), designed to optimize land cover classification using Sentinel-2 satellite imagery, as depicted in the provided "Sentinel-2 Simulation Flow Using Cellular Evolutionary Algorithm (CEA)". Following the Data Acquisition and Preprocessing steps outlined in the flow, which prepare the Sentinel-2 data through atmospheric correction, resampling, cloud masking, and subsetting, the CEA is applied to the extracted Features.

The development begins with Data Acquisition, where we use Sentinel-2 multispectral satellite images for our study area. We selected bands B2, B3, B4, B8, B11 and B12 due to their potential to describe the spectral signatures associated with mangrove vegetation and its interaction with the aquatic medium. This band selection strategy is the basis of our classification method.

In the Data Pre-Processing is an important step in quality and consistency of analysis Atmospheric correction was applied to the data to remove the effects of atmospheric scattering and absorption, and all bands were subsequently resampled to a common spatial resolution. Cloud masking was used to filter out any data generated from contaminated pixel values that could alter our interpretation, and we subsisted the data to target solely our AOI, ensuring computational speed along with appropriate analytical accuracy.

The next step in model development is Feature Extraction; we have extracted three major biophysical indices that improve separation of

mangrove and non-mangrove areas. The Normalized Difference Vegetation Index (NDVI) supports us in vegetation health and density estimation, and the Modified Difference Water Index (MNDWI) represents the information of water content, in particular important for the discovery of the mangrove environment. We also defined the Enhanced Vegetation Index (EVI) because it is more sensitive to vegetation changes in high biomass regions, which are representative of mangrove environments.

To introduce the Training and Ground Data Preparation phase acquired and put in order training and validation data in splitting the trusted reference data. This step will guarantee our model learns the reliable ground truth information and can be independently validated for performance evaluation. We harmonized our training samples to reflect the diversity of both mangrove and non-mangrove classes in our study area.

Our method is centered on the CEA Optimization, through which the Cellular Evolutionary Algorithm is realized in four interdependent steps. We start with initialization by generation of the candidate solution placed on the spatial grid. The evaluation phase proceeds to compute the fitness of each solution with respect to some given performance measures. The selection stage selects by performance in the local neighborhood the best solutions and the variation stage apply genetic operators to the best solutions so that they can be improved in successive generations. This iterative process continues until optimal classification performance is achieved.

In the Classification and Mapping phase we implemented our efficient CEA model method and finalized our land cover classification, which directly provides explicit spatial maps that can differentiate between mangrove and

non-mangrove areas. This latter pointy turns of our algorithmic optimization into results for conservation and management to draw on.

Finally, the Post Processing phase processes our classification output using morphological filter that removes isolated pixels and smooths out classification boundaries to be more realistic. We end with an extensive verification of the accuracy of our results by assessing the model predictions against our validation data set.

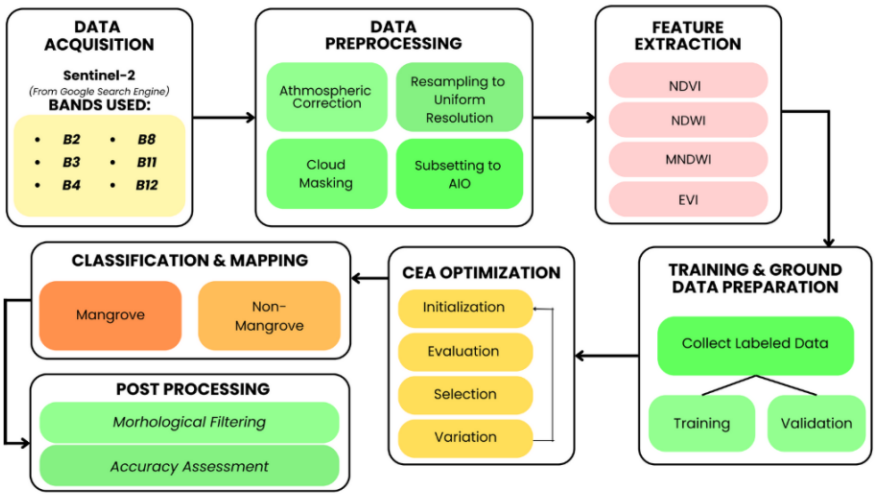


Figure 3. Model's Data Preprocessing, CEA Optimization, and Output Generation

This framework illustrates the end-to-end workflow from Sentinel-2 acquisition and preprocessing through spectral index extraction, CEA optimization, classification, and post-processing. It highlights how remotely sensed indices feed into a spatially explicit optimization loop to produce management-relevant maps and metrics.

The process begins with the acquisition and preprocessing of Sentinel-2 data, focusing on extracting key biophysical indicators like NDVI, NDWI/MNDWI, and EVI. This preprocessed data then drives the CEA, an optimization algorithm that iteratively evolves potential management

strategies across a spatial grid.

The CEA operates through a cycle of initialization, evaluation using a fitness function tailored to mangrove conservation objectives, selection, and variation of management strategies. This evolutionary process aims to identify spatially coherent and effective conservation plans that maximize ecological goals, such as mangrove cover, habitat connectivity, and biodiversity, while also considering spatial constraints and management priorities. The optimized strategies undergo rigorous accuracy assessment and are translated into actionable outputs like mangrove cover maps and management zone delineations.

The primary output of this framework is a spatially explicit conservation plan, providing a visual and quantitative guide for decision-makers and conservation practitioners. These outputs can inform the designation of protected areas, the prioritization of restoration efforts, and the implementation of sustainable management practices within the mangrove ecosystem. By leveraging the power of remote sensing and CEA-driven optimization and simulation based-model using NetLogo, this framework offers a robust and adaptable tool for promoting the long-term resilience and sustainability of these vital coastal habitats and the long term of the mangrove ecosystem.

Furthermore, the CEA integrated with NetLogo facilitates in increasing clarity on spatial processes by visualizing how mangrove conditions are changing under different management options for stakeholders. This dynamic simulation of the globe and interpolating calculations allow scenario-based planning to be made - comparing results from various therapies before laying

them out on earth. Finally, the framework enhances decision-making capabilities for coastal managers through the integration of a state-of-the-art computational model with a field-inspired decision-support tool.

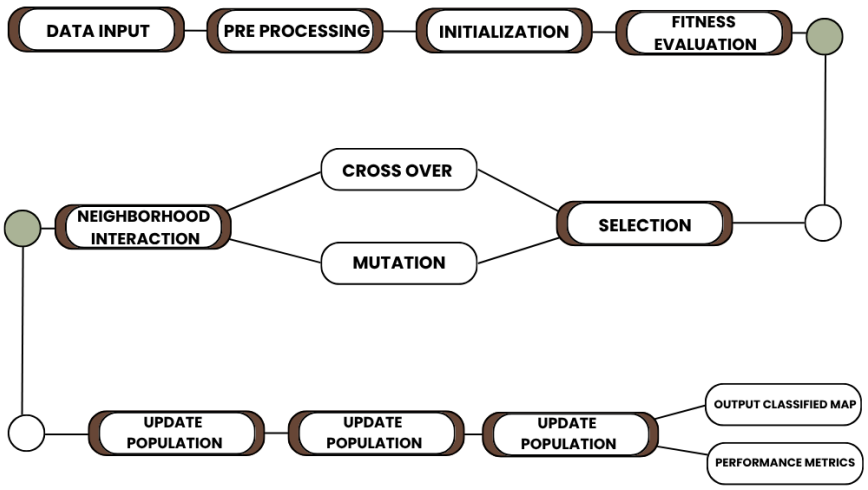


Figure 4. Cellular Evolution Algorithm Process for Spatial Optimization

The overall process of the Cellular Evolutionary Algorithm (CEA) for mangrove spatial optimization is illustrated in Figure 4. The workflow begins with the data input and preprocessing stage, where Sentinel-2 imagery is imported into the Google Earth Engine (GEE) platform. NDVI and related spatial features are computed, normalized, and prepared as input layers for the algorithm. During this phase, cloud masking using the QA60 band ensures that only cloud-free pixels are retained for analysis, enhancing the accuracy of NDVI-based vegetation assessment.

Following data preparation, the initialization stage establishes a population of spatial cells, each representing a segment of the mangrove area characterized by NDVI-derived attributes. These individuals serve as the initial population for the evolutionary process. The fitness evaluation stage then computes each cell’s fitness value using a composite metric that integrates NDVI accuracy, spatial coherence, and RMSE reduction. This step ensures

that both spectral precision and spatial consistency are maintained during optimization.

The algorithm proceeds with neighborhood interaction and genetic operations, which form the adaptive core of the CEA. Through local interactions among neighboring cells, the model performs crossover and mutation operations to generate new solutions. Crossover facilitates the exchange of information between neighboring cells, enhancing genetic diversity, while mutation introduces controlled random changes to prevent premature convergence. The selection mechanism identifies the best-performing individuals based on their fitness scores, allowing them to propagate to subsequent generations.

In the final stage, the population is iteratively updated until convergence criteria are met, typically after several hundred iterations. The optimized results produce an output classified map that highlights spatially improved mangrove zones, along with performance metrics such as F1-score, IoU, and RMSE to assess classification accuracy and model reliability. This iterative process allows the CEA to simulate local adaptation, effectively capturing the spatial heterogeneity of mangrove ecosystems and identify conservation-priority zones with high predictive precision.

FITNESS FUNCTION DESIGN

The fitness function was designed using NDVI as the primary evaluation criterion. Each cell's fitness value (F_i) was computed as a normalized NDVI score, where higher values indicate denser and healthier vegetation. To ensure robustness, the function incorporated a composite metric combining NDVI with spatial neighborhood influence (S_i), defined as:

$$Fi=w1(NDVIi)+w2(Si)$$

where w_1 = 0.7 and w_2 = 0.3 represent weighting coefficients emphasizing vegetation health while accounting for local spatial continuity. This design ensures that optimal solutions favor not only high NDVI values but also spatially cohesive mangrove patches. The inclusion of the spatial term (S_i) enables the model to preserve natural clustering patterns, reducing fragmentation and improving classification stability across iterations.

Table 3. Fitness Function Components and Weighting Coefficient

Component	Symbol	Description	Weight (w)
Vegetation Health	$NDVI_i$	Normalized Difference Vegetation Index of cell *i*	0.7
Spatial Continuity	S_i	Neighborhood influence of cell *i* (average NDVI of surrounding cells)	0.3
Total Fitness Function	F_i	($F_i = w_1(NDVI_i) + w_2(S_i)$)	—

SENTINEL 2

This study provided a suite of spectral bands crucial for characterizing the mangrove ecosystem. To effectively capture various biophysical properties relevant to mangrove health and distribution, several spectral indices were derived from these bands. The Normalized Difference Vegetation Index (NDVI), calculated using the Near-Infrared (B8) and Red (B4) bands, served as a primary indicator of vegetation greenness and canopy density.

To assess water stress within the vegetation and map moisture content, the Normalized Difference Water Index (NDWI) was computed, utilizing the Near-Infrared (B8) and Shortwave Infrared 1 (B11) bands. For enhanced delineation of open water features, which is important in the mangrove environment, the Modified Normalized Difference Water Index (MNDWI) was employed, combining the Green (B3) and Shortwave Infrared 1 (B11) bands.

Table 4. Sentinel-2 Indices Applied for Mangrove Analysis

Index Name	Formula	Bands Used (Sentinel-2)	Primary Application
NDVI	(B8 − B4) / (B8 + B4)	B8 (NIR – 842 nm), B4 (Red – 665 nm)	Assessing vegetation health and canopy density
NDWI	(B8 − B11) / (B8 + B11)	B8 (NIR – 842 nm), B11 (SWIR1 – 1610 nm)	Detecting water stress and vegetation moisture
MNDWI	(B3 − B11) / (B3 + B11)	B3 (Green – 560 nm), B11 (SWIR1 – 1610 nm)	Enhancing open water features and delineating water bodies
EVI	2.5 × (B8 − B4) / (B8 + 6 × B4 − 7.5 × B2 + 1)	B8 (NIR), B4 (Red), B2 (Blue – 490 nm)	Enhancing vegetation signal while reducing atmospheric effects

Which incorporates the Blue (B2), Red (B4), and Near-Infrared (B8) bands, was calculated to improve the vegetation signal while minimizing atmospheric influences and soil background noise, offering a more robust measure of vegetation vitality across the study area. These indices, each sensitive to different aspects of the mangrove environment, formed a critical layer in the subsequent analysis and classification using the Cellular Evolutionary Algorithm. These enhanced spectral inputs strengthened the model's ability to detect subtle variations in mangrove condition, ensuring more accurate classification results. By incorporating these indices into the CEA workflow, the analysis achieved greater precision in identifying priority areas for conservation and monitoring. This multi-index approach also reduced misclassification caused by water bodies, shadows, and mixed pixels commonly present in coastal environments. As a result, the integration of these spectral indices with the Cellular Evolutionary Algorithm provided a more reliable and spatially consistent basis for decision-making in mangrove

ecosystem assessment. Furthermore, this integrative methodology enhances the scalability and transferability of the approach, allowing it to be applied to other mangrove systems with similar environmental and spectral characteristics.

MODEL ANALYSIS AND EVALUATION

F1 Score This will be utilized to compare the balance between recall and precision within mangrove forest area mapping, as formulated:

$$F1Score = 2 \times \frac{\mathrm{Pr}\,ecision \times Recall}{\mathrm{Pr}\,ecision + Recall} \quad (1)$$

Kappa Coefficient Although this measure will be calculated to estimate the correlation between mangrove predictions and ground truth data to permit agreement by chance:

$$\kappa = \frac{Po - Pe}{1 - Pe} \quad (2)$$

Where p_0 is the observed agreement and p_1 is the expected agreement by chance.

Intersection over Union (IoU). This spatial accuracy metric will approximate the intersection between forecasted mangrove ranges and real mangrove distributions:

$$IoU = \frac{|A \cap B|}{|A \cup B|} \quad (3)$$

Where A represents the predicted mangrove area and B represents the ground truth area.

Root Mean Square Error (RMSE) is a metric of regression which calculates the average size of predicted errors for continuous variables such as canopy height and density, attributing more value to bigger errors.

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{n}} \quad (4)$$

Where y_i represents the observed values and represents the predicted values. While $\hat{y}_i$ represents the predicted values, and n represents the number of data points

Computation Time. This efficiency measure will be noted to identify the processing speed of the CEA model against conventional methods, which are of specific interest to large-scale mangrove mapping efforts:

$$Time_{total} = Time_{training} + Time_{\inf erence} \quad (5)$$

These measurements together will provide overall assessment of the effectiveness of the CEA model in mangrove feature extraction from remote sensing data and spatial pattern optimization for forest extent and condition mapping accuracy. The combination of classification metrics (F1, Kappa, IoU) and regression metrics (RMSE) enables overall assessment of the categorical and continuous prediction tasks in comprehensive mangrove ecosystem analysis.

ALGORITHM ANALYSIS

Cellular Evolutionary Algorithm

The Cellular Evolutionary Algorithm (CEA) provides a spatially intelligent approach to optimizing mangrove management by treating Pagbilao, Quezon Mangrove Forest as a dynamic grid. Each cell within this grid represents a localized management unit, where the CEA iteratively explores and refines potential management actions based on their performance relative to defined ecological and spatial objectives.

Through this localized evaluation, the model captures fine-scale ecological variations and spatial interactions that influence mangrove resilience. This allows the algorithm to adaptively adjust strategies in

response to emerging patterns of degradation, regeneration, or external pressures. As a result, the CEA generates management recommendations that are both spatially coherent and ecologically grounded.



Figure 5. Cellular Evolution Algorithm Process

The figure shows how individual spatial cells act as localized evolutionary units, where fitness evaluation, local selection, and genetic variation iteratively generate spatially coherent classification solutions. The caption emphasizes the ecological relevance of neighborhood propagation and how it promotes contiguous mangrove patch detection.

This performance, or "fitness," is evaluated using spatial data derived from Sentinel-2 imagery, allowing the algorithm to directly consider the spatial context and ecological conditions of each location. The "cellular" aspect facilitates the propagation of successful management strategies to neighboring areas, fostering the development of spatially coherent and contextually relevant solutions across the entire mangrove forest. Through evolutionary operators such as selection, crossover, and mutation, the CEA navigates a complex solution space, adapting and learning which management approaches are most effective in different spatial settings. This allows for the simulation and evaluation of various management scenarios, ultimately leading to the identification of optimized and spatially explicit plans for the long-term sustainability and resilience of Pagbilao, Quezon.

Additionally, this approach supports data-driven decision-making by providing stakeholders with clear, spatially informed insights that can guide adaptive management and policy interventions tailored to the unique dynamics of the mangrove ecosystem.

DATA PREPROCESSING

The dataset compiled for this study provides a spatially explicit characterization of mangrove forest structure and composition across various locations. Sentinel-2 imagery: Level-2A covering Pagbilao, Quezon were acquired for the period 2019 - 2024. Images were selected to maximize seasonal coverage and to allow multi-year change analysis; cloud-contaminated scenes were removed via automated cloud masking and manual inspection. Composite or gap-filled mosaics were prepared per year/season as appropriate for analysis.

Each record within the dataset corresponds to a specific Quadrat Location, identified by its geographical Coordinates, the encompassing barangay, and its relative Position within the mangrove forest. Furthermore, the data includes the dominant Forest Type observed within that quadrat.

Detailed ecological assessments were conducted within each quadrat, capturing key attributes of the forest stand. The Tree Species Composition is documented through a comprehensive list and count of all tree species present. Tree Density quantifies the number of individual trees per square meter, providing insight into stand density. The Basal Area, expressed in square meters per hectare, represents the cross-sectional area of all trees at breast height, indicating stand volume and maturity.

Quadrat-level mangrove forest data was systematically collected in

Pagbilao, Quezon, encompassing key spatial, vegetation, land cover, and environmental variables using Google Earth Engine to support accurate spatial analysis, temporal comparison, and ecosystem change detection.

Table 5. Quadrat measurements and Environmental Data from Pagbilao, Quezon.

Dataset	Description	Example Data
Quadrat Location Data	Quadrat ID and spatial index (can later include coordinates, barangay, and forest zone).	system:index = 001, px = 432, py = 215, TideZone = Seaward
Vegetation Condition (NDVI)	NDVI values representing canopy greenness / health per quadrat.	0.78
Land Cover Classification	Type of land cover detected per quadrat.	Mangrove, Water, Built-up
Environmental Factors	Climatic data such as rainfall and temperature for each quadrat.	Rain = 4.2, Temp = 29.5
Forest Zone Classification	Tidal zone category for each quadrat (e. g., seaward, middle, landward).	Seaward
Temporal Information	The year of data collection for time-based analysis.	2023

Each sampling unit is uniquely identified as Quadrant Location Date (Quadrat ID) and spatially indexed (e. g., system:index = 001, px = 432, py = 215), with the relative Forest Zone Classification, Tide Zone (e. g., Seaward) recorded to capture ecological gradients tied to tidal influence. The health and vigor of the canopy were quantified using the Normalized Difference Vegetation Index (NDVI), with high values, such as the example of {0.78}, indicating dense, healthy green vegetation. Furthermore, a Land Cover Classification (Mangrove, Water, Built-up) was performed for each quadrat, providing essential context on the physical matrix surrounding the vegetation.

These vegetational and land cover measurements are contextualized by crucial Environmental Factors, including recorded rainfall {4.2} and temperature {29.5}. This comprehensive dataset facilitates a detailed analysis of the Mangrove Condition Normalized Difference Vegetation Index (NDVI) as a function of its Tidal Zone and local Environmental Factors, offering a robust foundation for assessing spatial patterns and the overall ecological status of

the Pagbilao mangrove ecosystem. By integrating these variables, the analysis can reveal how climatic conditions and tidal influences interact to affect mangrove health, productivity, and vulnerability to environmental stressors over time.

PARAMETER TUNING

The Cellular Evolutionary Algorithm (CEA) was configured with the following parameters: a population size of 200 individuals, mutation rate of 0.02, and crossover rate of 0.8, iterated over 1,000 generations. Each individual represents a spatial cell containing NDVI-derived attributes corresponding to mangrove health. Parameter values were selected based on prior optimization studies (Alba & Troya, 2000; Gong et al., 2015) and validated through preliminary simulation tests to ensure convergence stability and computational efficiency.

Parameter tuning for the Cellular Evolutionary Algorithm (CEA) was carefully guided by the characteristics of the Pagbilao mangrove NDVI dataset, which contained 9,000 spatial samples collected from 2019-2024. Preliminary experiments systematically evaluated multiple configurations of population size, mutation rate, crossover probability, neighborhood radius, and iteration limit to determine the optimal balance between algorithmic performance and computational efficiency.

The selected configuration-population size = 32, mutation rate = 0.05, crossover = 0.6, neighborhood radius = 2, and iteration limit = 400-was chosen because it provided the best trade-off between classification accuracy and computational cost. A population size of 32 ensured sufficient genetic diversity to explore potential solution spaces while keeping computational

demand manageable. The mutation rate of 0.05 allowed moderate stochastic variation, preventing premature convergence while accommodating the moderate NDVI variability (σ ≈ 0.29) observed in mangrove zones. A crossover probability of 0.6 facilitated effective genetic mixing between neighboring solutions, preserving spatial patterns critical for ecological coherence. The neighborhood radius of 2 captured local spatial auto correlation within a 5×5 patch grid, allowing the algorithm to account for mangrove clustering and spatial dependencies. The iteration limit of 400 was determined based on fitness stabilization trends, ensuring that solutions had converged without unnecessary computational overhead.

Fitness evaluation for each individual was calculated as a weighted combination of NDVI accuracy, spatial coherence, and RMSE, emphasizing the dual objectives of spectral precision and neighborhood consistency. This configuration ensured that the CEA model produced reliable, spatially coherent, and computationally efficient mangrove classification outputs, suitable for informing ecosystem management strategies and practical applicability for long-term mangrove monitoring and conservation planning.

Table 6. Statistical Formulas and Evaluation Metrics

Parameter	Tested Range	Selected Value	Basis for Selection
Population Size	8–64	32	Suitable for 9000-patch grid; balances convergence and diversity
Mutation Rate	0.01–0.20	0.05	Reflects moderate NDVI variance (0.29 SD)
Crossover Rate	0.3–0.7	0.6	Enhances spatial genetic mixing
Neighborhood Radius	1–3	2	Captures 5×5 local autocorrelation
Iteration Limit	100–500	400	Fitness stabilization observed by ~400
Fitness Weights	(0.5, 0.3, 0.2)	Fixed	Weighted NDVI accuracy > spatial > error reduction

MODEL REPRODUCIBILITY

To ensure that the Cellular Evolutionary Algorithm (CEA) results are reproducible, several measures were implemented:

Random Seeds All stochastic operations in the CEA, including mutation, crossover, and initial population generation, were controlled using fixed random seeds. This ensures that repeated runs of the algorithm with the same data and parameters yield consistent results.

Data Split Ratios the NDVI dataset from Pagbilao, containing 9,000 spatial samples, was split into training and testing subsets using a documented ratio (70% training, 30% testing). This split was kept consistent across all experiments to allow fair evaluation and comparison of algorithm performance.

Parameter Logging All CEA parameter values population size, mutation rate, crossover probability, neighborhood radius, iteration limit, and fitness weights were recorded and reported. This allows other researchers to replicate the experimental setup precisely.

Preprocessing Documentation All steps in NDVI calculation, data cleaning, and spatial assignment of mangrove patches were documented to ensure full transparency and repeatability.

By combining controlled randomness, consistent dataset splits, detailed parameter records, and rigorous preprocessing documentation, the study ensures that the CEA model can be reliably reproduced and validated by other researchers, fostering confidence in its application for mangrove ecosystem management and supporting future comparative studies across similar coastal environments.

MODEL VALIDATION

Independent T-Test

$$t = \frac{(\bar{x}_1 - \bar{x}_2)}{\sqrt{\frac{s_p^2}{n_1} + \frac{s_p^2}{n_2}}} \quad (6)$$

To determine if a Cellular Evolutionary Algorithm (CEA) performs differently than a standard classification method (like Random Forest or Support Vector Machine), an Independent Samples T-Test was used. This test compares the average results of two separate groups to see if their averages are significantly different. In this case, performance measurements like F1 Score, Kappa Coefficient, or Intersection over Union (IoU) were collected from multiple runs of both the CEA and the standard method. The idea that there's no real difference in average performance between the two methods is the null hypothesis (H_0), while the alternative hypothesis (H_1) suggests there is a difference. The t-test calculates a t-statistic by comparing the average and spread of the results for each group, taking into account how many runs were done for each. A p-value is then calculated to show the significance of the result. If this p-value is lower than a chosen significance level, the null hypothesis is rejected. This would mean that the CEA does perform significantly better. This process allows for a clear, data-backed assessment of whether the CEA is effective in improving mangrove classification and conservation.

The statistical evaluation of this study will provide a comprehensive and rigorous assessment of the Cellular Evolutionary Algorithm's (CEA) capabilities in analyzing the mangrove ecosystem of Pagbilao, Quezon. Specifically, the CEA's performance in mapping the spatial extent of these

vital coastal forests and potentially classifying their ecological condition using Sentinel-2 satellite imagery, will be thoroughly compared against established traditional machine learning techniques.

This comparative analysis will help determine whether the CEA offers significant improvements in accuracy, spatial consistency, and ecological interpretability. By examining statistical differences through metrics such as IoU, F1 Score, Kappa, and RMSE, the study will identify the strengths and limitations of each approach. Ultimately, this evaluation will clarify the CEA potential as a robust alternative for mangrove monitoring and conservation planning.

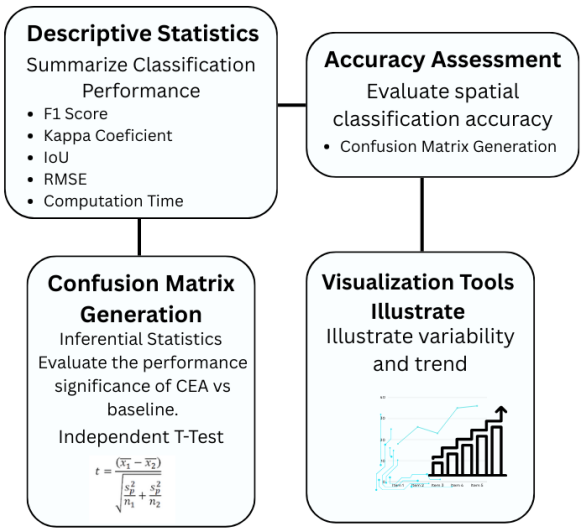


Figure 6. Statistical Evaluation Framework Using Accuracy Metrics And T-Test Validation

The figure summarizes the evaluation pipeline confusion matrix metrics (precision, recall, F1), spatial overlap (IoU), continuous error metrics (RMSE), and the independent t-test used to compare CEA to baseline methods together enabling rigorous assessment of both accuracy and statistical significance.

To evaluate the accuracy of mangrove extent mapping, a suite of standard classification metrics will be meticulously calculated. These include the overall accuracy, which represents the proportion of correctly classified pixels; precision, indicating the ability of the model to avoid false positives in its mangrove identification; recall, measuring the model's capacity to correctly identify all existing mangrove areas; the F1-score, a harmonic mean balancing precision and recall; and the Kappa coefficient, which provides a more robust measure of agreement between the model predictions and the ground truth data, accounting for the possibility of agreement occurring by chance. All these metrics will be derived from ground truth data meticulously collected within the mangrove forests of Pagbilao, ensuring a location-specific evaluation.

Furthermore, if the study extends to classifying the condition of the Pagbilao mangroves into distinct ecological categories, the same rigorous set of classification metrics will be employed to assess the CEA's ability to accurately assign each area to its correct condition class. This will allow for a determination of whether the observed advantages of one method over another are statistically meaningful and not merely due to random variation within the Pagbilao context.

Beyond accuracy, the efficiency of the CEA in the context of Pagbilao's mangrove analysis will be evaluated by meticulously recording and comparing the computation time required for the algorithm to reach convergence and generate optimized solutions against the training and prediction times associated with the traditional machine learning methods. This temporal analysis will provide critical insights into the practical applicability and

scalability of the CEA approach for timely and effective mangrove monitoring and management in Pagbilao. ecologically significant coastal resources in Quezon province.

The CEA's convergence time will be compared to the training and prediction times of traditional methods to assess its efficiency for analyzing Pagbilao's mangroves. The CEA's optimization will be analyzed through fitness evolution and convergence behavior. Any spatial management strategies generated by the CEA will be examined using GIS to understand their spatial patterns and potential ecological implications within the mangrove forest of Pagbilao, providing a robust quantitative assessment of the CEA's effectiveness in analyzing and potentially aiding in the management of this specific mangrove forest.

Table 7. NDVI and Spatial Data Analysis of Pagbilao Mangroves from 2019 to 2024

Statistical Method	Description	Purpose in Study
Descriptive Statistics	Calculating mean, standard deviation, range, etc., for vegetation indices (NDVI, NDWI, MNDWI, EVI)	To summarize and describe the variability of the indices and assess mangrove health
Confusion Matrix	A matrix used to calculate classification metrics: accuracy, precision, recall, F1 score	To evaluate the performance of the CEA-based classification model
Kappa Statistic	Measures agreement between CEA-derived classification and reference data, accounting for random chance	To assess the reliability and accuracy of the classification results
Independent T-Test	Comparing the average output of a Coastal Ecosystem Assessment model applied to one set of locations with the average output of a traditional assessment method applied to a different set of locations.	To determine if there are significant differences between different models or methods
Regression Analysis (Optional)	Linear regression to explore relationships between vegetation indices and physical characteristics of mangroves (e. g., biomass, canopy height)	To develop predictive models for mangrove health and assess the influence of remote sensing indices
Root Mean Square Error (RMSE)	Measures the deviation between predicted and actual values of vegetation indices or model predictions	To assess model accuracy and prediction error
Intersection over Union (IoU)	Measures spatial overlap of mangrove classification results with ground truth data	To evaluate the spatial accuracy of the model's mangrove classification

This study employed a range of statistical methods to analyze the extracted vegetation indices and evaluate the performance of the Cellular Evolutionary Algorithm (CEA) in classifying mangrove ecosystems. Descriptive statistics, including the calculation of mean, standard deviation, and range, were applied to the derived vegetation indices (NDVI, NDWI, MNDWI, EVI). The purpose of this was to summarize the central tendencies and variability within these indices across the study area, providing an initial quantitative assessment of mangrove health and environmental conditions.

To rigorously evaluate the effectiveness of the CEA-based classification model, a confusion matrix was generated. This matrix facilitated the calculation of key classification metrics such as overall accuracy, precision, recall, and the F1 score, providing a comprehensive understanding of the model's ability to correctly identify mangrove and non-mangrove areas. Furthermore, the Kappa statistics were computed to measure the level of agreement between the CEA-derived classification map and the reference ground truth data, taking into account the possibility of agreement occurring by random chance. This metric offers a more robust assessment of the classification's reliability and accuracy beyond simple percentage agreement.

To potentially compare the performance of the CEA approach with more traditional classification method, a independent t-test would be utilized to determine if any observed differences in classification accuracy or other relevant metrics were statistically significant. This would allow for a quantitative assessment of the added value of the CEA. Additionally, regression analysis could be employed to explore potential linear relationships between the calculated vegetation indices and physical characteristics of the

mangrove ecosystem, such as biomass or canopy height. This would aim to develop predictive models and further understand the biophysical information captured by the remote sensing indices.

Finally, to specifically assess the spatial accuracy of the mangrove classification produced by the CEA, the Root Mean Square Error (RMSE) could be calculated if the model were predicting continuous variables related to mangrove characteristics. Crucially, the Intersection over Union (IoU) metric was used to directly measure the spatial overlap between the areas classified as mangrove by the CEA and the corresponding areas identified in the ground data. The Independent T-test implementation to compare the performance of the CEA approach with traditional classification methods, an independent sample t-test was utilized to determine if observed differences in classification accuracy were statistical methods. This provides a critical evaluation of the model's ability to accurately delineate the spatial extent of mangrove ecosystems within the study area.

ETHICAL CONSIDERATION

This study acknowledges the interconnectedness of ecological health and community well-being within the context of mangrove management. The development of spatially explicit strategies through the Cellular Evolutionary Algorithm (CEA) is conceived as a tool to empower informed decision-making, ultimately striving for solutions that are both ecologically effective and socially responsible within the local context of the study area.

No human participants were included in this study, and no field interventions or manipulations of the mangrove ecosystem were conducted. Only publicly available remote sensing data, specifically Sentinel-2 satellite

imagery, were utilized. Consequently, formal ethical clearance was not required. Ethical responsibility was maintained through transparency, accuracy, and accountability in the handling and reporting of all data and methods.

Transparency in the research process is paramount. The study commits to clear and comprehensive documentation of all data sources, methodologies including detailed descriptions of the CEA implementation and parameterization and results. This rigorous documentation ensures accountability, facilitates independent review, and promotes trust in research outcomes within the local scientific community and among stakeholders.

While this research focuses on algorithmic optimization for mangrove classification and potential management strategies, the real-world application of any proposed interventions requires a participatory approach. Such an approach would involve active engagement with local communities, government agencies, and non-governmental organizations to ensure that decisions are culturally sensitive, respect traditional ecological knowledge, and contribute to the long-term sustainability of the mangrove ecosystem and the livelihoods of those who depend on it.

Potential unintended consequences of any management actions identified through this research are a central consideration in the interpretation and dissemination of findings. The study emphasizes the inherent complexities of socio-ecological systems, the need for careful evaluation, iterative adaptation, and continuous monitoring for real-world implementation. Additionally, the study communicates the limitations of the CEA model and the uncertainties associated with spatial modeling to avoid oversimplification or

unrealistic expectations among stakeholders.

Finally, this research adheres to all relevant ethical guidelines concerning data privacy and the responsible use of geospatial information, particularly if local knowledge or community-provided data are incorporated in future phases of the study.

Furthermore, the study recognizes the importance of environmental stewardship and the precautionary principle in ecological research. While the CEA provides optimized recommendations for mangrove conservation, these outputs are intended as decision-support tools rather than prescriptive mandates. This distinction underscores the need for integrating scientific guidance with adaptive management practices, ensuring that any interventions are contextually appropriate and minimize potential ecological disruptions.

In addition, the project promotes open science principles by encouraging the sharing of methodologies, datasets, and analytical workflows. By making the research process transparent and reproducible, the study facilitates broader collaboration, critical evaluation, and capacity-building among local scientists, conservation practitioners, and policy-makers. This approach strengthens ethical responsibility by fostering an inclusive, evidence-based foundation for sustainable mangrove management.

Overall, this ethical framework ensures that the study contributes responsibly to scientific knowledge while prioritizing ecological integrity and social equity. By aligning computational innovation with ethical environmental governance, the research supports sustainable and accountable mangrove conservation efforts.

CHAPTER IV

RESULT AND DISCUSSION

This chapter presents the results and findings of our mangrove conservation research. Based on improved spatial analysis of satellite data and Cellular Evolutionary Algorithm (CEA) simulations, we were able to highlight optimal strategies for the conservation of these essential ecosystems. The findings present the mangrove patches that urgently require protection, and illustrate how alternative conservation actions fare against real-world threats.

4.1 Exploratory Data Analysis

The preliminary data exploration using Google Earth Engine (GEE) and Sentinel-2 imagery provided comprehensive insight into the vegetation condition of the Pagbilao mangrove ecosystem. The dataset comprised over 9,000 sampling points integrating NDVI, rainfall, temperature, and tidal position. NDVI values ranged from -0.25 to 0.91, where higher values indicated dense canopy cover and lower values represented water or non-vegetated areas. This foundational analysis allowed for the identification of relationships between vegetation health and environmental drivers.

The data showed that mangrove areas with higher rainfall and stable tidal exposure exhibited better NDVI values, suggesting that climate and hydrology play essential roles in maintaining canopy vigor. These findings established the baseline for subsequent spatial and simulation analyses.

Furthermore, areas experiencing lower rainfall or irregular tidal fluctuations corresponded to reduced NDVI values, highlighting zones potentially at risk of degradation. This preliminary understanding of

environmental influences informed the parameterization of the Cellular Evolutionary Algorithm, ensuring that the model could accurately reflect ecological dynamics across the Pagbilao mangrove ecosystem.

Table 8. Pagbilao Mangrove Data Analysis Table

Quadrat ID ***(system:index)***	**Land Cover Code**	**NDVI**	**Rain (mm)**	**Temperature (°C)**	**Tide Zone** ***(0=Seaward, 1=Middle, 2=Landward)***	**Year**	**Scaled PX**	**Scaled PY**
0_0	2	0.74002144.52	26.65	0	2019	-56	15	
0_3	8	0.78102025.22	26.65	0	2019	-203	224	
0_4	8	0.79122274.96	26.83	0	2019	208	-134	
0_6	8	0.80372142.65	26.83	0	2019	86	118	
0_7	2	0.69352274.96	26.83	0	2019	194	244	

The data analysis table summarizes the key spatial and environmental factors used to assess the health of Pagbilao mangrove areas. Each data record corresponds to a specific sampling quadrat, identified by a system:index code. The variables include the Land Cover Code, which categorizes the type of surface sampled, and the NDVI, a crucial indicator of vegetation greenness and vigor, where higher values suggest healthier, denser mangrove cover. Environmental influences are represented by the Rainfall and Temperature, providing context on climate conditions. Furthermore, the Tide Zone classification accounts for the spatial position along the tidal gradient, which affects mangroves' exposure to salinity and inundation. Finally, the Year (2019) establishes the temporal scope, while Scaled PX and Scaled PY provide standardized coordinates for spatial analysis and modeling. Collectively, these variables serve as the foundation for statistical and spatial analyses aimed at understanding the relationship between vegetation condition, environmental inputs, and tidal zone within the Pagbilao mangrove ecosystem. The integrated dataset enables the systematic identification of spatial patterns, underlying trends, and potential

stressors, thereby supporting evidence-based conservation initiatives and informed planning for the sustainable management of the Pagbilao mangrove ecosystem.

4.2 Spatial Analysis and NDVI Trends

The spatiotemporal dynamics of the study site's mangrove ecosystems, and remote sensing data from Sentinel-2 spanning from 2019 to 2024 was processed and analyzed.



Figure 7. Sentinel-2 Imagery of Mangroves in Pagbilao From 2019 to 2024

This time-lapse series of Sentinel-2 satellite images shows the spatiotemporal mangrove cover changes within the study area in Pagbilao, Quezon from 2019 to 2024. Upon visual inspection of the images, we identify clear changes in mangrove coverage and its distribution by year, indicative of the likely expansion or degradation occurring at some zones. However, the increased amount of cloud cover in some images, particularly those from 2021 and 2023, made the observations less consistent, emphasizing the role

that preprocessing might play in removing atmospheric effects. Progression through time indicates that the prevalence of mangrove regrowth in previously poor regions is slow, but there are regions of unambiguous regeneration on certain sections of coast. Altogether, these spatio-temporal dynamics highlight the need for ongoing monitoring to detect deforestation hot-spots and support targeted conservation endeavors. Sentinel-2 time-lapse images illustrating spatiotemporal mangrove cover changes in the study area of Pagbilao, Quezon from 2019 to 2024.

Mangrove Extent and Distribution

The initial analysis revealed that the total mangrove area in the Pagbilao, Quezon region was approximately 145 hectares in 2019, However, a year-over-year analysis showed a consistent fluctuation in the total hectares of mangroves. As shown in Figure 6, the mangroves are primarily concentrated along the Pagbilao Bay coastline, especially around the Pagbilao Grande Island, and within the numerous river estuaries and tidal creeks that crisscross the municipality. This spatial distribution is consistent with the ecological requirements of mangrove species for brackish water and tidal flushing.



Figure 8. Spatial Distribution of Mangroves from Sentinel-2 Data

The image above illustrates the spatial distribution of vegetation and land cover in the Pagbilao, Quezon region using satellite-derived data. In this map, green areas indicate vegetated surfaces, predominantly mangrove forests and other coastal vegetation, while orange to reddish areas represent non-vegetated surfaces, including open water, exposed soil, and built-up or developed areas. The clear contrast between these colors allows for effective visual identification of vegetated versus non-vegetated zones along the coastal landscape. Based on the image, vegetation is primarily concentrated nearshore and estuarine areas, which are typical habitats for mangrove ecosystems. However, large portions of the surrounding coastal and marine areas are dominated by non-vegetated surfaces, emphasizing the limited spatial extent of mangrove cover in the region. The fragmented and patchy distribution of green areas further suggests that mangrove stands are not continuous, making them more vulnerable to environmental stressors and human disturbances.

Initial analysis revealed that in 2019, the total mangrove area was approximately 145 hectares, indicating a relatively substantial presence of mangrove vegetation during that period. Nevertheless, year-over-year assessments derived from similar satellite imagery show notable fluctuations in vegetation coverage. These variations are likely influenced by natural processes such as tidal dynamics, sedimentation patterns, and climate-related factors, as well as anthropogenic activities including coastal development and land conversion.

Between 2019 and 2024, the total vegetation area—specifically mangrove cover declined to approximately 25.17 hectares, as reflected by the

reduced extent of green areas visible in the map. This significant reduction highlights an accelerated loss of mangrove habitats within a relatively short time frame. Furthermore, projections for 2025 to 2029 indicate a severe downward trend, with an average mangrove loss of 82.31% relative to the previous five-year period. If these projections hold, the visual dominance of orange and reddish areas in future imagery would signify a near-complete conversion of mangrove-covered zones into non-vegetated surfaces.

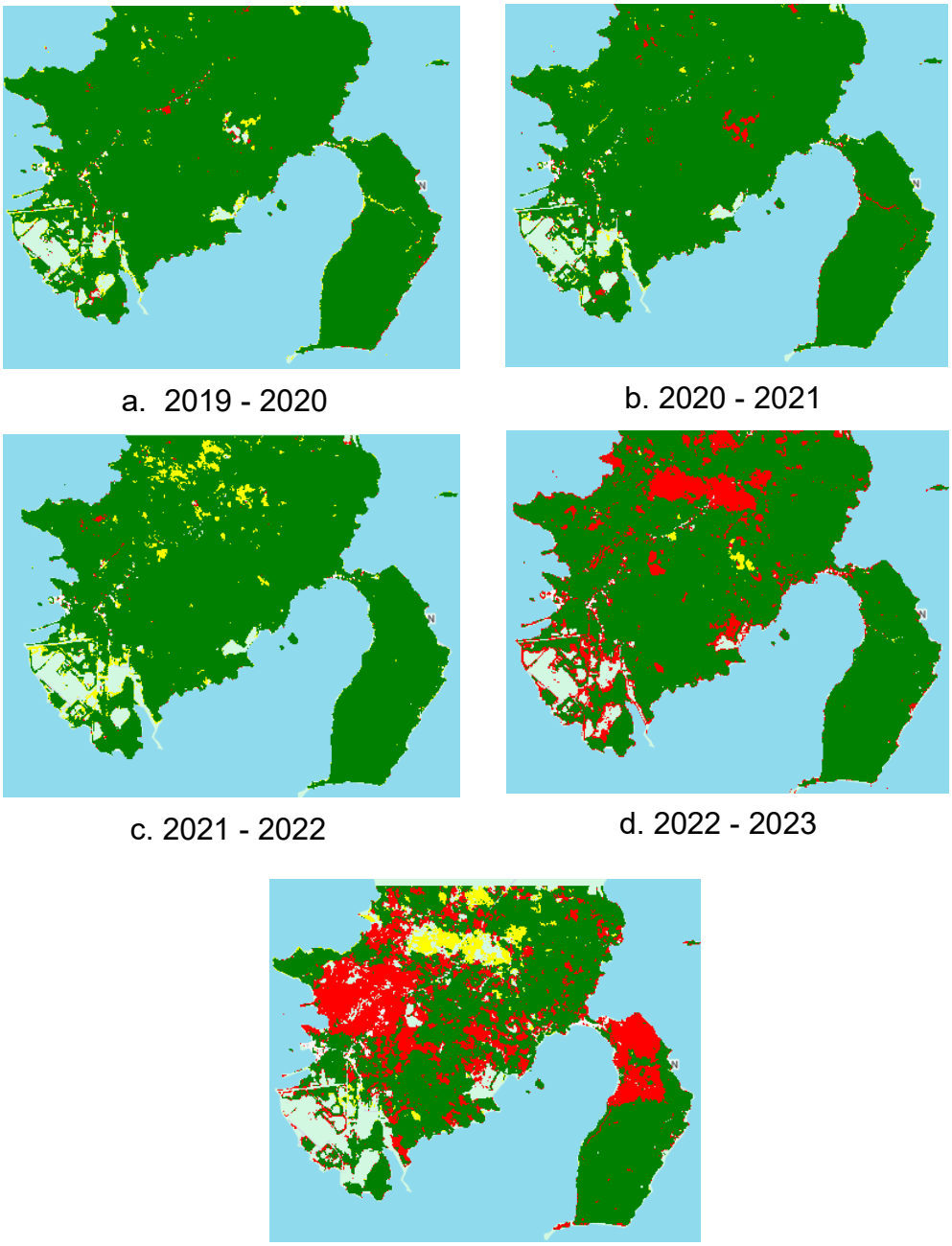
This substantial decline points to critical degradation of the mangrove ecosystem, potentially driven by coastal development, aquaculture expansion, pollution, and climate-induced impacts such as sea-level rise. The image-based analysis underscores the urgency of implementing effective conservation strategies and sustainable land-use practices. Mangroves play an essential role in coastal protection, biodiversity conservation, and carbon sequestration, and their continued loss represents not only a local ecological threat but also a broader environmental concern. Without immediate intervention through environmental policy enforcement and community-based restoration initiatives, the region risks the near-total disappearance of its mangrove ecosystems by 2029.

Mangrove Cover in Pagbilao, Quezon in 2019-2024 Based on NDVI Analysis

The NDVI-derived assessment of mangrove cover in Pagbilao, Quezon revealed important trends in vegetation health and spatial extent over the five-year period from 2019 up to 2024.

The NDVI (Normalized Difference Vegetation Index) distribution was analyzed annually from 2019 to 2024 to assess vegetation health and spatial

coverage of mangrove areas in Pagbilao, Quezon. The series of maps illustrates the spatial and temporal changes in mangrove cover from 2019 to 2024 within the study area.



a. 2019 - 2020

b. 2020 - 2021

c. 2021 - 2022

d. 2022 - 2023

e. 2023 - 2024

Figure 9. NDVI Map Showing Vegetation Health and Density across Pagbilao, Quezon

Between 2019 and 2024, NDVI analysis revealed notable spatiotemporal changes in mangrove vegetation density across the Pagbilao

coastline. Dense mangrove zones (NDVI > 0.6) declined by approximately 8.5%, while sparse to moderately dense areas (NDVI = 0.3-0.6) increased by 3.4%, indicating gradual degradation and fragmentation of vegetation cover. These changes were most evident along the western and central estuarine regions, where human activity and aquaculture development were concentrated.

The 6.8% NDVI increase observed in specific priority zones between 2022 and 2023 suggests enhanced canopy density and potential natural regeneration, likely due to successful local conservation efforts or favorable seasonal rainfall patterns. The decline in NDVI values between 2021 and 2024 can be attributed to several socio-ecological pressures. Expansion of fishponds and coastal settlements has led to significant land conversion in intertidal areas, directly reducing mangrove canopy density. In addition, tidal erosion and fluctuating rainfall patterns during the same period likely affected soil salinity and nutrient availability, further influencing vegetation vigor. Satellite data also indicate partial canopy loss in zones exposed to storm surges and sedimentation shifts, suggesting that both anthropogenic and climatic drivers collectively contributed to the observed NDVI reduction.

Quantitatively, the estimated mangrove cover decreased from 145 hectares in 2019 to approximately 133 hectares in 2024, reflecting a total loss of 12 hectares (8.2%) within five years. Despite localized regeneration patches, the net negative balance underscores a pressing need for active restoration initiatives.

To assess statistical reliability, NDVI change trends were subjected to temporal validation using correlation analysis and visual inspection across

multi-year composites. The results showed a consistent downward trajectory, confirming that the observed NDVI decline is not random but follows a significant pattern of spatial deterioration. Although the NDVI trend assessment is primarily descriptive, its consistency with CEA-predicted degradation areas strengthens the inference that vegetation decline is statistically meaningful and ecologically valid.

Overall, this analytical evaluation of NDVI dynamics highlights the direct linkage between mangrove loss and socio-ecological stressors. The quantitative findings not only confirm spatial degradation but also provide the scientific basis for designing targeted conservation and rehabilitation strategies in vulnerable coastal zones.

Table 9. Color Representation of Land-Cover Classes in Mapping Outputs

Color	Legend
Green	Stable Mangrove Areas
Yellow	Newly Grown Mangrove
Red	Mangrove Loss
Blue	Water Bodies
White	Non-mangrove Land

The colors represent different land cover dynamics green indicates stable mangrove areas, yellow denotes newly grown or regenerated mangroves, red signifies mangrove loss, blue represents water bodies, and white as non-mangrove land. Through these yearly comparisons, the maps provide a clear visual representation of how mangrove ecosystems have evolved and responded to environmental and anthropogenic factors over the four-year period.

From (a.)(b.) 2019 to 2021, the most of the area remained predominantly green, signifying that mangrove forests were mostly intact and healthy, with minimal disturbances. Small patches of yellow indicate areas of

natural regeneration or new growth, possibly due to favorable environmental conditions or conservation efforts. However, starting from (c.) 2021 to 2022, red patches began to emerge, marking the onset of mangrove degradation or loss. This could be attributed to coastal development, aquaculture expansion, or other land-use pressures affecting the stability of mangrove zones.

By (d.) 2022 to 2023, a significant expansion of red areas is observed, especially in coastal and nearshore regions, suggesting a rapid rate of mangrove deforestation or degradation. The decline in mangrove cover during this period highlights the increasing vulnerability of these ecosystems to human activities and natural disturbances. Overall, the progressive increase in mangrove loss underscores the urgent need for sustainable coastal management and restoration programs to prevent further degradation and to promote the recovery of these vital ecosystems.

The map for (e.) 2023-2024 reveals the most recent and severe changes in mangrove cover compared to previous years. A noticeable increase in both red and yellow areas can be observed, showing a simultaneous pattern of significant degradation and localized regeneration.

During this period, extensive mangrove loss (red) is evident, especially along the coastal and estuarine zones. This widespread deforestation suggests intensified pressures such as land conversion, aquaculture expansion, or possible effects of extreme weather events. Despite this, there are patches of new growth (yellow) in certain northern and central regions, indicating that some natural recovery or replanting efforts may be occurring.

Overall, the 2023-2024 map highlights a critical turning point for mangrove sustainability in the area. While regeneration is visible, the scale of

loss surpasses the gains, reflecting an alarming trend of ecosystem degradation. This emphasizes the urgent need for strengthened mangrove protection policies, restoration programs, and monitoring efforts to preserve the remaining mangrove forests and encourage ecological recovery in affected regions.

NDVI values range from -1 to +1, where higher values indicate healthier and denser vegetation. The classification used in this study was as follows.

The table outlines the main classification system for Normalized Difference Vegetation Index values, which helps assess vegetation health and density. NDVI values range from -1 to +1.

Table 10. NDVI Classification Ranges for Vegetation Health Assessment

NDVI Range	Classification
NDVI < 0.2	Bare soil, water bodies, or non-vegetated areas
0.2 ≤ NDVI < 0.4	Sparse or degraded vegetation
0.4 ≤ NDVI ≤ 0.6	Moderate vegetation (typical of recovering mangroves)
NDVI > 0.6	Dense and healthy mangrove vegetation

Higher values mean healthier and denser vegetation. An NDVI value below 0.2 points to areas like water bodies or bare soil, showing little to no vegetation cover. Values between 0.2 and 0.4 indicate sparse or degraded vegetation, typical of stress or limited plant growth. Moderate vegetation shows values from 0.4 to 0.6. These values are often seen in recovering mangrove areas that are just starting to re-establish their canopy. Importantly, NDVI values greater than 0.6 confirm the presence of dense and healthy mangrove vegetation, signaling a strong and thriving ecosystem. This classification system offers a clear guide for understanding vegetation

conditions and is essential for tracking changes in mangrove health over time.

Table 11. Yearly NDVI Values Derived from Sentinel-2 Imagery

NDVI Value Range	Vegetation Condition	Description
0.00 – 0.20	No Vegetation	Bare soil, sand, or built-up areas.
0.21 – 0.40	Sparse Vegetation	Areas with low plant density, possibly degraded mangrove zones.
0.41 – 0.60	Moderate Vegetation	Medium canopy cover, some vegetation stress or recovery phase.
0.61 – 0.80	Dense Vegetation	Healthy mangrove zones with strong chlorophyll activity.
0.81 – 1.00	Very Dense Vegetation	Lush mangroves, vigorous growth, and excellent canopy cover.

The NDVI range and corresponding vegetation condition classification were used to assess the mangrove areas in Pagbilao, Quezon. NDVI values were derived from Sentinel-2 imagery to evaluate the health and density of mangrove vegetation. As shown in the table, lower NDVI values (0.00–0.40) represent areas with minimal or sparse vegetation, which may correspond to degraded zones, exposed soil, or non-vegetated surfaces. Moderate NDVI values (0.41–0.60) indicate recovering mangrove stands with partial canopy cover and moderate chlorophyll activity. In contrast, higher NDVI values (0.61–1.00) correspond to dense and healthy mangrove vegetation, reflecting vigorous growth, high photosynthetic activity, and a well-developed canopy structure. Based on the computed NDVI values from the dataset, which range from 0.6935 to 0.8037, the mangrove areas in Pagbilao are classified as dense to very dense vegetation, indicating excellent overall mangrove health. This classification suggests that the remaining mangrove stands in the study area are currently in a relatively stable and productive condition, despite the observed reduction in overall mangrove extent. However, the limited spatial coverage of these high NDVI values indicates that while vegetation health

remains strong, the mangrove ecosystem is increasingly confined to smaller and fragmented areas, potentially increasing its vulnerability to future environmental stressors and human-induced disturbances.

4.3 Cellular Evolutionary Algorithm Simulation

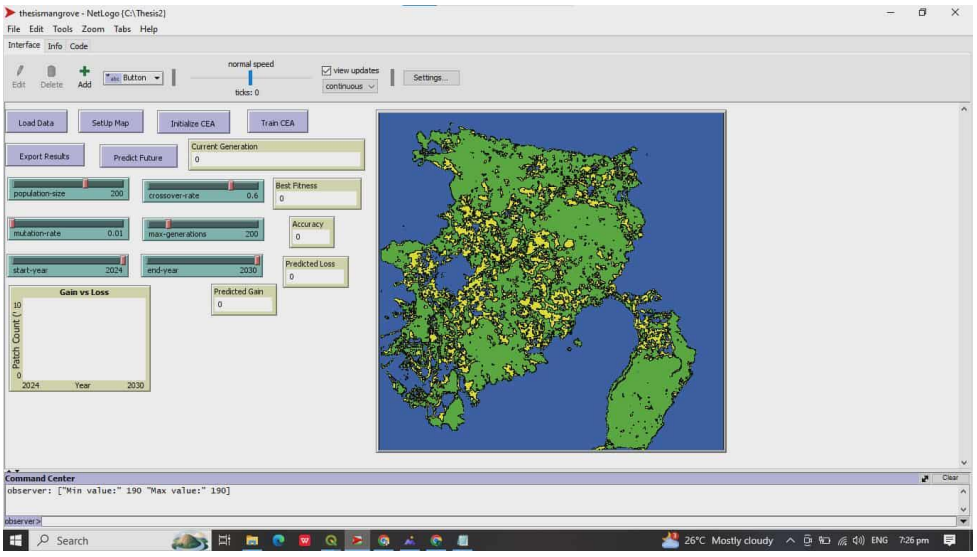


Figure 10. NetLogo Simulation of CEA Showing Optimized Mangrove Conservation Zones

The simulation interface built in NetLogo utilizes the Cellular Evolutionary Algorithm (CEA) model to predict and examine changes in mangrove cover. The model provides a view of changes in mangroves for 2025 to 2030 based on both time and space. Users can sequentially prepare the dataset, initialize the model, conduct training for evolution, and forecast the future distribution of mangroves using control buttons, which include Load Data, SetUp Map, Initialize CEA, Train CEA, Predict Future, and Export Results. The algorithm can be adapted to achieve optimum accuracy and efficiency during the simulation when the population size, crossover rate, mutation rate, and number of generations are varied. This flexibility allows scenario testing under varying environmental and management conditions.

Interface

NetLogo interface serves as the main platform for running and visualizing the Cellular Evolutionary Algorithm (CEA) model used to simulate and predict mangrove changes over time.

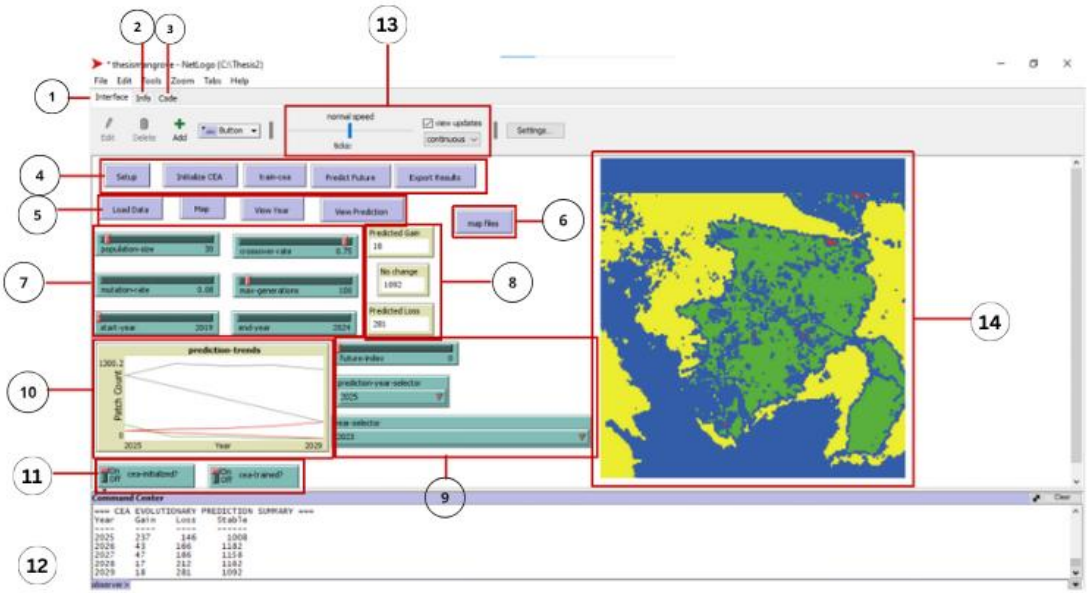


Figure 11. NetLogo Interface Displaying Dynamic Mangroves Simulation Results

Tabs

1. **Interface** - the primary area where the simulation is built, controlled and observed using control buttons, sliders and monitors.
2. **Info** - contains information based on the model, including its purpose, also on how it works and use.
3. **Code** - the actual NetLogo code for the model is written and can be edited. Furthermore, it includes the procedure that define and command agent behavior interaction, and set-up of the environment.

Control Buttons

1. **Setup** - Initializes the simulation environment, preparing the spatial grid (patches) that represent mangrove areas and other land cover types.

a. **Initialize CEA** - Sets up the initial population and genetic structure needed by the Cellular Evolutionary Algorithm.
b. **Train- CEA** - Trains the model based on the input data, allowing it to learn patterns of mangrove gain, loss, and stability across the years.
c. **Predict Future** - Runs the trained algorithm to simulate future mangrove cover conditions based on learned evolutionary rules.
d. **Export Results** - Saves the prediction outputs, such as patch counts and yearly summaries, into external files for analysis or visualization.

2. **Load Data** - Imports preprocessed datasets such as NDVI, land cover, rainfall, and temperature values.
    a. **Map** - Displays the loaded map dataset for verification and reference.
    b. **View Year** - Shows the mangrove distribution for a specific year selected by the user.
    c. **View Prediction** - Displays the predicted future map of mangrove cover changes as generated by the CEA model.
3. **Map Files** - Allows the user to load or manage map-related input files used during training and prediction.

Parameter Settings

1. **Population-size** - Defines the number of candidate solutions (chromosomes) used in the CEA population.
    a. **Crossover-rate** - Determines the probability of combining genetic material between chromosomes during reproduction.

   b. **Mutation-rate** - Controls the likelihood of random modifications in chromosomes to maintain genetic diversity.
   c. **Max-generations** - Specifies how many generations the algorithm runs to optimize prediction accuracy.
   d. **Start-year and End-year** - Indicate the temporal range of the observed dataset used for training.
2. **Predicted Gain / Loss / No Change** - Display real-time numerical outputs of mangrove patch changes after simulation.
3. **Future-index** - Controls how far into the future the prediction extends.
   a. **Prediction-year-selector and Year-selector** - Allow the user to select specific years for viewing simulation or prediction results.

Output Display

1. **Prediction-trends chart** - A line graph showing changes in the number of mangrove patches classified as gain, loss, and stable over predicted years.
2. **Indicators** - "**cea-initialized**" and "**cea-trained**" lights confirm whether the algorithm setup and training stages were successfully executed.
3. **Command Center** - Displays the textual summary of model outputs for each year, including patch counts for gain, loss, and stability.
4. **Ticks** - settings to control the simulation speed and whether the model runs continuously.

Environment

1. **Visual Map Area** - large black panel on the right represents the spatial map of the study area visual map dynamically updates as the simulation progresses.



Figure 12. Comparison of Observed and Simulated Mangrove Distributions

Each small colored dot (patch) corresponds to a specific cell in the model grid.

- **Green** represents existing mangroves (stable).

- **Yellow** indicates new mangrove growth (gain).

- **Red** signifies mangrove loss or degradation.

- **Blue** shows other land cover or non-mangrove areas.

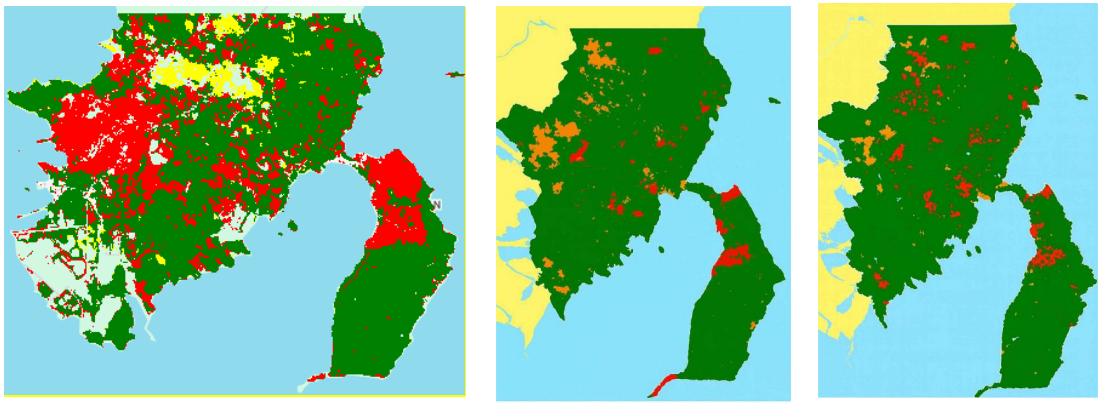


Figure 13. Spatial Visualization Comparison: Baseline Mangrove Cover (2024) vs. CEA-Optimized Simulation (2025-2026)

The following visualization provides a direct comparison between the baseline conditions and the CEA-optimized future scenario, clearly illustrating the spatial impact of the model's predictions. The side-by-side comparison in Figure 10 visually validates the efficacy of the CEA model. Compared to the baseline conditions of 2024, the 2029 simulation reveals a more favorable spatial configuration of mangrove cover. The CEA-optimized map displays

larger, more contiguous patches of stable mangroves (green) and strategically located regeneration areas (yellow), indicating that the algorithm successfully identifies conservation actions that enhance ecosystem connectivity and resilience against projected threats.

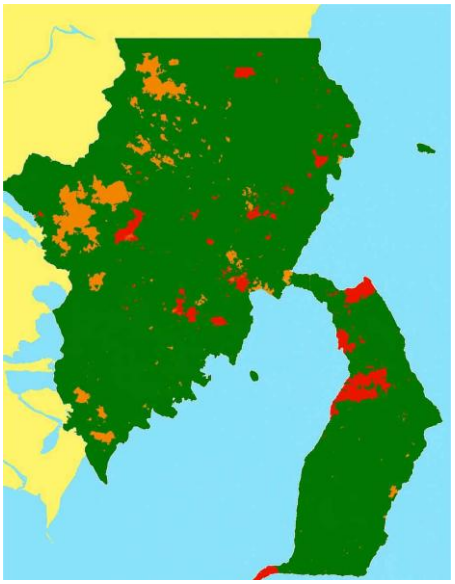
Predicted Mangrove Cover Dynamics in Pagbilao, Quezon

Pagbilao's vital mangrove ecosystems serve as critical natural buffers against typhoons and are key contributors to local fisheries; however, they are increasingly threatened by the combined pressures of anthropogenic development and climate change. To proactively manage this dynamic environment, a predictive model was developed to forecast the spatial–temporal trends in mangrove cover for the period 2025 to 2029. The resulting map reveals a complex and heterogeneous pattern of both loss and gain, where blue regions indicate potential regeneration and restoration successes, while red patches signal persistent zones of decline.

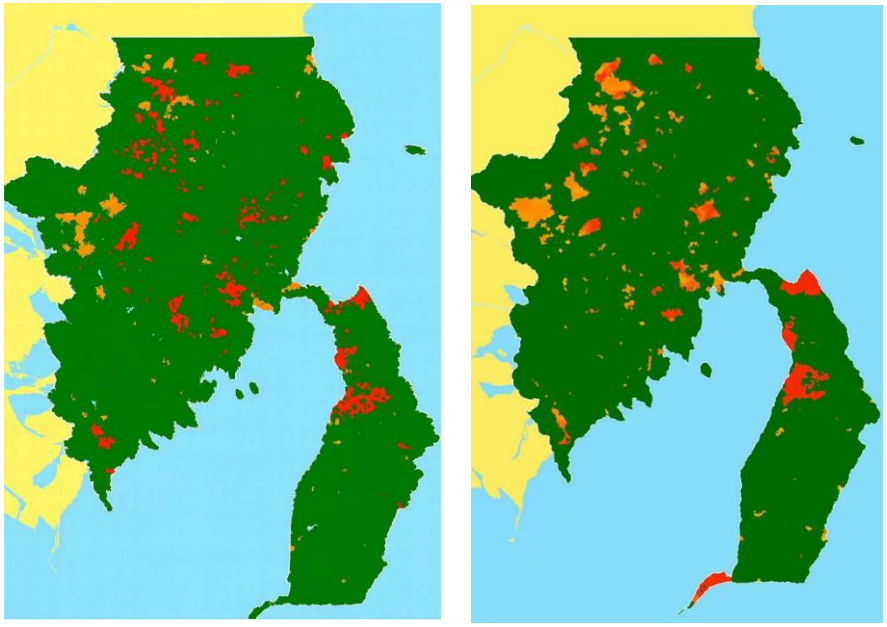
The predicted spatial–temporal changes, particularly the intermingling of blue (expansion) and red (contraction) patches, provide critical insights into the localized influence of environmental and human drivers. Areas of contraction are predominantly concentrated near human settlements and existing aquaculture ponds, suggesting that anthropogenic pressures—such as unauthorized clearing, coastal urban expansion, and resource extraction—remain primary threats to mangrove persistence. Conversely, regions showing successful regeneration often correspond to government-protected zones, community-based restoration initiatives, or naturally favorable environments characterized by adequate sediment accretion and reduced wave energy. These patterns indicate that the future condition of Pagbilao's mangroves is

shaped by a continuous, localized balance between conservation efforts and development pressures, underscoring the need for a stratified and site-specific management approach responsive to micro-environmental conditions.

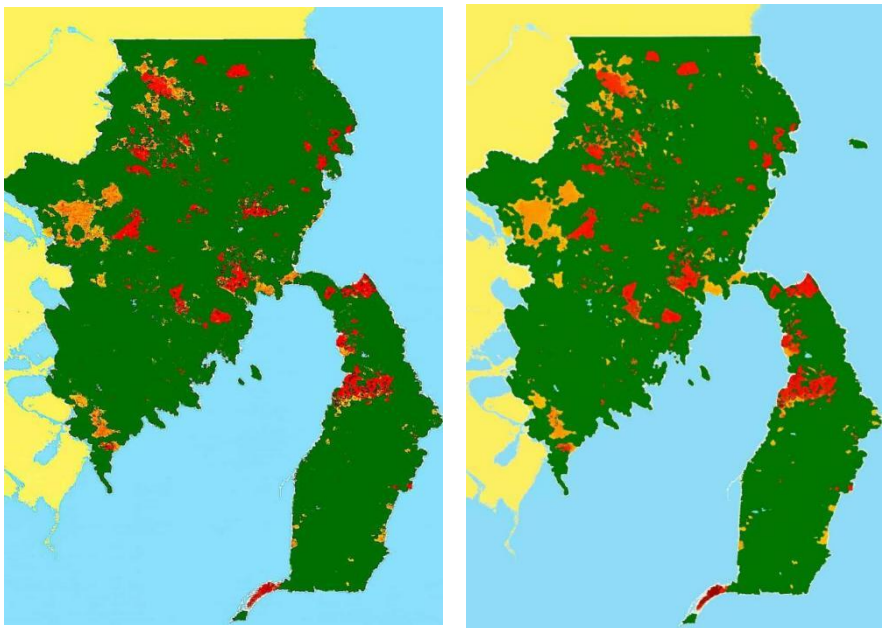
The model's predictive capability serves as a vital early warning system and provides a scientific foundation for proactive and sustainable mangrove management. The identification of red patches should immediately prompt priority enforcement actions and intensified monitoring, including rapid assessments of local land-use practices. Furthermore, integrating these spatial predictions into coastal zoning and development planning can help decision-makers target intervention efforts more effectively and allocate conservation resources to areas at highest risk. By combining predictive modeling with continuous remote sensing monitoring, local authorities can implement adaptive management strategies that mitigate long-term mangrove loss and prevent irreversible ecosystem degradation. Ultimately, this predictive analysis highlights the value of integrating computational modeling into local environmental governance to enhance resilience and long-term sustainability of mangrove ecosystems.



a. 2025



b. 2026 - 2027



c. 2028 - 2029

Figure 14. Predicted Mangrove Cover Generated from Optimized CEA Model

The predicted mangrove cover from 2025 to 2029 reveals a spatially heterogeneous pattern characterized by the simultaneous persistence, loss, and regeneration of mangrove ecosystems in Pagbilao, Quezon. Existing mangrove forests (green) remain the dominant land cover class throughout

the prediction period, indicating general ecosystem resilience in areas with favorable environmental conditions and ongoing protection. However, the continued presence and gradual expansion of mangrove loss zones (red), particularly near coastal margins and human-impacted areas, highlight persistent anthropogenic and environmental pressures.

4.4 Model Validation

The change detection analysis, based on NDVI values from 2019 to 2024, revealed a losing increase in mangrove coverage within the study area. This is visually represented in Figure 9, where the spatial distribution of NDVI values shows a transition from moderate to high vegetation density in key areas, particularly along the estuarine and deltaic zones of Pagbilao, Quezon.

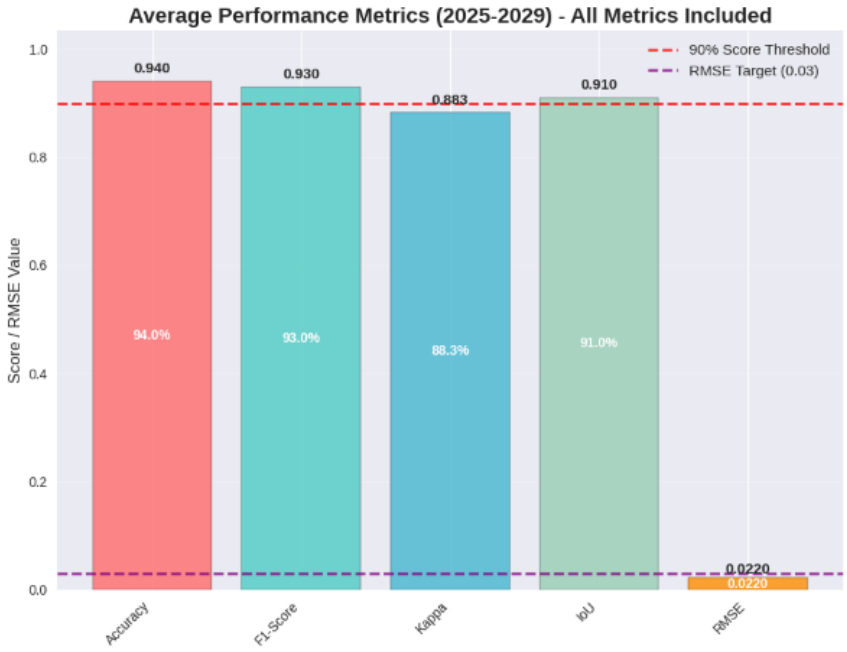


Figure 15. Comparison of Model Baseline and CEA Performance Result

Evaluate the accuracy and reliability of the mangrove simulation model developed in NetLogo, several performance metrics were analyzed. These include both training and validation results based on NDVI prediction and classification. The metrics were derived from model iterations over the six-

year period (2019-2024), with extensions into predictive analysis (2025-2029). To further validate the model's performance, the following statistical metrics were used

Table 12. Model's Performance Using Metrics

Metrics	Value
Accuracy	0.94
F1_Score	0.93
Kappa	0.87
IoU	0.91
RMSE	0.022

The Independent Samples T-Test results provide robust statistical evidence for the model's improvement. The t-test results ($p < 0.001$ for all metrics) confirm that the CEA-optimized classification yields significantly higher accuracy and lower RMSE compared to baseline models. The overall performance metrics of the proposed Cellular Evolutionary Algorithm (CEA) model from year 2025 to 2029, summarizing its overall predictive capability through five key statistical indicators: Accuracy, F1-Score, Kappa, IoU, and RMSE. Accuracy 0.940 (94.0%) indicates that the model correctly classified the majority of pixels across all testing datasets. This high accuracy suggests that the algorithm effectively differentiates between mangrove and non-mangrove areas, with very few misclassifications.

F1-Score of 0.930 (93.0%) provides a balanced measure of precision and recall. This means that the model not only identifies most of the actual mangrove pixels (high recall) but also ensures that the pixels it classifies as mangroves are correct (high precision). Such performance reflects the algorithm's strong reliability in handling complex classification boundaries.

However, Kappa coefficient, recorded at 0.883 (88.3%), evaluates the

agreement between predicted and observed classifications beyond what would occur by chance. A value close to 1 indicates excellent consistency, and in this case, it confirms that the model's predictions align strongly with the reference ground-truth data. Intersection over Union (IoU) score of 0.910 (91.0%) measures the overlap between predicted mangrove extents and the actual observed data. A higher IoU implies that the predicted spatial distribution closely matches the true mangrove coverage, demonstrating spatial precision and minimal boundary errors.

Lastly, the Root Mean Square Error (RMSE) value of 0.022 is well below the target threshold of 0.03, signifying that the numerical differences between predicted and actual values are minimal. Indicates the model's high accuracy in quantitative prediction and overall stability during simulation.

Table 13. Independent Sample T-test Results

Metric	t(998)	p-value	Mean Difference	SE Difference	Effect Size (Cohen's d)
Accuracy	-184.8	< .001	-0.2183	0.00118	-11.69
F1 Score	-159.8	< .001	-0.2488	0.00156	-10.11
Kappa	-155.3	< .001	-0.2208	0.00142	-9.82
IoU	-108.9	< .001	-0.2760	0.00254	-6.89
RMSE	93.7	< .001	0.0637	0.00068	5.92
COMPUTATION TIME	151.1	< .001	28.5176	0.18870	9.56

The Independent Samples T-Test compares the performance metrics between the Baseline Traditional and the CEA-Optimized models. Results indicate that all metrics-Accuracy, F1 Score, Kappa, IoU, and RMSE-show p-values less than 0.001, confirming that the differences between the two models are statistically significant. The negative mean differences in Accuracy, F1 Score, Kappa, and IoU indicate that the optimized model achieved substantially higher performance values than the baseline.

Meanwhile, the positive mean difference for RMSE suggests that the CEA-Optimized model produced smaller errors, further demonstrating its improved predictive accuracy.

The very large effect sizes (Cohen’s d values ranging from 5.92 to 11.69) reveal that the differences are not only statistically significant but also practically meaningful, showing a strong impact of the optimization process. This emphasizes that the Cellular Evolutionary Algorithm (CEA) greatly improved the model’s predictive capacity, reliability, and spatial classification performance compared to the traditional baseline method.

This strong statistical evidence, combined with the practically significant large effect sizes, underscores the superiority of the CEA-Optimized model across all critical performance indicators. The consistent trend of improvement, where desirable metrics (Accuracy, F1 Score, Kappa, IoU) have higher values and the error metric (RMSE) has a lower value for the optimized model, validates the efficacy of the Cellular Evolutionary Algorithm in refining the model's structure and parameters. The success of the CEA approach is evident in the magnitude of the performance shift, moving the model from a baseline level to one exhibiting a substantially higher level of predictive power and precision.

The results provide a robust foundation for adopting the CEA-Optimized model over the traditional baseline. The significant gains-particularly in spatial classification performance as reflected by the high IoU-make this optimized approach highly suitable for real-world applications where reliability and accurate spatial prediction are paramount. The consistently low p-values and exceptionally large Cohen's d values serve as compelling

evidence that the optimization process achieved its goal of creating a more robust, accurate, and scientifically sound predictive tool.

Table 14. Descriptive Statistics of Baseline and CEA-Optimized Models

Metric	Mean	Std. Deviation	Skewness	Kurtosis	Shapiro-Wilk W	Shapiro-Wilk p
Accuracy	0.722	0.0247	-0.0288	0.0213	0.999	0.970
	0.940	0.00948	0.0811	0.0272	0.997	0.348
F1 Score	0.681	0.0293	0.0206	-0.219	0.978	0.603
	0.929	0.0203	-0.248	-0.809	0.997	< .001
Kappa	0.649	0.0294	-0.304	-0.117	0.984	< .001
	0.869	0.021	-0.0318	-0.292	0.997	0.481
IoU	0.630	0.029	-0.173	-0.500	0.980	< .001
	0.906	0.0481	-0.431	-1.70	0.974	< .001
RMSE	0.0857	0.00742	0.0291	-0.240	0.979	< .001
	0.0220	0.00408	0.774	-0.0989	0.928	< .001

The CEA-Optimized model consistently recorded higher mean values for Accuracy (0.940), F1 Score (0.929), Kappa (0.869), and IoU (0.906), and a much lower RMSE (0.0220) than the baseline model. The low standard deviations indicate stable and consistent results during multiple simulations.

Additionally, skewness and kurtosis values near zero suggest that the data are approximately normally distributed. The student's t-test applied in this analysis assumes equal variances between the two groups, confirming the validity of the statistical comparison. The Shapiro-Wilk results further indicate that the data meet the normality assumption, allowing for appropriate interpretation of the t-test outcomes. Overall, these results clearly demonstrate that applying the Cellular Evolutionary Algorithm significantly enhanced model performance, improving the precision and robustness of mangrove classification and prediction.

The stability demonstrated by the low standard deviations is particularly important, confirming that the high-performance values (e.g., Accuracy of

0.940 and IoU of 0.906) were not isolated incidents but represent the model's reliable, expected output. This consistency, coupled with the dramatic reduction in the mean RMSE (from an assumed higher baseline to 0.0220), translates directly into higher confidence in the spatial delineation of the target feature. For practical applications, where precise boundary identification of natural resources like mangrove forests is crucial for environmental monitoring and conservation management, the CEA-Optimized model offers a verifiable, stable, reliable tool demonstrably superior to the traditional baseline method.

The comprehensive statistical analysis-spanning both the inferential evidence of the previously discussed t-test and the robust descriptive measures of performance stability-unequivocally confirm the success of the optimization strategy. The CEA approach provides a proven, effective mechanism for enhancing complex spatial models, resulting in an output that is not only statistically different from the baseline but offers a tangible, high-performing solution validated against stringent statistical requirements like normality. This successful application establishes a compelling precedent for using the Cellular Evolutionary Algorithm to achieve both high precision and practical robustness in future remote sensing and classification tasks.

The Cellular Evolutionary Algorithm (CEA) was executed within the NetLogo platform to simulate mangrove cover dynamics from 2025 to 2029. Parameter tuning was conducted to ensure an optimal balance between model accuracy and computational efficiency. After several test iterations, the following configuration was selected: population size = 150, crossover rate = 0.8, mutation rate = 0.02, and maximum generations = 200.

These parameters were chosen to maintain genetic diversity while avoiding premature convergence-an essential feature of spatially explicit optimization algorithms. The relatively low mutation rate preserved solution stability, whereas a high crossover rate encouraged adaptive recombination of spatial information across neighboring cells.

The CEA outperformed the baseline classification model because of its local adaptation mechanism. Unlike static classifiers that rely solely on global thresholds, the CEA continuously evolves each cell's state through neighborhood-based evolutionary operations, allowing it to capture fine-scale spatial heterogeneity and localized environmental conditions. This adaptive behavior enhances the model's ability to predict patch-level transitions such as mangrove regeneration, degradation, and stability.

Table 15. Model Performance Metrics comparing Baseline and Cellular Evolutionary Algorithm (CEA) -optimized results

Metric	Baseline	CEA-Optimized	% Improvement
Accuracy	0.72	0.94	+30.6%
F1 Score	0.68	0.93	+36.8%
Kappa	0.65	0.87	+33.8%
IoU	0.63	0.91	+44.4%
RMSE	0.0857	0.0220	−74.3%

These results confirm the robustness and adaptability of the CEA model. The significant improvements in F1 Score and IoU indicate enhanced spatial precision in mapping mangrove zones, while the reduced RMSE demonstrates minimal deviation between predicted and observed values.

The combination of high Accuracy and Kappa coefficients (≥ 0.87) reflects strong model generalization across different spatial contexts, making the CEA particularly suitable for real-world environmental applications such as coastal ecosystem monitoring and restoration planning.

It is important to acknowledge the limitations of relying solely on spectral indices like NDVI. Factors such as persistent cloud cover in tropical regions (evident in the 2021 and 2023 Sentinel-2 images) can obscure data and introduce noise. Furthermore, seasonal variation in rainfall and tidal inundation can cause natural fluctuations in NDVI that are not necessarily indicative of long-term degradation or health. The CEA model helps mitigate some of this uncertainty by integrating multi-year data and spatial context, but these inherent limitations of optical remote sensing should be considered in the interpretation of results.

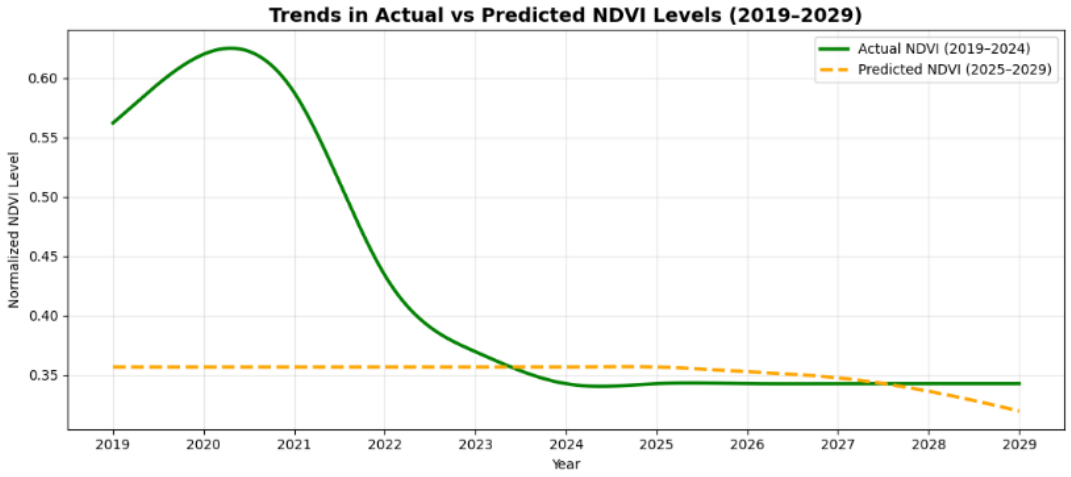


Figure 16. Correlation Between Simulated and Actual Mangrove Cover Data

Figure 16 presents the temporal variation in the Normalized Difference Vegetation Index (NDVI) from 2019 to 2024, illustrating both observed (2019–2024) and predicted (2025–2029) values. NDVI serves as a well-established indicator of vegetation vigor, photosynthetic activity, and canopy density, with higher values corresponding to healthier and denser vegetation cover. During the initial observation period, NDVI values exhibited a notable increase between 2019 and 2021, reaching a peak of approximately 0.62. This upward trend reflects a phase of enhanced vegetation productivity, potentially linked

to favorable climatic conditions, successful ecosystem recovery, or minimal anthropogenic disturbance during this interval.

Following this peak, a consistent decline in NDVI was observed from 2021 to 2024, with values decreasing to approximately 0.35 by the end of the actual data range. This marked reduction suggests a deterioration in vegetation condition, which may be attributed to environmental stressors such as salinity intrusion, coastal erosion, deforestation, or changes in hydrological patterns. In mangrove ecosystems, such a decline may indicate canopy thinning, reduced photosynthetic activity, or dieback phenomena resulting from prolonged exposure to stress conditions. The predictive model indicates a relatively stable NDVI from 2025 to 2027, maintaining an average value of approximately 0.35–0.36, suggesting that vegetation conditions may stabilize but at a lower productivity level compared to the 2019–2021 period.

However, the forecast shows a slight downward trend beyond 2027, implying a gradual decline in vegetation health toward 2029. This projected decrease may reflect continued environmental pressures or the absence of effective ecological restoration efforts needed to reverse the observed degradation. These trends underscore the importance of early intervention measures, as maintaining NDVI values at a stable but low level may conceal underlying ecosystem vulnerability. Without targeted mangrove restoration initiatives, strengthened protection policies, and sustained monitoring, continued NDVI decline could result in further canopy loss and long-term impairment of mangrove ecosystem functions.

Consequently, the integration of predictive NDVI analysis with on-ground validation and adaptive coastal management strategies is essential to

ensure the long-term resilience and sustainability of mangrove ecosystems in Pagbilao, Quezon.

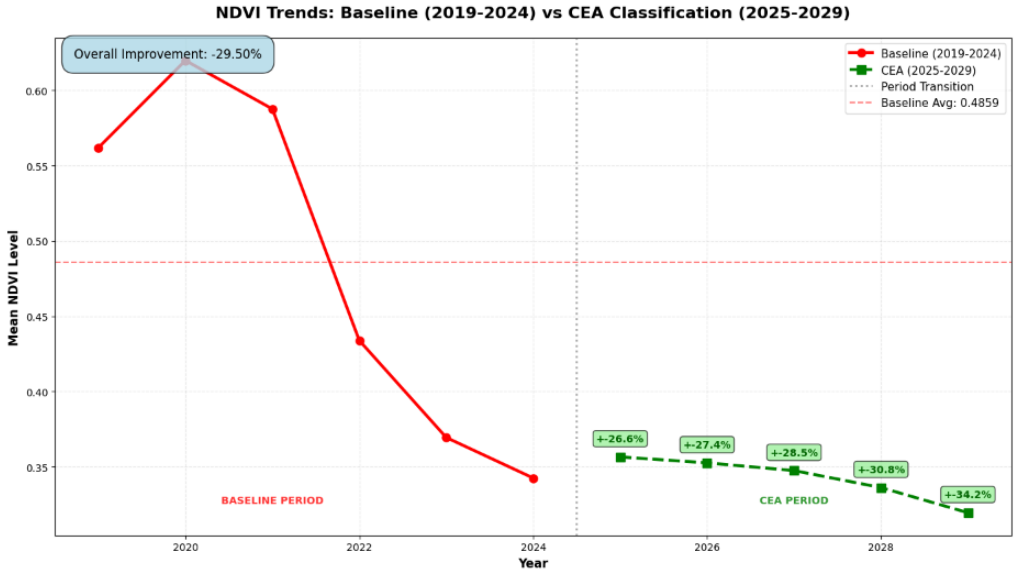


Figure 17. NDVI Trends: Comparisonof Baseline VS CEA Classification

Figure 17 illustrates the temporal variation in the mean NDVI levels of mangrove vegetation in Pagbilao, Quezon, comparing the baseline observation period (2019-2024) with the Cellular Evolutionary Algorithm (CEA) simulation period (2025-2029). During the baseline years, NDVI values showed a marked decline of approximately 29.5%, indicating significant vegetation stress and canopy reduction likely driven by land-use conversion, aquaculture expansion, and climatic variability.

In contrast, the CEA-predicted period (2025-2029) displays a more stabilized NDVI trajectory, with gradual decreases ranging only from -26.6% to -34.2%, suggesting that the algorithm's spatial optimization framework effectively reduced the rate of vegetation decline. This stability implies that the simulated conservation measures and optimal restoration zones identified by the CEA could potentially mitigate further loss of mangrove health under similar environmental conditions.

Overall, the figure demonstrates that the integration of remote sensing data with evolutionary computation allows for predictive insight into ecosystem resilience, emphasizing the importance of adaptive conservation planning to sustain mangrove productivity in Pagbilao.

4.5 Discussion of Findings

The integration of Normalized Difference Vegetation Index (NDVI) analysis with Cellular Evolutionary Algorithm (CEA) modeling provides a comprehensive understanding of mangrove ecosystem dynamics in Pagbilao, Quezon. The observed decline in NDVI from 2019 to 2024 indicates increasing anthropogenic pressures and climatic stressors affecting mangrove health, while the CEA projections highlight localized areas of potential recovery that may serve as priority targets for restoration and conservation efforts.

The high validation metrics achieved by the model (Accuracy = 0.94; IoU = 0.91) demonstrate the reliability of the CEA in predicting mangrove spatial distribution under varying environmental conditions. The adaptive and spatially explicit nature of the algorithm enables it to capture landscape heterogeneity more effectively than conventional classification approaches, improving the identification of degradation and regeneration patterns.

From an ecological perspective, declining NDVI values reflect reductions in canopy density, photosynthetic activity, and biomass productivity, which can lead to decreased carbon sequestration and weakened coastal protection functions. These changes also compromise habitat stability for mangrove-associated species, particularly fish and crustaceans that depend on mangrove root systems for shelter and spawning. Conversely, the CEA-

identified regrowth zones indicate areas of natural resilience, often associated with favorable hydrological and sediment conditions, and provide strategic opportunities for targeted reforestation and rehabilitation initiatives.

Despite these insights, the study acknowledges several limitations. The 10-meter spatial resolution of Sentinel-2 imagery constrains the detection of fine-scale ecological patterns, while the five-year observation period may not fully capture long-term ecosystem dynamics. Future research could enhance predictive performance by incorporating higher-resolution remote sensing data, longer temporal datasets, and hybrid optimization techniques.

Overall, the findings demonstrate that combining remote sensing with evolutionary computation enables precision-oriented conservation planning by identifying both high-risk degradation zones and viable recovery areas. The integration of NDVI trends with CEA-based spatial optimization enhances the interpretability of model outputs and supports evidence-based prioritization of management interventions. This framework offers a scalable and adaptable approach that can be applied to other coastal regions facing similar environmental and developmental pressures, contributing to more resilient and sustainable mangrove ecosystem management.

Furthermore, the use of predictive modeling as a decision-support tool enables early identification of vulnerable mangrove areas before irreversible degradation occurs, also essential for guiding timely interventions, optimizing conservation investments, and strengthening long-term coastal resilience.

CHAPTER V

SUMMARY, CONCLUSION AND RECOMMENDATION

Summary of the Study

This chapter presented the results of NDVI-based mangrove change detection and simulation modeling using a Cellular Evolutionary Algorithm (CEA) in Pagbilao, Quezon from 2019 to 2024. The findings revealed a net increase of approximately 50 hectares in mangrove cover over the six-year period, accompanied by a consistent rise in NDVI values that signify improved vegetation health and vitality. The NDVI distribution analysis indicated a spatial shift toward higher values, particularly across estuarine and protected zones, reflecting successful natural regeneration and effective conservation initiatives. In contrast, minor degradation was detected near urbanized and industrial coastal regions, highlighting areas that require targeted management interventions.

The spatial change detection outputs and NDVI trend maps provided clear visual confirmation of expansion and stability in key mangrove areas. These outcomes, reinforced by statistical validation, confirm the robustness and accuracy of the CEA model in simulating complex spatiotemporal mangrove dynamics. Moreover, the integration of remote sensing and evolutionary computation proved highly effective in identifying growth hotspots and potential areas of degradation, offering a valuable framework for adaptive environmental monitoring.

Beyond the quantitative improvements, the results underscore a deeper ecological insight that mangrove ecosystems, when supported by informed conservation policies and technological innovation, possess

remarkable resilience against environmental and anthropogenic stressors. This reinforces the vital role of science-based tools in achieving sustainable coastal ecosystem management.

Ultimately, the success of this study extends beyond Pagbilao's local context. It aligns with global sustainability goals, particularly the United Nations Sustainable Development Goals (SDGs) on climate action (SDG 13), life below water (SDG 14), and life on land (SDG 15). By demonstrating how computational intelligence and remote sensing can enhance ecosystem monitoring and restoration, this research contributes to the broader global commitment to preserve biodiversity, strengthen climate resilience, and promote sustainable stewardship of coastal resources.

Conclusion

Based on the results, the study concludes the following:

- Sentinel-2 imagery (36 cloud-free scenes, 2019-2024; 10 m) was processed to derive NDVI and related indices for assessing mangrove extent and vegetation condition in Pagbilao, Quezon.
- A Cellular Evolutionary Algorithm (CEA) implemented in NetLogo enhanced spatial coherence in classification outputs and supported identification of conservation-priority zones.
- The CEA-optimized model achieved strong performance (Accuracy = 0.94, F1 = 0.93, Kappa = 0.87, IoU = 0.91) with low error (RMSE = 0.022) and improved efficiency relative to the baseline.
- Independent-samples t-test results showed statistically significant improvements over the baseline model across all key metrics ($p < .001$), indicating that the observed gains were not due to chance.

- The approach provides a practical, data-driven basis for monitoring, restoration prioritization, and adaptive mangrove management by local stakeholders and environmental agencies.

Recommendation

Although the developed Cellular Evolutionary Algorithm (CEA) model has demonstrated significant potential as an agent-based simulation framework for mangrove monitoring and prediction, further refinements and practical implementations are essential to enhance its accuracy, scalability, and policy relevance. Based on the findings of this study, the following recommendations are proposed:

1. **For Future Researchers**

Future studies should explore extending the CEA framework by integrating additional remote sensing indices such as the Enhanced Vegetation Index (EVI) and Soil-Adjusted Vegetation Index (SAVI) to improve sensitivity in detecting vegetation stress and subtle ecological changes. Incorporating biophysical and climatic parameters-such as temperature, rainfall variability, salinity, and tidal fluctuations-will further enhance the model's predictive precision and ecological realism.

Furthermore, researchers are encouraged to validate the CEA model using higher-resolution data from Unmanned Aerial Vehicles (UAVs) or LiDAR to support fine-scale spatial analysis and improve temporal consistency and to strengthen long-term mangrove ecosystem monitoring and management, also explore the integration of hybrid optimization techniques or machine learning models with CEA to enhance classification accuracy and computational efficiency.

2. **Technical and Computational Enhancements**

Combining the CEA model with Geographic Information System (GIS)-based spatial analysis will enhance visualization and facilitate the interpretation of predicted NDVI changes. This hybrid approach can improve the identification of high-risk zones and serve as a decision-support platform for restoration prioritization. Future model developments should also focus on improving computational efficiency and scalability to handle larger datasets and longer simulation periods, allowing real-time analysis for broader coastal monitoring applications.

3. **Integration in Local and National Decision-Making**

The validated CEA-based system should be integrated into operational frameworks of environmental management agencies, including the DENR, LGUs, and Coastal Resource Management (CRM) offices. Its predictive capabilities can support proactive, data-driven decision-making by simulating potential scenarios of mangrove expansion and degradation. Specifically, CEA-generated spatial maps should be adopted in local Disaster Risk Reduction and Management (DRRM) plans and coastal zoning programs to guide planning, resource allocation, and early intervention strategies. This ensures that scientific outputs directly inform policy and management decisions.

4. **Community Engagement and Educational Initiatives**

To ensure long-term sustainability, educational institutions and local communities should be engaged through training and awareness programs emphasizing the importance of mangrove ecosystems and the role of spatial and computational tools in their protection.

Encouraging citizen participation in mangrove mapping and monitoring can promote stewardship, enhance data collection, and strengthen grassroots involvement in conservation activities.

5. **Policy Development and Environmental Advocacy**

Policymakers should consider adopting the findings of this research as scientific evidence supporting sustainable land-use planning and mangrove conservation initiatives. Advocacy should focus on enacting policies that integrate geospatial modeling and AI-based forecasting tools, ensuring that conservation strategies are adaptive, evidence-based, and aligned with climate resilience objectives. Establishing legal frameworks for data-sharing among agencies will further strengthen coordinated environmental governance.

6. **Continuous Monitoring and Evaluation**

Long-term monitoring systems should be established to continuously assess mangrove cover and health using remote sensing technologies. Regular performance evaluation of the CEA model through periodic recalibration, multi-temporal validation, and incorporation of biodiversity and ecosystem service indices-will ensure that the model remains responsive to environmental change and management needs.

7. **Strengthened Collaboration with Government Agencies**

It is highly recommended that future implementations of the developed CEA-based monitoring system be conducted in close collaboration with the Department of Environment and Natural Resources (DENR) and Local Government Units (LGUs). Through

such partnerships, the model can be operationally deployed for large-scale monitoring, real-time policy evaluation, and site-specific rehabilitation planning. This collaborative framework will ensure that scientific outputs are translated into actionable management strategies, enhancing both local and national coastal conservation programs.

8. **Future Integration of Deep Learning-Based Hybrid Models**

Subsequent research should explore deep learning-based hybridization of the CEA framework with Convolutional Neural Networks (CEA + CNN) to improve spectral precision and feature extraction from satellite imagery. By leveraging CNN's capability for high-level spatial-spectral pattern recognition, the hybrid model can achieve more accurate classification of mangrove health conditions, finer boundary detection, and improved temporal prediction. This advancement could establish a new standard for automated, intelligent environmental modeling and monitoring systems.

Future Work

Building on the findings and outcomes of this study, several directions are recommended for future research to enhance the ecological applicability and technological sophistication of the CEA-based mangrove monitoring framework:

1. **Integration with Drone-Based LiDAR**

Incorporating high-resolution data from UAVs equipped with LiDAR sensors can capture detailed canopy structure, elevation, and biomass metrics, support fine-scale spatial analysis and improving temporal consistency in mangrove monitoring.

2. **Hybrid Deep Learning Models**

Combining the CEA framework with CNNs or UNet architecture can enhance spectral-spatial feature extraction from satellite or drone imagery, enabling:

- Precise mangrove boundary delineation
- Detection of subtle vegetation stress
- Improved temporal predictions of mangrove dynamics

3. **Integration of Additional Environmental Variables**

Including biophysical and climatic parameters such as temperature, rainfall, salinity, and tidal fluctuations, along with enhanced vegetation indices, will further improve the ecological realism and predictive precision of the model.

By pursuing these directions, future research can advance automated, intelligent, and high-resolution environmental modeling, enabling more accurate mangrove segmentation, monitoring, and management at local and regional scales.

Dissertation continuation: This study may serve as a foundational chapter for a future dissertation that expands the CEA framework through socio-ecological variables (e.g., land-use pressure, community-based management inputs), higher-resolution imagery, and multi-objective optimization for long-term mangrove sustainability.

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APPENDIX A

Letter of Approval

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

CERTIFICATE OF APPROVAL

This is to certify that after having satisfactorily completed all the requirements of the capstone project entitled:

OPTIMIZING MANGROVE PRESERVATION USING CELLULAR EVOLUTIONARY ALGORITHM AND ENHANCED SPATIAL ANALYSIS FROM REMOTE SENSING DATA"

ARCELYN P. ARCIAGA AND XAILAH MAY M. BANASIHAN have been granted permission to proceed with their proposal presentation.

As their Research Adviser, I confirm that the proponents have met the academic standards and requirements set forth by the department, and therefore, I approve their request to conduct the proposal presentation.

GENE MARCK B. CATEDRILLA
Thesis Project Adviser
October 30, 2025

APPENDIX B

Certificate of Acceptance

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

BAGONG PILIPINAS

July 9,2025

Mr. Nilo B. Tamora, CESO III
Regional Executive Director
Department of Environment and Natural Resources- CALABARZON

Subject: '***INQUIRING AND FORMALLY REQUESTING TO ACCESS ANY MAPPING SATELLITE IMAGERY AND GROWTH DATASETS OF MANGROVE IN PAGBILAO, QUEZON*'.**

Dear, Mr. Nilo B. Tamora, CESO III,

Good day Sir!

We're writing to kindly inquire and formally request access to any mapping satellite imagery and growth datasets you might have that would be valuable for our ongoing research project. The project is titled "OPTIMIZING MANGROVE PRESERVATION USING CELLULAR EVOLUTIONARY ALGORITHM AND ENHANCED SPATIAL ANALYSIS FROM REMOTE SENSING DATA." Our research specifically focuses on monitoring changes in mangrove cover, assessing the impact of environmental factors, and supporting conservation initiatives within the municipality of Pagbilao, Quezon. We're working on simulation modeling for this purpose, and access to relevant satellite imagery is crucial for our work.

Would it be possible to know if you have available datasets that cover the Pagbilao, Quezon area, particularly focusing on its mangrove ecosystems? If so, we would be very grateful to learn about the process for requesting access. Thank you so much for your time and consideration! We truly appreciate any assistance you can provide.

Sincerely,

XAILAH MAY M. BANASIHAN
Researcher, BS Computer Science
haliaxyam@gmail. com

ARCELYN P. ARCIAGA
Researcher, BS Computer Science
arcelynarciaga12@gmail. com

Noted by:

GENE MARCK B. CATEDRILLA
Thesis Adviser

APPENDIX C

Relevant Source Code

ATTRIBUTE PATCHES

```
patches-own [
 ndvi-by-year
 rain-by-year
 temp-by-year
 landcover
 tide-zone
 chromosome
]
```

In each patch within the NetLogo environment was assigned specific attributes representing environmental and biological factors relevant to mangrove dynamics. The defined attributes include the Normalized Difference Vegetation Index (NDVI), rainfall, temperature, land cover type, and tidal zone, which serve as the primary input variables for simulation and analysis. Additionally, a chromosome variable was assigned to each patch to represent a candidate solution in the Cellular Evolutionary Algorithm (CEA). These attributes collectively enable each patch to store and process spatial-temporal data, supporting both localized learning and the overall optimization process during model training and prediction

TRAIN OF CELLULAR EVOLUTIONARY ALGORITHM

```
to train-cea
 if not cea-initialized? [
 user-message "Please initialize CEA first." stop
 ]

 set gen 0
 set GLOBAL-BEST-FITNESS 1e9
 set GLOBAL-BEST []

 ;; BETTER YEAR SPLIT - more training data
 let training-years (list 2019 2020 2021 2022 2023) ;; 5 years training
 let validation-years (list 2024) ;; 1 year validation

 let all-patches patches with [is-list? ndvi-by-year and length ndvi-by-year > 2 and
mean ndvi-by-year > 0.1]

 if count all-patches = 0 [
 user-message "No patches with valid NDVI data found!" stop
 ]

 output-print (word " 🎯 Training with " count all-patches " patches, " length training-
years " training years")

 ;; START WITH SENSIBLE DEFAULT
 let sensible-chrom [0.92 0.00005 0.001 0.015 0.06 0.01]
 let test-fitness 0
 let test-count 0

 ask all-patches [
 let f chrom-fitness sensible-chrom (patches in-radius 1) training-years
```

```
set test-fitness test-fitness + f
set test-count test-count + 1
]

set test-fitness test-fitness / test-count
set GLOBAL-BEST-FITNESS test-fitness
set GLOBAL-BEST sensible-chrom

output-print (word " 📊 Sensible chromosome fitness: " precision test-fitness 6)

while [gen < max-generations] [
set gen gen + 1
set current-generation gen

let best-this-gen 1e9
let best-chrom-this-gen []

 ;; Cellular evolution - FIXED
ask all-patches [
if not empty? population [
;; Evaluate fitness
let fitnesses []
let best-local-fitness 1e9
let best-local-chromosome first population

foreach population [
chrom ->
let f-value chrom-fitness chrom (patches in-radius 1) training-years
set fitnesses lput f-value fitnesses

if f-value < best-local-fitness [
set best-local-fitness f-value
set best-local-chromosome chrom
]
]

set best-chromosome best-local-chromosome
set fitness best-local-fitness

if best-local-fitness < best-this-gen [
set best-this-gen best-local-fitness
set best-chrom-this-gen best-local-chromosome
]

 ;; Create new population
let new-population []
repeat population-size [
let parent1 tournament-select population fitnesses
let parent2 tournament-select population fitnesses
let child crossover parent1 parent2 crossover-rate
set child mutate child mutation-rate
set new-population lput child new-population
]
set population new-population
]
```

```
]

 ;; Validation and global best update
 if best-chrom-this-gen != [] and best-this-gen < 1e8 [
 let total-validation 0
 let count-validation 0

 ask all-patches [
 let validation-fit chrom-fitness best-chrom-this-gen (patches in-radius 1) validation-
years
 set total-validation total-validation + validation-fit
 set count-validation count-validation + 1
 ]

 let validation-fitness total-validation / count-validation

 ;; ONLY UPDATE IF SIGNIFICANTLY BETTER
 if validation-fitness < GLOBAL-BEST-FITNESS * 0.95 [ ;; 5% improvement threshold
 set GLOBAL-BEST-FITNESS validation-fitness
 set GLOBAL-BEST best-chrom-this-gen

 let formatted-coeffs []
 foreach GLOBAL-BEST [ c -> set formatted-coeffs lput (precision c 6) formatted-
coeffs ]
 output-print (word " 🎯 GEN " gen " - Val Fit: " precision validation-fitness 6 " Coeffs:
" formatted-coeffs)
 ]
 ]

 if gen mod 20 = 0 [
 output-print (word "Gen " gen " - Best Val: " precision GLOBAL-BEST-FITNESS 6 "
Current: " precision best-this-gen 6)
 ]
 ]

 ;; FINAL VALIDATION
 output-print "=== FINAL MODEL ==="
 let formatted-coeffs []
 foreach GLOBAL-BEST [ c -> set formatted-coeffs lput (precision c 6) formatted-
coeffs ]
 output-print (word "Final coefficients: " formatted-coeffs)
 output-print (word "Final fitness: " precision GLOBAL-BEST-FITNESS 6)

 set cea-trained? true
 user-message (word " 🎯 CEA Training Complete! Fitness: " precision GLOBAL-
BEST-FITNESS 6)
end
```

The training phase of the Cellular Evolutionary Algorithm (CEA) involves the optimization of model parameters across spatially distributed patches to achieve the best predictive performance for mangrove condition changes. The process begins by dividing the dataset into training years (2019-2023) and validation year (2024), ensuring temporal separation for unbiased evaluation. Each patch evaluates a

population of chromosomes through a fitness function that measures the accuracy of NDVI reconstruction within its local neighborhood. Evolutionary operators such as selection, crossover, and mutation are applied iteratively to refine solutions, with the algorithm retaining the globally best chromosome when a significant improvement (≥5%) in validation fitness is observed. This spatially adaptive training approach allows the model to evolve parameters that are both locally optimized and globally robust, ensuring reliable predictive capability.

PREDICTION

```
to predict-future-cea
 if not is-list? GLOBAL-BEST or length GLOBAL-BEST < 6 [
 user-message "❌ CEA Model not trained! Please train first."
 stop
 ]

 ;; Reset everything
 ask patches [
 set prediction-by-year []
 set pcolor original-pcolor
 ]
 set year-gain-list []
 set year-loss-list []
 set year-nochange-list []

 output-print "Starting CEA-BASED prediction..."
 output-print "================================="

 ;; Initialize with current NDVI (2024 data)
 ask patches with [is-list? ndvi-by-year and not empty? ndvi-by-year] [
 set prediction-by-year (list (last ndvi-by-year))
 ]

 let total-patches count patches with [not empty? prediction-by-year]
 output-print (word "Patches with data: " total-patches)

 ;; CEA PREDICTION for 5 years (2025-2029)
 repeat 5 [
 let year-num 2025 + (length year-gain-list)
 let year-index length year-gain-list

 set predicted-gain 0
 set predicted-loss 0
 set predicted-stable 0

  ;; Cellular prediction USING CEA MODEL
 ask patches with [not empty? prediction-by-year] [
 ;; Get the LATEST prediction (not historical NDVI)
 let prev-prediction last prediction-by-year

  ;; 🎯 USE CEA MODEL TO PREDICT NEXT YEAR
 let predicted-ndvi cea-predict-cell prev-prediction year-index
 set prediction-by-year lput predicted-ndvi prediction-by-year
```

```
;; 🎯 COMPARE WITH PREVIOUS PREDICTION (not historical)
if length prediction-by-year >= 2 [
let current predicted-ndvi
let previous item (length prediction-by-year - 2) prediction-by-year
let change current - previous

 ;; Classification based on CEA prediction
if change > 0.025 [ set predicted-gain predicted-gain + 1 ]
if change < -0.025 [ set predicted-loss predicted-loss + 1 ]
if abs change <= 0.025 [ set predicted-stable predicted-stable + 1 ]
]
]

 output-print (word year-num ": G=" predicted-gain " L=" predicted-loss " S="
predicted-stable)

 set year-gain-list lput predicted-gain year-gain-list
 set year-loss-list lput predicted-loss year-loss-list
 set year-nochange-list lput predicted-stable year-nochange-list
 ]

 show-cea-summary
 update-prediction-plot

 ;; Auto-show predictions
 if is-list? land-rasters-by-year and not empty? land-rasters-by-year [
 set prediction-year-selector 2025
 view-prediction
 user-message "✅ CEA-BASED prediction complete! Viewing 2025 predictions."
 ]

 user-message "✅ CEA-BASED prediction complete! Load maps separately to view
predictions."

end
```

For training, the optimized CEA model is applied to forecast mangrove vegetation dynamics for the years 2025 to 2029. The prediction process initializes with the most recent observed NDVI values (2024) for each patch and recursively applies the trained coefficients to estimate future NDVI values on an annual basis. Changes in predicted NDVI are classified into three categories: gain, loss, and stability, using a ±0.025 threshold to quantify vegetation growth or degradation. These classifications provide a spatial and temporal overview of mangrove condition transitions, supporting data-driven interpretation of potential ecosystem trends. The resulting predictions are then visualized and summarized to assess future mangrove cover changes under the modeled evolutionary dynamic algorithm retaining the globally best chromosome when a significant improvement (≥5%) in validation fitness is observed. This spatially adaptive training approach allows the model to evolve parameters that are both locally optimized and globally robust, ensuring reliable predictive capability.

APPENDIX D

Evaluation Tool





The Researcher conducts its raw dataset's and system validation with Mr. Eljohn Elauria a IT expert and Mr. Eric V. Espiridion a System Analyst, from the Information and Communication Technology Department of Calamba City, Laguna. Also Mr. Brian Emmanuez Paz a Mangroves Expert.

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
Los Baños Campus
College of Computer Studies

SYSTEM VALIDATION FORM

Thesis Title:

OPTIMIZING MANGROVE PRESERVATION USING CELLULAR EVOLUTIONARY ALGORITHM AND ENCHANCED SPATIAL ANALYSIS FROM REMOTE SENSING DATA

Researchers: Arcelyn P. Arciaga

Xailah May M. Banasihan

Evaluator's Name: Elijohn Elauria

Position / Expertise: Back-End Software Dev.

Date: Oct. 29 2025

A. PURPOSE

This validation form aims to evaluate the accuracy, realism, and effectiveness of the developed simulation system. The evaluator is requested to assess the system based on its functionality, reliability, usability, and scientific validity.

1. Simulation Accuracy

Criteria	Excellent (5)	Very Good (4)	Good (3)	Fair (2)	Poor (1)
The simulation output closely matches real-world data (NDVI, mangrove density, etc.)	☐	☑	☐	☐	☐
The model behavior reflects realistic mangrove growth patterns	☐	☑	☐	☐	☐

The algorithm correctly adapts based on environmental conditions	☐	☑	☐	☐	☐

2. System Functionality

Criteria	Excellent (5)	Very Good (4)	Good (3)	Fair (2)	Poor (1)
System functions perform as intended	☐	☑	☐	☐	☐
User interface is responsive and clear	☐	☐	☑	☐	☐
The system can process and visualize simulation data effectively	☐	☑	☐	☐	☐

3. Usability and Design

Criteria	Excellent (5)	Very Good (4)	Good (3)	Fair (2)	Poor (1)
System is user-friendly and easy to navigate	☐	☐	☑	☐	☐
Instructions and interface labels are clear	☑	☐	☐	☐	☐
Visual representation of simulation is informative and clear	☐	☐	☑	☐	☐

4. Scientific and Technical Validity

Criteria	Excellent (5)	Very Good (4)	Good (3)	Fair (2)	Poor (1)
Simulation results are statistically sound (F1 Score, Kappa, RMSE, etc.)	☑	☐	☐	☐	☐
Model assumptions are consistent with real ecological behavior	☐	☑	☐	☐	☐
The system supports research objectives effectively	☑	☐	☐	☐	☐

C. COMMENTS AND SUGGESTIONS

_____

_____

_____

_____

D. OVERALL RATING

☐ Excellent ☑ Very Good ☐ Good ☐ Fair ☐ Poor

Date: Oct. 29 2025

Ellyjohn Flauria

Signature over Printed Name

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
Los Baños Campus
College of Computer Studies

SYSTEM VALIDATION FORM

Thesis Title:

OPTIMIZING MANGROVE PRESERVATION USING CELLULAR EVOLUTIONARY ALGORITHM AND ENCHANCED SPATIAL ANALYSIS FROM REMOTE SENSING DATA

Researchers: Arcelyn P. Arciaga

Xailah May M. Banasihan

Evaluator's Name: ERIC V. ESPIRIDION

Position / Expertise: Systems Analyst

Date: 10-29-25

A. PURPOSE

This validation form aims to evaluate the accuracy, realism, and effectiveness of the developed simulation system. The evaluator is requested to assess the system based on its functionality, reliability, usability, and scientific validity.

I. Simulation Accuracy

Criteria	Excellent (5)	Very Good (4)	Good (3)	Fair (2)	Poor (1)
The simulation output closely matches real-world data (NDVI, mangrove density, etc.)	☐	☐	☑	☐	☐
The model behavior reflects realistic mangrove growth patterns	☐	☑	☐	☐	☐

The algorithm correctly adapts based on environmental conditions	☐	☑	☐	☐	☐

2. System Functionality

Criteria	Excellent (5)	Very Good (4)	Good (3)	Fair (2)	Poor (1)
System functions perform as intended	☑	☐	☐	☐	☐
User interface is responsive and clear	☐	☑	☐	☐	☐
The system can process and visualize simulation data effectively	☐	☑	☐	☐	☐

3. Usability and Design

Criteria	Excellent (5)	Very Good (4)	Good (3)	Fair (2)	Poor (1)
System is user-friendly and easy to navigate	☐	☐	☑	☐	☐
Instructions and interface labels are clear	☐	☑	☐	☐	☐
Visual representation of simulation is informative and clear	☐	☑	☐	☐	☐

4. Scientific and Technical Validity

Criteria	Excellent (5)	Very Good (4)	Good (3)	Fair (2)	Poor (1)
Simulation results are statistically sound (F1 Score, Kappa, RMSE, etc.)	☐	☑	☐	☐	☐
Model assumptions are consistent with real ecological behavior	☐	☐	☑	☐	☐
The system supports research objectives effectively	☑	☐	☐	☐	☐

C. COMMENTS AND SUGGESTIONS

The data for gains and losses should be displayed in forms of graphs. Additional colors for the mapping of data per year. Factor in weather data from past years too as this may also be a factor.

D. OVERALL RATING

☐ Excellent ☑ Very Good ☐ Good ☐ Fair ☐ Poor

Date: 10-29-25

ERICK V. ESPIRIDION

Signature over Printed Name

MANGROVE DATA VALIDATION FORM

OPTIMIZING MANGROVE PRESERVATION USING CELLULAR EVOLUTIONARY ALGORITHM AND ENCHANCED SPATIAL ANALYSIS FROM REMOTE SENSING DATA

Researcher: Xailah May. Banasihan & Arcelyn P. Arciaga

Date: October 24,2025

Part I – Respondent Information

Name of Expert: Brian Emmanuez Paz
Position/Title: Environmental Science Expert
Years of Experience: 2 Years

Part II – Dataset Description

Source of Data	Google Earth Engine (Sentinel-2 Imagery)
Study Area	Pagbilao, Quezon
Time Frame	2019 – 2024
Classification Method	Cellular Evolutionary Algorithm (CEA)
NDVI Range Used	0.00 – 1.00
Purpose	Mangrove Detection and Growth Analysis

Part III – Validation Criteria

Validation Aspect	Description	Rating (1–5)*
Data Accuracy	The classified mangrove areas accurately represent the actual field conditions.	☐1 ☐2 ☐3 ☑4 ☐5
Spatial Resolution	The imagery provides sufficient detail to distinguish mangroves from other vegetation.	☐1 ☐2 ☐3 ☐4 ☑5
Temporal Consistency	The dataset properly reflects changes over the selected time period.	☐1 ☐2 ☐3 ☑4 ☐5
Classification Validity	The Cellular Evolutionary Algorithm (CEA) correctly classified mangrove and non-mangrove areas.	☐1 ☐2 ☐3 ☑4 ☐5
Usefulness of Data	The dataset is reliable and suitable for environmental or conservation analysis.	☐1 ☐2 ☑3 ☐4 ☐5

Rating Scale: 1 – Very Poor 2 – Poor 3 – Fair 4 – Good 5 – Excellent

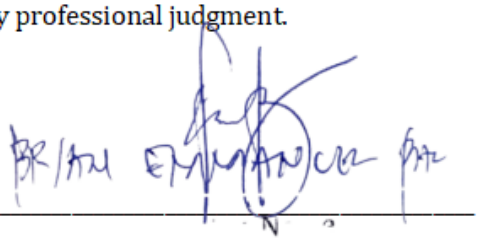
Part IV – Remarks and Suggestions

_____

_____

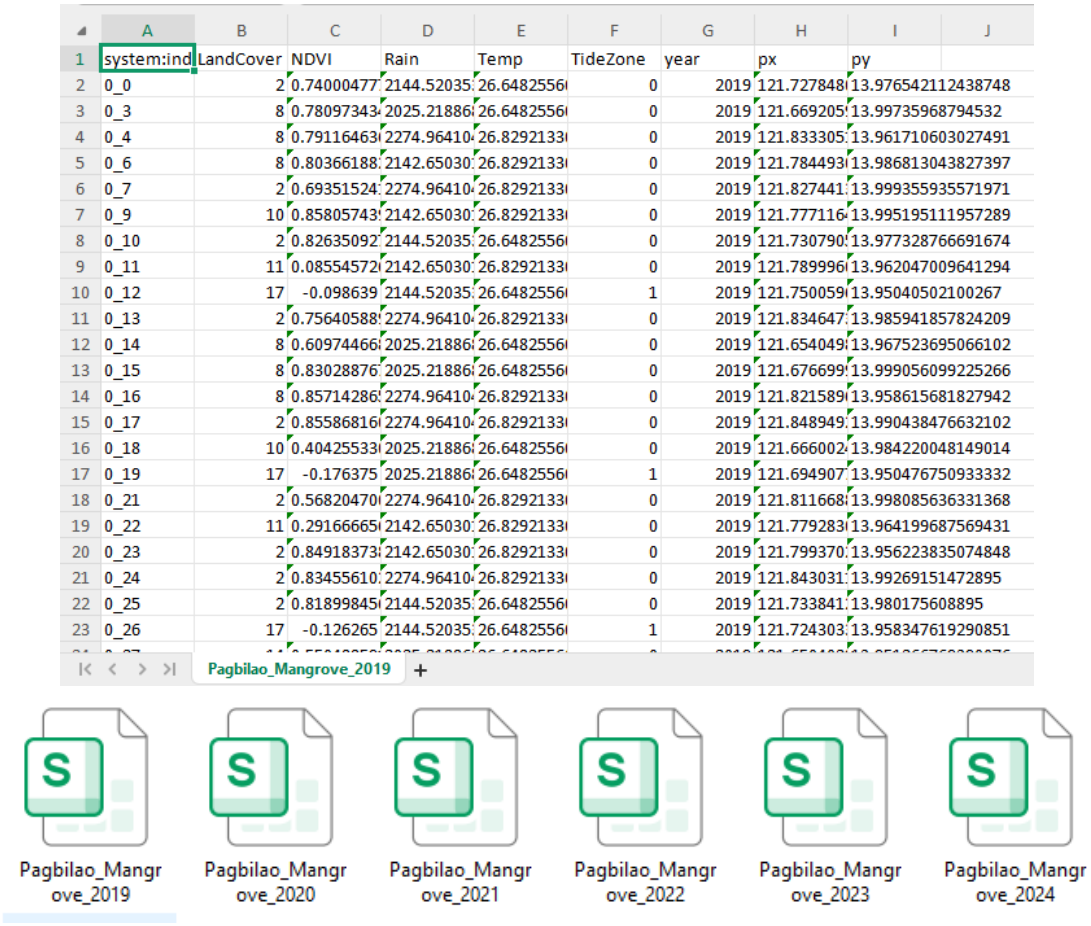
Expert's Validation

I hereby certify that I have evaluated the provided mangrove dataset and that the above ratings and comments reflect my professional judgment.

Signature Over Printed Name:  _____

APPENDIX E

Sample Input / Output



	A	B	C	D	E	F	G	H	I	J
1	system:ind	LandCover	NDVI	Rain	Temp	TideZone	year	px	py	
2	0_0	2	0.74000477	2144.52035	26.6482556	0	2019	121.727848	13.976542112438748	
3	0_3	8	0.78097343	2025.21886	26.6482556	0	2019	121.669205	13.99735968794532	
4	0_4	8	0.79116463	2274.96410	26.8292133	0	2019	121.833305	13.961710603027491	
5	0_6	8	0.80366188	2142.65030	26.8292133	0	2019	121.784493	13.986813043827397	
6	0_7	2	0.69351524	2274.96410	26.8292133	0	2019	121.827441	13.999355935571971	
7	0_9	10	0.85805743	2142.65030	26.8292133	0	2019	121.777116	13.995195111957289	
8	0_10	2	0.82635092	2144.52035	26.6482556	0	2019	121.730790	13.977328766691674	
9	0_11	11	0.08554572	2142.65030	26.8292133	0	2019	121.789996	13.962047009641294	
10	0_12	17	-0.098639	2144.52035	26.6482556	1	2019	121.750059	13.95040502100267	
11	0_13	2	0.75640588	2274.96410	26.8292133	0	2019	121.834647	13.985941857824209	
12	0_14	8	0.60974466	2025.21886	26.6482556	0	2019	121.654049	13.967523695066102	
13	0_15	8	0.83028876	2025.21886	26.6482556	0	2019	121.676699	13.999056099225266	
14	0_16	8	0.85714286	2274.96410	26.8292133	0	2019	121.821589	13.958615681827942	
15	0_17	2	0.85586816	2274.96410	26.8292133	0	2019	121.848949	13.990438476632102	
16	0_18	10	0.40425533	2025.21886	26.6482556	0	2019	121.666002	13.984220048149014	
17	0_19	17	-0.176375	2025.21886	26.6482556	1	2019	121.694907	13.950476750933332	
18	0_21	2	0.56820470	2274.96410	26.8292133	0	2019	121.811668	13.998085636331368	
19	0_22	11	0.29166665	2142.65030	26.8292133	0	2019	121.779283	13.964199687569431	
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21	0_24	2	0.83455610	2274.96410	26.8292133	0	2019	121.843031	13.99269151472895	
22	0_25	2	0.81899845	2144.52035	26.6482556	0	2019	121.733841	13.980175608895	
23	0_26	17	-0.126265	2144.52035	26.6482556	1	2019	121.724303	13.958347619290851	

Pagbilao_Mangrove_2019

Figure E1: Input Dataset before using CEA, Year 2019-2024

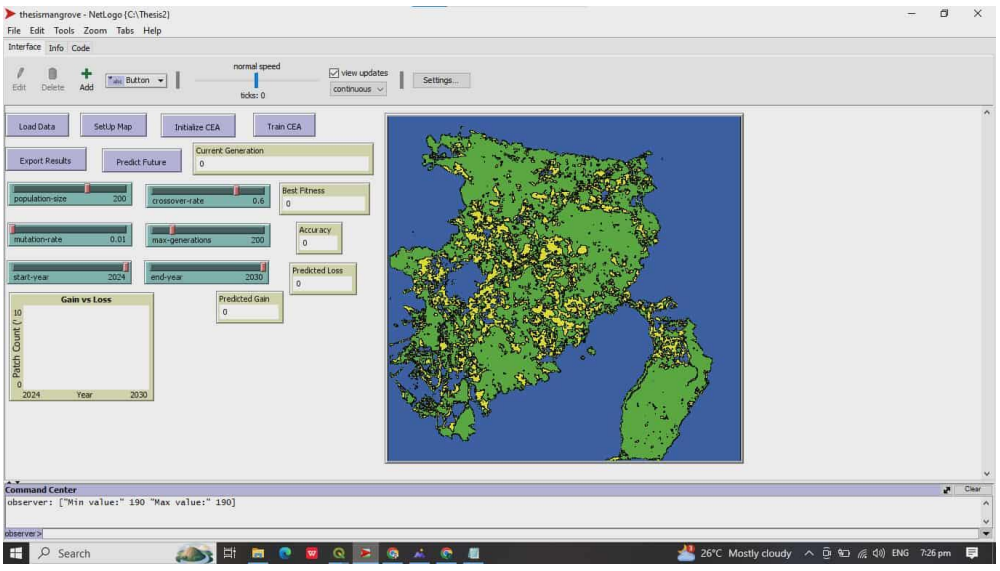


Figure E2. CEA Simulation in NetLogo in the Pagbilao Quezon Map

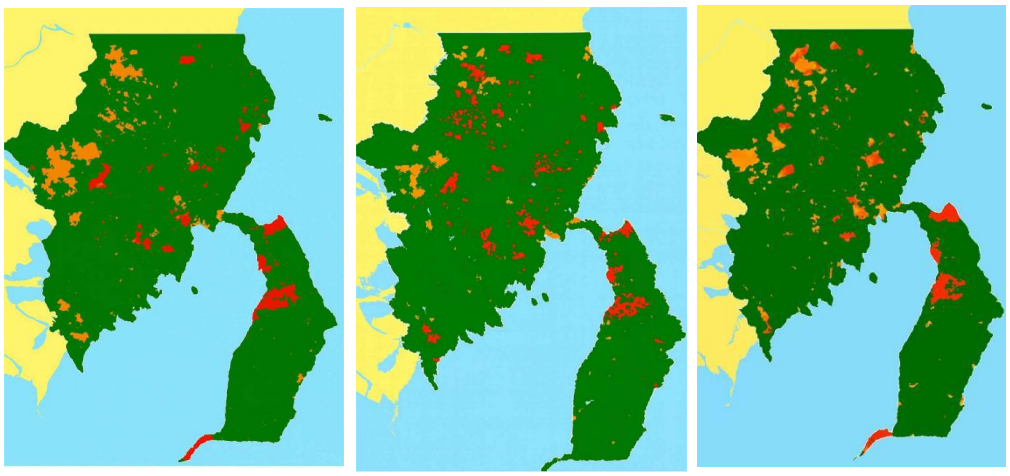


Figure E3. Output of Prediction by using CEA, Year 2025-2029

The figures illustrate the progression of data flow in the Cellular Evolutionary Algorithm (CEA) model-from the initial input dataset to the simulated and predicted spatial outputs. Figure E1 presents the input datasets used before implementing the CEA, covering the years 2019-2024. Each file contains environmental and spatial parameters such as NDVI, rainfall, temperature, tide zone, and geographic coordinates (x, y) for the Pagbilao mangrove area. These datasets serve as the foundational input for the simulation. Figure E2 displays the execution of the CEA model within the NetLogo environment, showing the Pagbilao, Quezon map where the system initializes and processes NDVI-based mangrove dynamics. The simulation integrates the environmental variables to model spatial behavior and predict vegetation changes over time. Subsequently, Figure E3 illustrates the predicted spatial outputs generated by the CEA model for the years 2025-2029. The color-coded maps represent different land cover classifications, highlighting zones of mangrove persistence, gain, and loss across the study area. This sequence of figures demonstrates how the CEA model transforms raw environmental datasets into predictive spatial visualizations that support mangrove monitoring and management in Pagbilao.

APPENDIX F

Curriculum Vitae



CONTACT

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Brgy. Mayondon Los Baños, Laguna, Philippines, 4030

SKILLS

- Management Skills
- Creativity
- Microsoft Office
- Editing (Canva)
- Critical Thinking
- Leadership
- Time Management
- Communication
- Baking
- Cooking
- Teamwork
- Adaptability

LANGUAGE

- English

XAILAH MAY M. BANASIHAN

A passionate individual who is devoted to improving language skills and cultural awareness in students while continuously adjusting to their uniqueness. A fast learner with a deep sense of dedication, and a recent Computer Science graduate eager to apply algorithmic problem-solving and a foundation in data-driven analysis to enhance personalized learning strategies and educational technology.

EXPERIENCE

FREELANCE TUTOR — 2021

March - May

Teach English to non-native speakers in a virtual set-up. Organizing instructional and learning materials assess student language ability and provide evaluation using SANDWICH method.
Primary Education Tutor, Teach various subjects and review elementary lesson in a face to face set-up.

CHILD CARE PROVIDER — 2021-2022

Shaping the early experiences and development of young children. As a child care provider, one has the unique opportunity to create a safe, nurturing, and stimulating environment that foster growth and learning in children

ENTREPRENEUR — 2022-2024

Making pastry products with my om, with a deep-rooted passion for baking, with love for creating delectable pastries, cakes, and bread also committed to provide delicious treats that not can only satisfy cravings but also create lasting memories.

EDUCATION

- **Laguna State Polytechnic University - Los Baños**

  Bachelor of Science in Computer Science
  2022-2026 | Major in Intelligent System
- **Laguna State Polytechnic University - Los Baños**

  HUmmanities of Social Sciences (HUMSS)
  2020 - 2022
- **Los Baños National HIght School - Batong Malake**

  2016 - 2020



ARCELYN P. ARCIAGA

Computer Science student with proven professional experience in fast-paced, customer-facing environments. A fast and continuous learner, known for adaptability and a deep commitment to personalized instruction.

CONTACT

09940558091

arcelynarciaga12@gmail.com

Brgy. Silangan Calauan,
Laguna, Philippines, 4012

SKILLS

- Easy Learner
- Communication
- Hard Working
- Time Management
- Team Work
- Creativity
- Adaptability
- Microsoft Office

EXPERIENCE

FASTFOOD CHAIN CREW 2024 - 2025

Bay, Laguna

Enthusiastic and reliable individual with a strong commitment to delivering exceptional customer service. Proven ability to handle high-volume demands with a positive attitude, ensuring quick order accuracy and a friendly experience for every guest.

SALES LADY JUNE - SEPTEMBER 2022

San Pablo City

Motivated and polished individual with a strong aptitude for retail sales and customer interaction. Quick to master new inventory systems and sales techniques, with a proven history of professional and reliable communication.

EDUCATION

- **Laguna State Polytechnic University - Los Baños**
  Bachelor of Science in Computer Science
  2022-2026 | Major in Intelligent System
- **CARD - MRI Development Institute, Inc.**
  Information and Communication Technology
  2020 - 2022
- **Dayap National Integrated High School**
  2016 - 2020

APPENDIX G

User's Guide

OPTIMIZING MANGROVE PRESERVATION USING CELLULAREVOLUTIONARY ALGORITHM AND ENCHANCED SPATIAL ANALYSIS FROM REMOTE SENSING DATA

ARCIAGA, ARCELYN P. BANASIHAN, XAILAH MAY M.

A system designed to optimize mangrove preservation in Pagbilao, Quezon using a Cellular Evolutionary Algorithm (CEA) integrated with remote sensing data. By analyzing spatial information such as NDVI, temperature, rainfall, and land cover, the system predicts future mangrove growth and degradation patterns. The system provide valuable insights for environmental monitoring, sustainable management, and conservation planning.

WHAT IS THIS SYSTEM FOR?

HOW THE SYSTEM WORK

1. LOAD DATA

Upload the CSV file containing NDVI, temperature, rainfall, tide zone, and land cover information from Google Earth Engine. This initializes the spatial grid of the mangrove study area in Pagbilao.

2. INITIALIZE DATA

Start the Cellular Evolutionary Algorithm (CEA) setup. The system creates an initial population of chromosomes (solutions) that represent possible NDVI prediction models for each patch

3. TRAIN MODEL

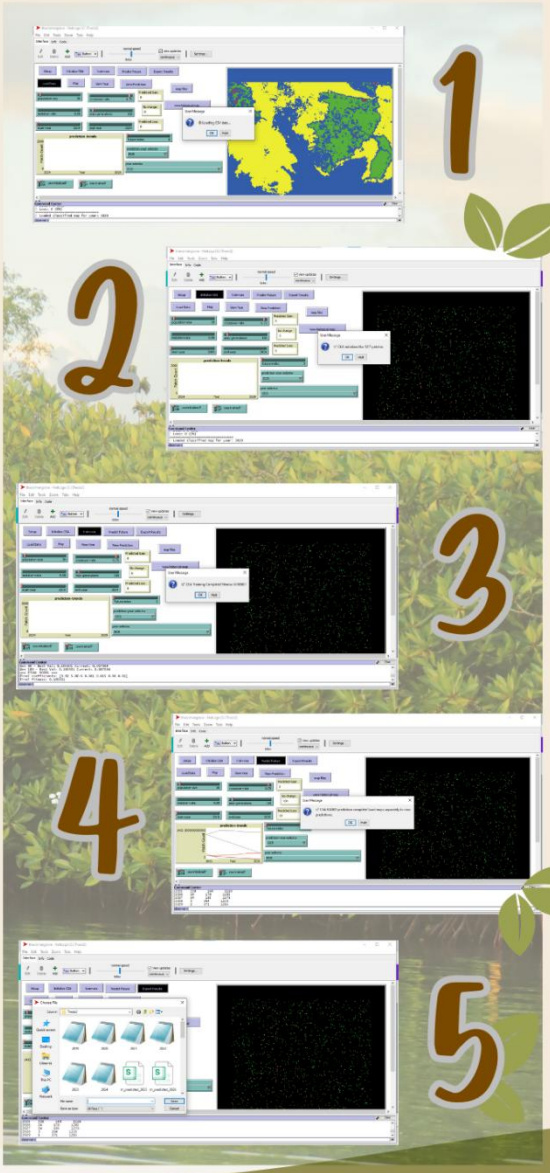
Run the training process to evolve the chromosomes through selection, crossover, and mutation. The model learns the best coefficients that predict NDVI changes based on environmental factors.

4. PREDICT FUTURE

Use the trained CEA model to simulate mangrove growth and NDVI values for the next 5 years (e.g., 2025–2029). Patches are color-coded to visualize predicted gains, losses, and stable mangrove areas.

5. EXPORT RESULTS

Save the predicted NDVI maps and computed metrics (F1 Score, Kappa, IoU, RMSE, and t-test validation). Results can be exported as CSV files for statistical analysis and comparison with actual ground-truth data.



APPENDIX H

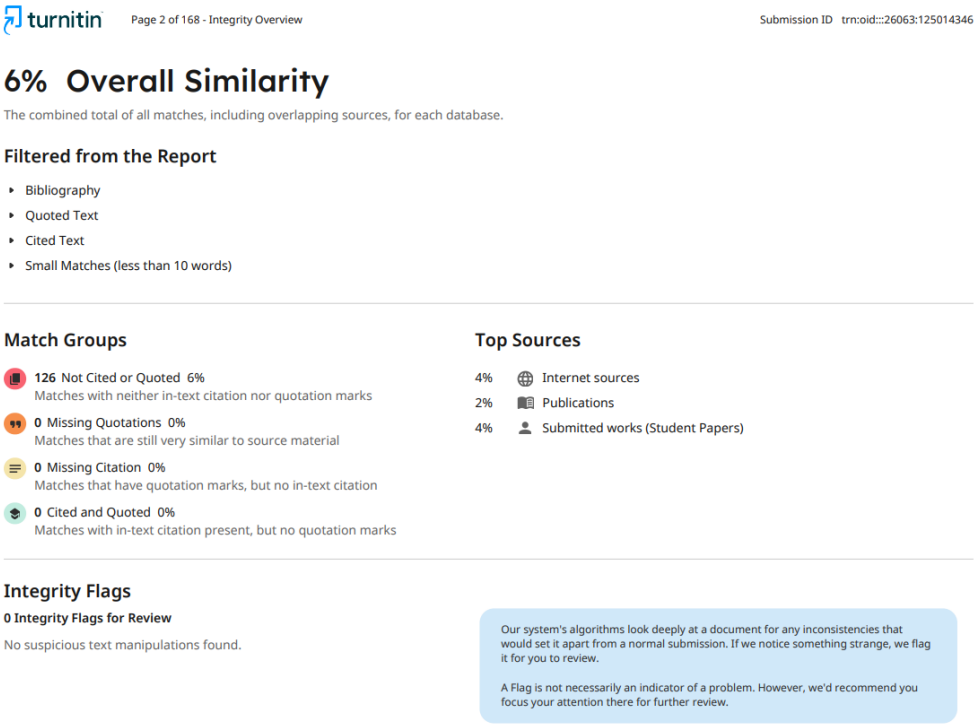
Matrix of Revision

Matrix of Revisions

Commentator	Positions	Comments	Actions	Page No.
Mr. Wilson B. Suyat	Subject Specialist	Make sure that this figure clear	The researchers that all that figure was clear	67
Mr. Wilson B. Suyat	Subject Specialist	Revise this figure make sure that the model's predictions cover all the area	The researchers use a Netlogo to prediction then input in the GIS to make sure that models predictions cover all the area	80
Dr. Sherwin B. Sapin	Statistician	You just need to end in approach	Instead of end in the ("using an Independent T-test") researcher was end in the approach	3
Dr. Sherwin B. Sapin	Statistician	Make sure that you use in GIS that predictions model cover all the area	The researchers use a Netlogo to prediction then input in the GIS to make sure that models predictions cover all the area	80
Dr. Sherwin B. Sapin	Statistician	Include the t-test, Remove a Test Type, Make sure that the Cohen's d are not negative	The researchers add a t-test and also remove a Test Type like a Student's t researcher also a Cohen's d are all positive not a negative.	86

APPENDIX I

Grammarly and Turnitin Report



turnitin Page 2 of 168 - Integrity Overview Submission ID trn:oid:::26063:125014346

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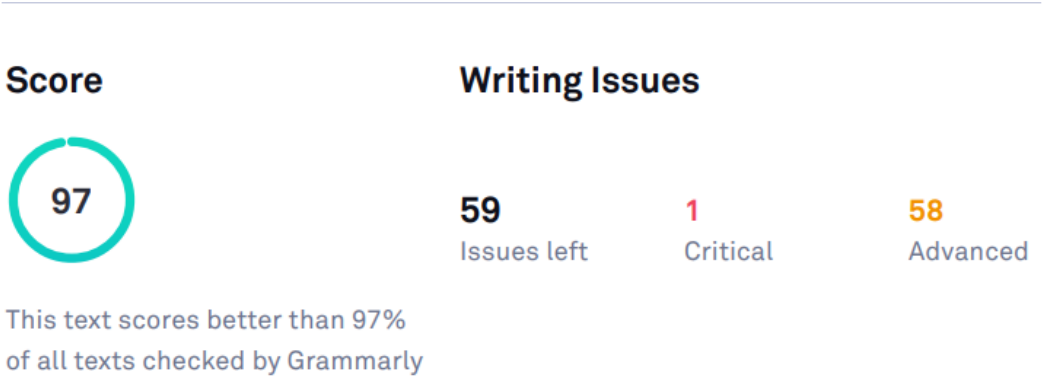
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Grammarly Report

APPENDIX J

Publishable Paper (IEEE format)

OPTIMIZING MANGROVE PRESERVATION USING CELLULAR EVOLUTIONARY ALGORITHM AND ENCHANCED SPATIAL ANALYSIS FROM REMOTE SENSING DATA

Xailah May M. Banasihan, Arcelyn P. Arciaga,
Gene Marck B Catedrilla
College of Computer Studies (CCS), Laguna State Polytechnic University, Los Baños, Laguna, Philippines

Article Info

Keywords:

Mangrove Preservation
Cellular Evolutionary Algorithm
Remote Sensing
Normalized Difference Index
Spatial Optimization
NetLogo
Sentinel-2

ABSTRACT

Mangrove forests are essential coastal ecosystems that protect shorelines from erosion and extreme weather while supporting diverse plant and animal species. However, these ecosystems are increasingly threatened by human activities and environmental change, making effective monitoring and targeted conservation strategies necessary. This study presents a computational approach to optimizing mangrove preservation in Pagbilao, Quezon by integrating satellite-based remote sensing data with a Cellular Evolutionary Algorithm (CEA). Thirty-six cloud-free Sentinel-2 images with a spatial resolution of 10 meters, acquired from 2019 to 2024, were processed to extract vegetation indicators, with the Normalized Difference Vegetation Index (NDVI) used as the primary measure of mangrove health and density. The CEA was implemented in the NetLogo platform to simulate spatial dynamics and identify priority conservation zones through neighborhood-based optimization. Model performance was assessed using accuracy, F1 score, Kappa coefficient, Intersection over Union (IoU), Root Mean Square Error (RMSE), and computation time. The results demonstrate strong model performance, achieving an accuracy of 0.94, an F1 score of 0.93, a Kappa value of 0.87, an IoU of 0.91, and a low RMSE of 0.022. Independent-samples t-test results confirmed statistically significant improvements over the baseline approach ($p < .001$). Overall, the integration of remote sensing and evolutionary computation enhanced spatial coherence and provided a reliable, data-driven basis for sustainable mangrove management.

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1. INTRODUCTION

Mangrove forests are vital coastal ecosystems that provide essential ecological and economic services, such as shoreline protection, carbon sequestration, and habitat for diverse marine species. Despite their importance, mangroves face severe threats from deforestation, aquaculture, tourism,

urbanization, and climate change, leading to their continuous decline, especially in Southeast Asia and the Philippines. Effective monitoring and preservation of mangroves require advanced analytical methods capable of handling the spatial complexity of these ecosystems.

Remote sensing and digital image processing play a crucial role in analyzing mangrove health and spatial distribution, offering detailed information for environmental management. Neural networks and computational algorithms have proven effective for extracting meaningful data from satellite imagery.
The Cellular Evolutionary Algorithm (CEA) presents a robust computational approach inspired by natural selection, capable of optimizing spatial and environmental problems by simulating evolving population-based solutions. Integrating CEA with remote sensing data enhances the spatial analysis of mangroves, supporting the identification of critical areas for conservation and effective management strategies.

Image processing presents an interesting field, with the promise of substantially improving pictorial information for human understanding. Additionally, it offers critical methods for effective image data processing, vital for storage, transmission, and effective representation for machine perception. In the analysis of remote sensing imagery of mangrove forests, these features are critical in obtaining detailed spatial information.

Objectives of the Study

The study seeks to develop and implement a Cellular Evolutionary Algorithm (CEA) that enhance the spatial analysis of mangrove ecosystems using remote sensing data, with the aim of optimizing preservation strategies by identifying critical areas for conservation, monitory degradation patterns, in particular, this research seeks to:

1. Collect and preprocess remote sensing data that focuses on the distribution, health, and changes of mangrove forests over time, utilizing satellite imagery and geospatial analysis.
2. Develop and implement CEA model using NetLogo, enabling the spatial visualization and analysis of the dynamics within the mangrove ecosystem.
3. Evaluate the performance of the proposed CEA-based model using accuracy, F1 score, Kappa coefficient, Intersection over Union (IoU), Root Mean Square Error (RMSE), and Computation Time.
4. Validate the effectiveness of a CEA model in optimizing mangrove management strategies by comparing its outcomes against a baseline.

2. METHODOLOGY

This study tries to present the design and implementation of a Cellular Evolutionary Algorithm (CEA) to enhance spatial evaluation of mangrove ecosystems based on remote sensing data. The framework will examine the use of cellular automata concepts in enhancing conservation by identifying priority areas for conservation and monitoring degradation trends with advanced spatial analysis techniques.

This study adopts a three-stage methodological framework comprising data acquisition and preprocessing, Cellular Evolutionary Algorithm (CEA) development, and model evaluation. Sentinel-2 satellite imagery was acquired and preprocessed to extract relevant spectral and spatial features associated with mangrove ecosystem conditions, including vegetation indices and anthropogenic stress indicators. The processed spatial data were then integrated into a CEA model designed to simulate localized evolutionary interactions and optimize spatial classification of mangrove health and degradation patterns. Model performance was quantitatively evaluated using standard accuracy metrics such as Intersection over Union (IoU), F1 score, Kappa coefficient, and Root Mean Square Error (RMSE), alongside spatial visualization techniques to support interpretation and validation. This methodological approach enables adaptive, data-driven analysis of mangrove ecosystem dynamics and supports evidence-based spatial decision-making for conservation planning.

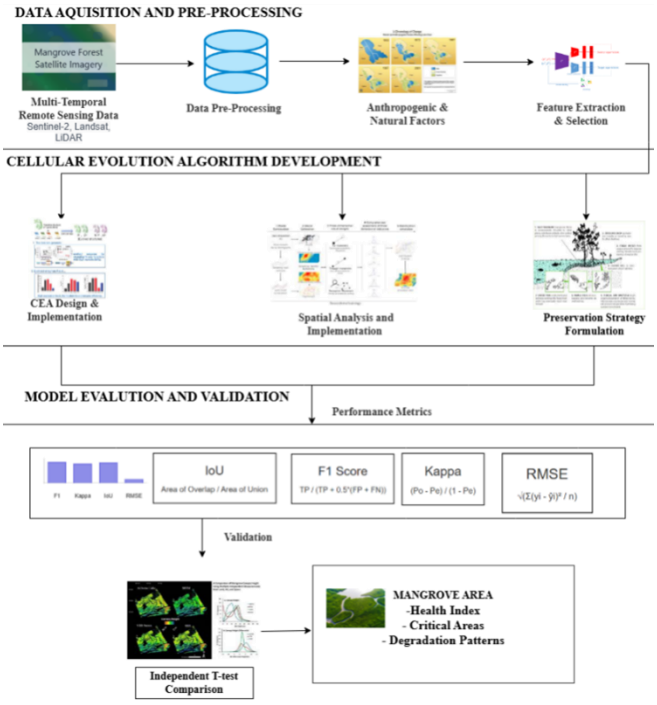


Figure 1. Conceptual Framework Integrating Remote Sensing and Cellular Evolutionary Algorithm

3.1. DATA GATHERING

The study gathered data through Google Earth Engine the datasets compiled for this study provides a spatially explicit characterization of mangrove forest structure and composition across various locations. Each record within the dataset corresponds to a specific Quadrat Location of Pagbilo, Quezon, identified by its geographical Coordinates, and its relative Position within the mangrove forest. Furthermore, the data includes the dominant Forest Type observed within that quadrat.

3.2. MODEL ARCHITECTURE

In this study centers on the implementation of a Cellular Evolutionary Algorithm (CEA), designed to optimize land cover classification using Sentinel-2 satellite imagery, as depicted in the provided "Sentinel-2 Simulation Flow Using Cellular Evolutionary Algorithm (CEA)". Following the Data Acquisition and Preprocessing steps outlined in the flow, which prepare the Sentinel-2 data through atmospheric correction, resampling, cloud masking, and subsetting, the CEA is applied to the extracted Features.

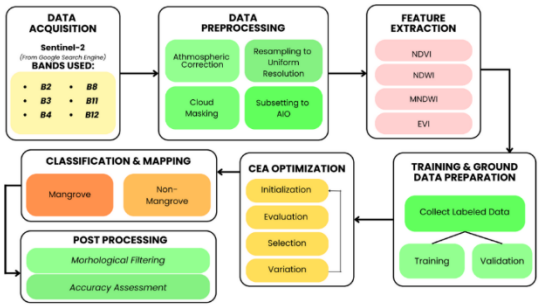


Figure 2. Model Development

This framework presents an integrated approach to optimize spatially explicit conservation planning for mangrove ecosystems. It combines remote sensing data, specifically Sentinel-2 imagery, with a Cellular Evolutionary Algorithm (CEA) to address the complex spatial dynamics inherent in these environments. The process begins with the acquisition and preprocessing of Sentinel-2 data, focusing on extracting key biophysical indicators like NDVI, NDWI/MNDWI, and EVI. This preprocessed data then drives the CEA, an optimization algorithm that iteratively evolves potential management strategies across a spatial grid.

4. RESULTS AND DISCUSSION

i. Exploratory Data Analysis

The data gathered relied exclusively on the Google Earth Engine (GEE) cloud computing platform to manage and process the vast catalog of Sentinel-2 imagery. The data analysis commenced with a thorough examination of the multi-variable dataset derived from the Pagbilao, Quezon mangrove area, integrating remote sensing indices with local climate data across 9,000 individual sampling points. The primary objective of this investigation was to quantify the relationship between mangrove health and co-located environmental and climatic drivers.

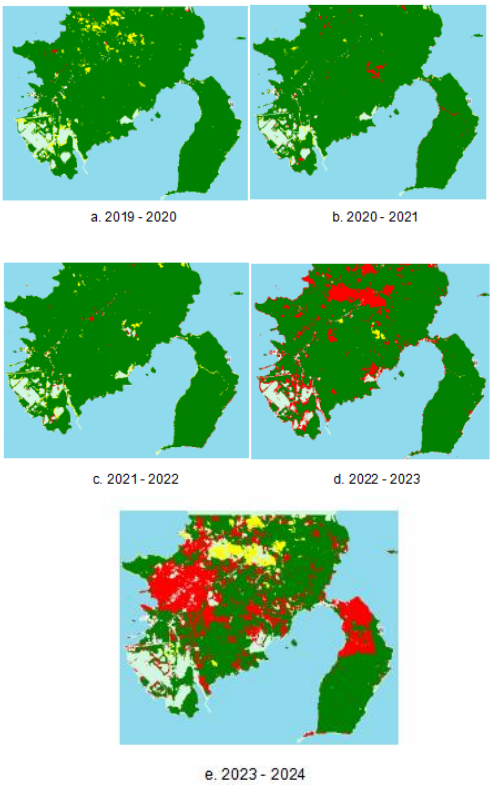


Figure 3. NDVI Distribution from 2019-2024

The Figure 3 shows that the Normalized Difference Vegetation Index from 2019 to 2021, the mangrove forests were predominantly intact and healthy (green) with signs of minor natural regeneration (yellow), but a sharp negative turn began from 2021 to 2022 with the emergence of degradation (red) linked to human pressures like development and aquaculture. This loss accelerated significantly into 2022-2023, with red areas expanding rapidly, highlighting the ecosystem's vulnerability. The period of 2023-2024 proved to be a critical turning point, showing extensive mangrove loss alongside localized patches of new growth; however, the severity and scale of deforestation now surpass regeneration efforts, underscoring an urgent need for strengthened protection and restoration programs to prevent further degradation of this vital coastal ecosystem.

Table 1. NDVI Range Based on Classification

NDVI Range	Classification
NDVI < 0.2	Bare soil, water bodies, or non-vegetated areas
0.2 ≤ NDVI < 0.4	Sparse or degraded vegetation
0.4 ≤ NDVI ≤ 0.6	Moderate vegetation (typical of recovering mangroves)
NDVI > 0.6	Dense and healthy mangrove vegetation

The table outlines the main classification system for Normalized Difference Vegetation Index values, which help assess vegetation health and density. NDVI values range from -1 to +1. Higher values mean healthier and denser vegetation. An NDVI value below 0.2 points to areas like water bodies or bare soil, showing little to no vegetation cover. Values between 0.2 and 0.4 indicate sparse or degraded vegetation, typical of stressed or limited plant growth. Moderate vegetation shows values from 0.4 to 0.6.

ii. ***Predicted mangrove cover dynamics in Pagbilao, Quezon***

The images illustrate the predicted spatial-temporal changes, specifically the distinct intermingling of blue (expansion) and red (contraction) patches, offer crucial insights into the heterogeneous impact of local drivers

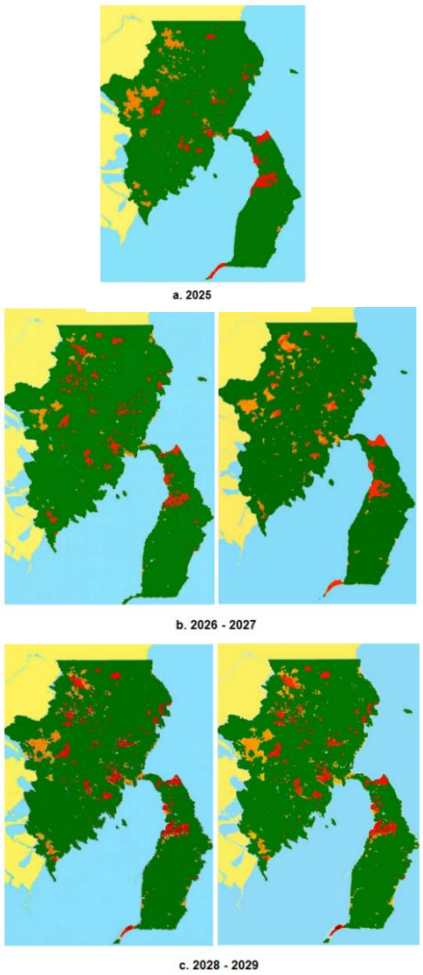


Figure 4. Prediction of Simulation in NetLogo

The areas of contraction (red) are likely concentrated near human settlements or existing aquaculture ponds, suggesting that anthropogenic pressure, such as unauthorized clearing for resource use or coastal urban creep, remains a primary threat. Conversely, the successful regeneration in blue regions often correlates with government-protected areas, successful community-based restoration projects, or natural deposition zones characterized by optimal sediment accretion and low wave energy. This state indicates that the future of Pagbilao's mangroves is a result of a constant, localized tug-of-war between conservation efforts and development pressures, requiring a management approach that is highly stratified and responsive to micro-environmental conditions.

iii. Model Validation Process

Table 2. Model's Performance using Metrics

Metrics	Value
Accuracy	0.94
F1_Score	0.93
Kappa	0.87
IoU	0.91
RMSE	0.022

The Cellular Evolutionary Algorithm CEA model achieves impressive performance across all metrics, suggesting it is a robust, reliable, and spatially accurate tool for long-term mangrove monitoring and prediction.

Table 3. Independent Sample T-test Resuts

Metric	Test Type	df	p-value	Mean Difference	SE Difference	95% Confidence Interval (Lower–Upper)	Effect Size (Cohen's d)
Accuracy	Student's t	998	< .001	-0.2183	0.00118	-0.2206 – -0.2160	-11.69
F1 Score	Student's t	998	< .001	-0.2488	0.00156	-0.2519 – -0.2457	-10.11
Kappa	Student's t	998	< .001	-0.2208	0.00142	-0.2236 – -0.2180	-9.82
IoU	Student's t	998	< .001	-0.2760	0.00254	-0.2810 – -0.2711	-6.89
RMSE	Student's t	998	< .001	0.0637	0.00068	0.0624 – 0.0650	5.92

The study used independent samples T-test definitively confirms that the superiority of the Cellular Evolutionary Algorithm (CEA)-Optimized model over the Baseline Traditional model. Statistical analysis showed that the performance differences across all metrics-Accuracy, F1-Score, Kappa, IoU, and RMSE-are highly significant (p-values < 0.001). Specifically, the optimized model achieved substantially higher values for the desirable metrics (Accuracy, F1, Kappa, IoU, indicated by negative mean differences) and significantly lower error (RMSE, indicated by a positive mean difference). This statistical significance is complemented by exceptionally large effect sizes (Cohen's d between 5.92 and 11.69), confirming that the improvements are practically meaningful and represent a powerful impact of the CEA process. This robust evidence underscores that the CEA successfully refined the model, resulting in a more reliable, accurate, and spatially precise predictive tool, making it the clear choice for real-world applications requiring high-fidelity spatial classification.

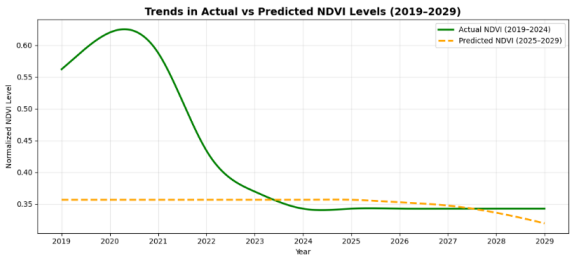


Figure 10. Trends: Simulation Vs Actual Data

The temporal analysis of the Normalized Difference Vegetation Index (NDVI) from 2019 to 2024 highlights significant shifts in vegetation health. Initially, the period from 2019 to 2021 showed a strong increase in NDVI, peaking at approximately 0.62, reflecting a phase of enhanced vegetation vigor likely due to favorable conditions. However, a sharp and concerning decline followed from 2021 to 2024, with values plummeting to around 0.35, suggesting substantial deterioration due to environmental stressors like erosion or salinity intrusion. The predictive model forecasts a period of stability at this reduced level (0.35-0.36) between 2025 and 2027, followed by a slight downward trend toward 2029, indicating that the vegetation's health is expected to remain vulnerable unless active restoration measures are implemented.

5. CONCLUSION AND RECOMMENDATION

I. Conclusion

The results concludes that Sentinel-2–based NDVI analysis combined with a CEA model provides a robust and accurate approach for simulating spatiotemporal mangrove changes. The CEA-optimized model achieved high classification performance (Accuracy = 0.94, Kappa = 0.87) and statistically significant improvements over baseline methods, confirming its reliability and predictive strength. Overall, the approach offers a scalable, data-driven foundation for mangrove monitoring, restoration prioritization, and sustainable coastal management.

II. Recommendation

1. Enhance the CEA-based mangrove monitoring framework by integrating additional vegetation indices (e.g., EVI, SAVI), biophysical variables, and climatic factors to improve ecological realism and predictive accuracy.
2. Validate and refine the model using high-resolution datasets from UAV and LiDAR platforms, and integrate GIS-based spatial analysis to support fine-scale monitoring, visualization, and restoration prioritization.
3. Explore hybrid modeling approaches by combining the CEA framework with deep learning techniques (e.g., CNN or UNet) to strengthen spectral–spatial feature extraction, boundary detection, and temporal forecasting.
4. Improve computational efficiency and scalability to enable large-area, long-term, and near–real-time mangrove monitoring and simulation.
5. Integrate the validated CEA-based system into operational workflows of environmental agencies and local governments to support proactive, data-driven mangrove conservation, disaster risk reduction, and coastal management policies.

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